Basic Stochastic Processes

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Contents

CHAPTER 1

Martingale Inequalities and Convergence

1. Doob's Martingale Inequalities

Proposition 1.1 (Doob's maximal inequality)**.** *Suppose that $\xi_n, n \in \mathbb{N}$, is a non-negative submartingale (with respect to a filtration $\mathcal{F}_n$). Then for any $\lambda > 0$*

$$\lambda P\left(\max_{k\leq n} \xi_k \geq \lambda\right) \leq E\left(\xi_n 1_{\left\{\max_{k\leq n} \xi_k \geq \lambda\right\}}\right),$$

where 1_A is the characteristic function of a set A.

Proof. *We put $\xi_n^* = \max_{k\leq n} \xi_k$ for brevity. For $\lambda > 0$ let us define*

$$\tau = \min\left\{k \leq n : \xi_k \geq \lambda\right\},$$

if there is a $k \leq n$ such that $\xi_k \geq \lambda$, and $\tau = n$ otherwise. Then τ is a stopping time such that $\tau \leq n$ a.s. Since ξ_n is a submartingale,

$$E(\xi_n) \geq E(\xi_\tau).$$

But

$$E(\xi_\tau) = E\left(\xi_\tau 1_{\{\xi_n^* \geq \lambda\}}\right) + E\left(\xi_\tau 1_{\{\xi_n^* < \lambda\}}\right).$$

Observe that if $\xi_n^* \geq \lambda$, then $\xi_\tau \geq \lambda$. Moreover, if $\xi_n^* < \lambda$, then $\tau = n$, and so $\xi_\tau = \xi_n$. Therefore

$$E(\xi_n) \geq E(\xi_\tau) \geq \lambda P(\xi_n^* \geq \lambda) + E(\xi_n 1_{\{\xi_n^* < \lambda\}}),$$

It follows that

$$\lambda P\left(\xi_n^* \geq \lambda\right) \leq E\left(\xi_n\right) - E\left(\xi_n 1_{\{\xi_n^* < \lambda\}}\right) = E\left(\xi_n 1_{\{\xi_n^* \geq \lambda\}}\right),$$

completing the proof. □

Theorem 1.2 (Doob's maximal L^2 inequality)**.** *If $\xi_n, n \in \mathbb{N}$, is a non-negative square integrable submartingale (with respect to a filtration $\mathcal{F}_n$), then*

$$E\left|\max_{k\leq n} \xi_k\right|^2 \leq 4E|\xi_n|^2. \tag{4.2}$$

Proof. *Put $\xi_n^* = \max_{k\leq n} \xi_k$. By Exercise 1.9, Proposition 1.1, the Fubini theorem and finally the Cauchy-Schwarz inequality*

$$\begin{aligned} E\left|\xi_n^*\right|^2 &= 2\int_0^\infty tP\left(\xi_n^* > t\right) dt \leq 2\int_0^\infty E\left(\xi_n 1_{\{\xi_n^* \geq t\}}\right) dt \\ &= 2\int_0^\infty \left(\int_{\{\xi_n^* \geq t\}} \xi_n dP\right) dt = 2\int_\Omega \xi_n \left(\int_0^{\xi_n^*} dt\right) dP \\ &= 2\int_\Omega \xi_n \xi_n^* dP = 2E(\xi_n \xi_n^*) \leq 2(E|\xi_n|^2)^{1/2}(E|\xi_n^*|^2)^{1/2}. \end{aligned}$$

Dividing by $(E|\xi_n^*|^2)^{1/2}$, we get (4.2). □

The proof of Doob's Convergence Theorem in the next section hinges on an inequality involving the number of upcrossings.

Definition 1.3. *Given an adapted sequence of random variables $\xi_1, \xi_2, \dots$ and two real numbers $a < b$, we define a gambling strategy $\alpha_1, \alpha_2, \dots$ by putting*

$$\alpha_1 = 0$$

and for $n = 1, 2, \dots$

$$\alpha_{n+1} = \begin{cases} 1 & \text{if } \alpha_n = 0 \text{ and } \xi_n < a, \\ 1 & \text{if } \alpha_n = 1 \text{ and } \xi_n \le b, \\ 0 & \text{otherwise.} \end{cases}$$

It will be called the ***upcrossings strategy****. Each $k = 1, 2, \dots$ such that $\alpha_k = 1$ and $\alpha_{k+1} = 0$ will be called an upcrossing of the interval $[a, b]$. The upcrossings form a (finite or infinite) increasing sequence*

$$u_1 < u_2 < \dots .$$

The number of upcrossings made up to time n, that is, the largest k such that $u_k \le n$ will be denoted by $U_n[a, b]$ (we put $U_n[a, b] = 0$ if no such k exists).

The meaning of the above definition is this. Initially, we refrain from playing the game and wait until ξ_n becomes less than a. As soon as this happens, we start playing unit stakes at each round of the game and continue until ξ_n becomes greater than b. At this stage we refrain from playing again, wait until ξ_n becomes less than a, and so on. The strategy α_n is defined in such a way that $\alpha_n = 0$ whenever we refrain from playing the n-th game, and $\alpha_n = 1$ otherwise. During each run of consecutive games with $\alpha_n = 1$ the process ξ_n crosses the interval $[a, b]$, starting below a and finishing above b. This is what is meant by an upcrossing. Observe that each upcrossing will increase our total winnings by at least $b - a$. For convenience, we identify each upcrossing with its last step k, such that $\alpha_k = 1$ and $\alpha_{k+1} = 0$. A typical sample path of the upcrossings strategy is shown in Figure 1.

Figure 1. Typical paths of ξ_n, α_n and ζ_n; upcrossings are indicated by bold lines

Exercise 1.1. *Verify that the upcrossings strategy α_n is indeed a gambling strategy.*

Hint: You want to prove that α_n is $\mathcal{F}_{n-1}$-measurable for each n. Since the upcrossings strategy is defined by induction, a proof by induction on n may be your best bet.

Lemma 1.4 (Upcrossings Inequality). *If $\xi_1, \xi_2, \dots$ is a supermartingale and $a < b$, then*

$$(b - a) E(U_n[a, b]) \le E((\xi_n - a)^-).$$

By x^- we denote the negative part of a real number x, i.e. $x^- = \max\{0, -x\}$.

Proof. *Let*

$$\zeta_n = \alpha_1(\xi_1 - \xi_0) + \dots + \alpha_n(\xi_n - \xi_{n-1})$$

be the total winnings at step $n = 1, 2, \dots$ if the upcrossings strategy is followed, see Figure 1. It will be convenient to put $\zeta_0 = 0$. By Proposition 3.1 (one cannot beat the system using a gambling strategy) ζ_n is a supermartingale.

Let us fix an n and put $k = U_n[a, b]$, so that

$$0 < u_1 < u_2 < \dots < u_k \le n.$$

Clearly, each upcrossing increases the total winnings by $b - a$,

$$\zeta_{u_i} - \zeta_{u_{i-1}} \ge b - a$$

for $i = 1, \ldots, k$. (We put $u_0 = 0$ for simplicity.) Moreover,

$$\zeta_n - \zeta_{u_k} \geq -(\xi_n - a)^-.$$

It follows that

$$\zeta_n \geq (b-a)U_n[a,b] - (\xi_n - a)^-.$$

Taking the expectation on both sides, we get

$$E(\zeta_n) \geq (b-a)E(U_n[a,b]) - E((\xi_n - a)^-).$$

But ζ_n is a supermartingale, so

$$0 = E(\zeta_1) \geq E(\zeta_n),$$

which proves the Upcrossings Inequality. □

2. Doob's Martingale Convergence Theorem

Theorem 1.5 (Doob's Martingale Convergence Theorem)**.** *Suppose that $\xi_1, \xi_2, \ldots$ is a supermartingale (with respect to a filtration $\mathcal{F}_1, \mathcal{F}_2, \ldots$) such that*

$$\sup_n E(|\xi_n|) < \infty.$$

Then there is an integrable random variable ξ such that

$$\lim_{n\to\infty} \xi_n = \xi \quad a.s.$$

Remark 1.1. *In particular, the theorem is valid for martingales because every martingale is a supermartingale. It is also valid for submartingales, since ξ_n is a submartingale if and only if $-\xi_n$ is a supermartingale.*

Remark 1.2. *Observe that even though all the ξ_n as well as the limit ξ are integrable random variables, it is claimed only that ξ_n trends to ξ a.s. Note that no convergence in L^1 is asserted.*

Proof (Proof of Doob's Martingale Convergence Theorem). *By the Upcrossings Inequality (Lemma 1.4)*

$$E(U_n[a,b]) \leq \frac{E\left((\xi_n - a)^-\right)}{b-a} \leq \frac{M + |a|}{b-a} < \infty,$$

where

$$M = \sup_n E(|\xi_n|) < \infty.$$

Since $U_n[a,b]$ is a non-decreasing sequence, it follows that

$$E\left(\lim_{n\to\infty} U_n[a,b]\right) = \lim_{n\to\infty} E(U_n[a,b]) \leq \frac{M+|a|}{b-a} < \infty.$$

This implies that

$$P\left\{\lim_{n\to\infty} U_n[a,b] < \infty\right\} = 1$$

for any $a < b$. Since the set of all pairs of rational numbers $a < b$ is countable, the event

$$A = \bigcap_{a<b \text{ rational}} \left\{\lim_{n\to\infty} U_n[a,b] < \infty\right\} \tag{4.3}$$

has probability 1. (The intersection of countably many events has probability 1 if each of these events has probability 1.)

We claim that the sequence ξ_n *converges a.s. to a limit* ξ. *Consider the set*

$$B = \left\{ \liminf_n \xi_n < \limsup_n \xi_n \right\} \subset \Omega$$

on which the sequence ξ_n *fails to converge. Then for any* $\omega \in B$ *there are rational numbers* a, b *such that*

$$\liminf_n \xi_n(\omega) < a < b < \limsup_n \xi_n(\omega),$$

implying that $\lim_{n\to\infty} U_n[a,b](\omega) = \infty$. *This means that* B *and the event* A *in (4.3) are disjoint, so* $P(B) = 0$, *since* $P(A) = 1$, *which proves the claim.*

It remains to show that the limit ξ *is an integrable random variable. By Fatou's lemma*

$$\begin{aligned} &E(|\xi|) = E\left(\lim_n \inf |\xi_n|\right) \\ &\leq \text{liminf}_n\, E(|\xi_n|) \\ &< \sup_n E(|\xi_n|) < \infty. \end{aligned}$$

This completes the proof. □

Exercise 1.2. *Show that if* ξ_n *is a non-negative supermartingale, then it converges a.s. to an integrable random variable.*

Hint: To apply Doob's Theorem all you need to verify is that the sequence ξ_n *is bounded in* L^1, *i.e. the supremum of* $E(|\xi_n|)$ *is less than* ∞.

3. Uniform Integrability and L^1 Convergence of Martingales

The conditions of Doob's theorem imply pointwise (a.s.) convergence of martingales. In this section we shall study convergence in L^1. To this end we introduce a stronger condition called uniform integrability. Proposition 1.7 shows that it is a necessary condition for L^1 convergence. In Theorem 1.9 we prove that uniform integrability is in fact sufficient for a martingale to converge in L^1. This enables us to show that each integrable martingale is of the form $E(\xi|\mathcal{F}_n)$. As an application we give a martingale proof of Kolmogorov's 0-1 law.

Exercise 1.3. *Show that a random variable* ξ *is integrable if and only if for every* $\varepsilon > 0$ *there exists an* $M > 0$ *such that*

$$\int_{\{|\xi|>M\}} |\xi| dP < \varepsilon.$$

Hint: Split Ω *into two sets:* $\{|\xi| > M\}$ *and* $\{|\xi| \leq M\}$. *The integrals of* $|\xi|$ *over these sets must add up to* $E(|\xi|)$. *As* M *increases, one of the integrals increases, while the other one decreases. Investigate their limits as* $M \to \infty$.

Thus, for any sequence ξ_n of integrable random variables and any $\varepsilon > 0$ there is a sequence of numbers $M_n > 0$ such that

$$\int_{\{|\xi_n|>M_n\}} |\xi_n| dP < \varepsilon.$$

If the M_n are independent of n, then we say that the sequence ξ_n is uniformly integrable.

Definition 1.6. *A sequence* $\xi_1, \xi_2, \dots$ *of random variables is called* ***uniformly integrable*** *if for every* $\varepsilon > 0$ *there exists an* $M > 0$ *such that*

$$\int_{\{|\xi_n|>M\}} |\xi_n| dP < \varepsilon$$

for all $n = 1, 2, \dots$

EXERCISE 1.4. *Let $\Omega = [0,1]$ with the σ-field of Borel sets and Lebesgue measure. Take*

$$\xi_n = n1_{(0,\frac{1}{n})}.$$

Show that the sequence $\xi_1, \xi_2, \dots$ is not uniformly integrable.

Hint: What is the integral of ξ_n over $\{\xi_n > M\}$ if $n > M$?

Proposition 1.7. *Uniform integrability is a necessary condition for a sequence $\xi_1, \xi_2, \dots$ of integrable random variables to converge in L^1.*

Lemma 1.8. *If ξ is integrable, then for every $\varepsilon > 0$ there is a $\delta > 0$ such that*

$$P(A) < \delta \Longrightarrow \int_A |\xi| dP < \varepsilon.$$

PROOF (Proof of Lemma 1.8). *Let $\varepsilon > 0$. Since ξ is integrable, by Exercise 1.3 there is an $M > 0$ such that*

$$\int_{\{|\xi|>M\}} |\xi| dP < \frac{\varepsilon}{2}.$$

Now

$$\begin{aligned} &\textstyle\int_A |\xi| dP = \int_{A\cap\{|\xi|\le M\}} |\xi| dP + \int_{A\cap\{|\xi|>M\}} |\xi| dP \\ &\le \textstyle\int_A M dP + \int_{\{|\xi|>M\}} |\xi| dP \\ &< MP(A) + \tfrac{\varepsilon}{2}. \end{aligned}$$

Let $\delta = \frac{\epsilon}{2M}$. Then

$$P(A) < \delta \Longrightarrow \int_A |\xi| dP < \varepsilon,$$

as required. $\square$

EXERCISE 1.5. *Let ξ be an integrable random variable and $\mathcal{F}_1, \mathcal{F}_2, \dots$ a filtration. Show that $E(\xi|\mathcal{F}_n)$ is a uniformly integrable martingale.*

Hint: Use Lemma 1.8.

PROOF (Proof of Proposition 1.7). *Suppose that $\xi_n \to \xi$ in L^1, i.e. $E|\xi_n - \xi| \to 0$. We take any $\varepsilon > 0$. There is an integer N such that*

$$n \ge N \Longrightarrow E|\xi_n - \xi| < \frac{\varepsilon}{2}.$$

By Lemma 1.8 there is a $\delta > 0$ such that

$$P(A) < \delta \Longrightarrow \int_A |\xi| dP < \frac{\varepsilon}{2}.$$

Taking a smaller $\delta > 0$ if necessary, we also have

$$P(A) < \delta \Longrightarrow \int_A |\xi_n| dP < \varepsilon \quad \text{for } n = 1, \dots, N.$$

We claim that there is an $M > 0$ such that

$$P\{|\xi_n| > M\} < \delta$$

for all n. Indeed, since

$$E(|\xi_n|) \ge \int_{\{|\xi_n|>M\}} |\xi_n| dP \ge MP\{|\xi_n| > M\},$$

it suffices to take

$$M = \frac{1}{\delta} \sup_n E(|\xi_n|).$$

(Because the sequence ξ_n converges in L^1, it is bounded in L^1, so the supremum is $< \infty$.)

Now, since $P\{|\xi_n| > M\} < \delta$,

$$\begin{aligned}&\textstyle\int_{\{|\xi_n|>M\}} |\xi_n| dP \leq \int_{\{|\xi_n|>M\}} |\xi| dP + \int_{\{|\xi_n|>M\}} |\xi_n - \xi| dP \\ &\leq \textstyle\int_{\{|\xi_n|>M\}} |\xi| dP + E(|\xi_n - \xi|) \\ &< \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon.\end{aligned}$$

for any $n > N$ and

$$\int_{\{|\xi_n|>M\}} |\xi_n| dP < \varepsilon$$

for any $n = 1, \ldots, N$, completing the proof.

Exercise 1.6. *Show that a uniformly integrable sequence of random variables is bounded in L^1, i.e.*

$$\sup_n E\left(|\xi_n|\right) < \infty.$$

Hint: Write $E(|\xi_n|)$ as the sum of the integrals of $|\xi_n|$ over $\{|\xi_n| > M\}$ and $\{|\xi_n| \leq M\}$.

Exercise 1.6 implies that each uniformly integrable martingale satisfies the conditions of Doob's theorem. Therefore it converges a.s. to an integrable random variable. We shall show that in fact it converges in L^1.

Theorem 1.9. *Every uniformly integrable supermartingale (submartingale) ξ_n converges in L^1.*

Proof. *By Exercise 1.6 the sequence ξ_n is bounded in L^1, so it satisfies the conditions of Theorem 1.5 (Doob's Martingale Convergence Theorem). Therefore, there is an integrable random variable ξ such that $\xi_n \to \xi$ a.s. We can assume without loss of generality that $\xi = 0$ (since $\xi_n - \xi$ can be taken in place of ξ_n). That is to say,*

$$P\left\{\lim_n \xi_n = 0\right\} = 1.$$

It follows that $\xi_n \to 0$ in probability, i.e. for any $\varepsilon > 0$

$$P\left\{|\xi_n| > \varepsilon\right\} \to 0$$

as $n \to \infty$. This is because by Fatou's lemma

$$\begin{aligned}&\limsup\nolimits_n P\left\{|\xi_n| > \varepsilon\right\} \leq P\left(\limsup\nolimits_n \left\{|\xi_n| > \varepsilon\right\}\right) \\ &\leq P\left(\Omega \setminus \left\{\lim\nolimits_n \xi_n = 0\right\}\right) \\ &= 0.\end{aligned}$$

Let $\varepsilon > 0$. By uniform integrability there is an $M > 0$ such that

$$\int_{\{|\xi_n|>M\}} |\xi_n| dP \leq \frac{\varepsilon}{3}$$

for all n. Since $\xi_n \to 0$ in probability, there is an integer N such that if $n > N$, then

$$P\left\{|\xi_n| > \frac{\varepsilon}{3}\right\} < \frac{\varepsilon}{3M}.$$

We can assume without loss of generality that $M > \frac{\epsilon}{3}$. *Then*

$$\begin{aligned}&E\left(|\xi_n|\right) = \int_{\{|\xi_n|>M\}} |\xi_n|\, dP + \int_{\left\{M \geq |\xi_n| > \frac{\varepsilon}{3}\right\}} |\xi_n|\, dP \\ &+ \int_{\left\{\frac{\varepsilon}{3} \geq |\xi_n|\right\}} |\xi_n| dP \\ &\leq \tfrac{\varepsilon}{3} + MP\left\{|\xi_n| > \tfrac{\varepsilon}{3}\right\} + \tfrac{\varepsilon}{3} P\left\{\tfrac{\varepsilon}{3} \geq |\xi_n|\right\} \\ &< \varepsilon\end{aligned}$$

for all $n > N$. *This proves that* $E(|\xi_n|) \to 0$, *that is,* $\xi_n \to 0$ *in* L^1. □

Theorem 1.10. *Let* ξ_n *be a uniformly integrable martingale. Then*

$$\xi_n = E(\xi|\mathcal{F}_n),$$

where $\xi = \lim_n \xi_n$ *is the limit of* ξ_n *in* L^1 *and* $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$ *is the filtration generated by* ξ_n.

Proof. *For any* $m > n$

$$E\left(\xi_m|\mathcal{F}_n\right) = \xi_n,$$

i.e. for any $A \in \mathcal{F}_n$

$$\int_A \xi_m dP = \int_A \xi_n dP.$$

Let n *be an arbitrary integer and let* $A \in \mathcal{F}_n$. *For any* $m > n$

$$\begin{aligned}&\left|\int_A (\xi_n - \xi) dP\right| = \left|\int_A (\xi_m - \xi) dP\right| \\ &\leq \int_A |\xi_m - \xi| dP \\ &\leq E\left(|\xi_m - \xi|\right) \to 0\end{aligned}$$

as $m \to \infty$. *It follows that*

$$\int_A \xi_n dP = \int_A \xi dP$$

for any $A \in \mathcal{F}_n$, *so* $\xi_n = E(\xi|\mathcal{F}_n)$. □

Exercise 1.7. *Show that if* ξ_n *is a martingale and* $\xi_n \to a$ *in* L^1 *for some* $a \in \mathbb{R}$, *then* $\xi_n = a$ *a.s. for each* n.

Hint: Apply Theorem 1.10.

Theorem 1.11 (Kolmogorov's 0–1 Law)**.** *Let* $\eta_1, \eta_2, \ldots$ *be a sequence of independent random variables. We define the tail* σ*-field*

$$\mathcal{T} = \mathcal{T}_1 \cap \mathcal{T}_2 \cap \ldots,$$

where $\mathcal{T}_n = \sigma(\eta_n, \eta_{n+1}, \ldots)$. *Then*

$$P(A) = 0 \text{ or } 1$$

for any $A \in \mathcal{T}$.

Proof. *Take any* $A \in \mathcal{T}$ *and define*

$$\xi_n = E\left(1_A|\mathcal{F}_n\right),$$

where $\mathcal{F}_n = \sigma(\eta_1, \ldots, \eta_n)$. *By Exercise 1.5* ξ_n *is a uniformly integrable martingale, so* $\xi_n \to \xi$ *in* L^1. *By Theorem 1.10*

$$E(\xi|\mathcal{F}_n) = E(1_A|\mathcal{F}_n)$$

for all n. *Both* $\xi = \lim_n \xi_n$ *and* 1_A *are measurable with respect to the* σ*-field*

$$\mathcal{F}_\infty = \sigma(\eta_1, \eta_2, \ldots).$$

The family $\mathcal{G}$ consisting of all sets $B \in \mathcal{F}$ such that $\int_B \xi dP = \int_B 1_A dP$ is a σ-field containing $\mathcal{F}_1 \cup \mathcal{F}_2 \cup \cdots$. As a result, $\mathcal{G}$ contains the σ-field $\mathcal{F}_\infty$ generated by the family $\mathcal{F}_1 \cup \mathcal{F}_2 \cup \cdots$. By Lemma 2.1 it follows that $\xi = 1_A$ a.s.

Since η_n is a sequence of independent random variables, the σ-fields $\mathcal{F}_n$ and $\mathcal{T}_{n+1}$ are independent. Because $\mathcal{T} \subset \mathcal{T}_{n+1}$, the σ-fields $\mathcal{F}_n$ and $\mathcal{T}$ are independent. Being $\mathcal{T}$-measurable, 1_A is therefore independent of $\mathcal{F}_n$ for any n. This means that

$$\xi_n = E(1_A|\mathcal{F}_n) = E(1_A) = P(A) \quad a.s.$$

Therefore the limit $\lim_{n\to\infty} \xi_n = \xi$ is also constant and equal to $P(A)$ a.s. This means that $P(A) = 1_A$ a.s., so $P(A) = 0$ or 1. □

EXERCISE 1.8. *Show that if $A_n \in \sigma(\xi_n)$ for each n, then the events*

$$\limsup_n A_n = \bigcap_{j\geq 1} \bigcup_{i\geq j} A_i$$

and

$$\liminf_n A_n = \bigcup_{j\geq 1} \bigcap_{i\geq j} A_i$$

belong to the tail σ-field $\mathcal{T}$.

Hint: You need to write $\limsup_n A_n$ and $\liminf_n A_n$ in terms of $A_k, A_{k+1}, \ldots$ for any k, that is, to show that $\limsup_n A_n$ and $\liminf_n A_n$ will not be affected if any finite number of sets are removed from the sequence $A_1, A_2, \ldots$.

EXERCISE 1.9. *Use Kolmogorov's 0–1 law to show that in a sequence of coin tosses there are a.s. infinitely many heads.*

Hint: Show that the event

$$\{\eta_1, \eta_2, \ldots \text{ contains infinitely many heads }\}$$

belongs to the σ-field $\mathcal{T}$. Can the probability of this event be 0? The probability of the event

$$\{\eta_1, \eta_2, \ldots \text{ contains infinitely many tails }\}$$

should be the same. Can both probabilities be equal to 0? Can you simultaneously have finitely many heads and finitely many tails in the sequence $\eta_1, \eta_2, \ldots$?

4. Solutions

SOLUTION 1.1. *Because $\alpha_1 = 0$ is constant, it is $\mathcal{F}_0 = \{\emptyset, \Omega\}$-measurable. Suppose that α_n is $\mathcal{F}_{n-1}$-measurable for some $n = 1, 2, \ldots$. Then*

$$\{\alpha_n = 0, \xi_n < a\} \cup \{\alpha_n = 1, \xi_n \leq b\} \in \mathcal{F}_n$$

because $\mathcal{F}_{n-1} \subset \mathcal{F}_n$ and ξ_n is $\mathcal{F}_n$-measurable. This means that

$$\alpha_{n+1} = 1_{\{\alpha_n=0,\xi_n<a\}\cup\{\alpha_n=1,\xi_n\leq b\}}$$

is $\mathcal{F}_n$-measurable. By induction it follows that α_n is $\mathcal{F}_{n-1}$-measurable for each $n = 1, 2, \ldots$, so $\alpha_1, \alpha_2, \ldots$ is a gambling strategy.

SOLUTION 1.2. *For a non-negative supermartingale*

$$\sup_n E(|\xi_n|) = \sup_n E(\xi_n) \leq E(\xi_1) = E(|\xi_1|) < \infty,$$

since

$$E(\xi_n) \leq E(\xi_1)$$

for each $n = 1, 2, \ldots$. *Thus Doob's Martingale Convergence Theorem implies that* ξ_n *converges a.s. to an integrable limit.*

Solution 1.3. ***Necessity.*** *Suppose that* ξ *is integrable. It follows that*

$$P\{|\xi| < \infty\} = 1.$$

The sequence of random variables $|\xi| 1_{\{|\xi|>M\}}$ *indexed by* $M = 1, 2, \ldots$ *is monotone and*

$$|\xi| 1_{\{|\xi|>M\}} \searrow 0 \quad \text{as } M \to \infty$$

on the set $\{|\xi| < \infty\}$, *i.e. a.s. By the monotone convergence theorem for integrals*

$$\int_{\{|\xi|>M\}} |\xi| dP \searrow 0 \quad \text{as } M \to \infty.$$

It follows that for every $\varepsilon > 0$ *there exists an* $M > 0$ *such that*

$$\int_{\{|\xi|>M\}} |\xi| dP < \varepsilon.$$

Sufficiency. *Take* $\varepsilon = 1$. *There exists an* $M > 0$ *such that*

$$\int_{\{|\xi|>M\}} |\xi| dP < 1.$$

Then

$$\begin{aligned} E(|\xi|) &= \textstyle\int_\Omega |\xi| dP \\ &= \textstyle\int_{\{|\xi|>M\}} |\xi| dP + \int_{\{|\xi|\leq M\}} |\xi| dP \\ &< 1 + MP\{|\xi| \leq M\} \\ &\leq 1 + M < \infty. \end{aligned}$$

Solution 1.4. *For any* $M > 0$ *and any* $n > M$ *we have*

$$(0, \frac{1}{n}) = \{\xi_n > M\},$$

so

$$\int_{\{\xi_n>M\}} \xi_n dP = \int_{(0,\frac{1}{n})} n dP = 1.$$

This means that there is no $M > 0$ *such that for all* n

$$\int_{\{\xi_n>M\}} \xi_n dP < \frac{1}{2}.$$

The sequence ξ_n *is not uniformly integrable.*

Solution 1.5. *In Example 3.4 it was verified that* $\xi_n = E(\xi|\mathcal{F}_n)$ *is a martingale. Let* $\varepsilon > 0$. *By Lemma 1.8 there is a* $\delta > 0$ *such that*

$$P(A) < \delta \Longrightarrow \int_A |\xi| dP < \varepsilon.$$

By Jensen's inequality $|\xi_n| \leq E(|\xi||\mathcal{F}_n)$ *a.s., so*

$$E(|\xi|) \geq E(|\xi_n|) \geq \int_{\{|\xi_n|\geq M\}} |\xi_n| dP \geq MP\{|\xi_n| > M\}.$$

If we take $M > E(|\xi|)/\delta$, then

$$P\{|\xi_n| > M\} < \delta.$$

Since $\{|\xi_n| > M\} \in \mathcal{F}_n$, it follows that

$$\int_{\{|\xi_n|>M\}} |\xi_n| dP \leq \int_{\{|\xi_n|>M\}} E(|\xi||\mathcal{F}_n) dP = \int_{\{|\xi_n|>M\}} |\xi| dP < \varepsilon,$$

proving that $\xi_n = E(\xi|\mathcal{F}_n)$ is a uniformly integrable sequence.

SOLUTION 1.6. *Because ξ_n is a uniformly integrable sequence, there is an $M > 0$ such that for all n*

$$\int_{\{|\xi_n|>M\}} |\xi_n| dP < 1.$$

It follows that

$$\begin{aligned} & E\left(|\xi_n|\right) = \textstyle\int_{\{|\xi_n|>M\}} |\xi_n|\, dP + \int_{\{|\xi_n|\leq M\}} |\xi_n|\, dP \\ & < 1 + MP\left\{|\xi_n| \leq M\right\} \\ & < 1 + M < \infty \end{aligned}$$

for all n, proving that ξ_n is a bounded sequence in L^1.

SOLUTION 1.7. *By Theorem 1.10, $\xi_n = E(a|\mathcal{F}_n)$ a.s. But $E(a|\mathcal{F}_n) = a$ a.s., which proves that $\xi_n = a$ a.s.*

SOLUTION 1.8. *Observe that*

$$\limsup_n A_n = \bigcap_{j\geq k} \bigcup_{i\geq j} A_i$$

for any k. Since

$$A_i \in \sigma(\xi_i) \subset \mathcal{T}_k$$

for every $i \geq k$, it follows that

$$\limsup_n A_n = \bigcap_{j\geq k} \bigcup_{i\geq j} A_i \in \mathcal{T}_k$$

for every k. Therefore

$$\sup_n A_n \in \mathcal{T}.$$

The argument for $\liminf_n A_n$ is similar.

SOLUTION 1.9. *Let $\eta_1, \eta_2, \ldots$ be a sequence of coin tosses, i.e. independent random variables with values $+1, -1$ (heads or tails) taken with probability $\frac{1}{2}$ each. Consider the following event:*

$$A = \{\eta_1, \eta_2, \ldots \text{ contains infinitely many heads }\}.$$

This event belongs to the tail σ-field $\mathcal{T}$ because

$$A = \limsup_n A_n,$$

where

$$A_n = \{\eta_n = \text{ heads }\} \in \sigma(\eta_n)$$

(see Exercise 1.8). Thus, by Kolmogorov's 0-1 law $P(A) = 0$ or 1. However, it cannot be 0 because the event

$$B = \{\eta_1, \eta_2, \ldots \text{ contains infinitely many tails }\}$$

has the same probability by symmetry and $\Omega = A \cup B$ *(there must be infinitely many heads or infinitely many tails).*

This chapter is concerned with an interesting class of sequences of random variables taking values in a finite or countable set, called the state space, and satisfying the so-called Markov property. One of the simplest examples is provided by a symmetric random walk ξ_n with values in the set of integers $\mathbb{Z}$. If ξ_n is equal to some $i \in \mathbb{Z}$ at time n, then in the next time instance $n+1$ it will jump either to $i+1$, with probability $\frac{1}{2}$, or to $i-1$, also with probability $\frac{1}{2}$. What makes this model interesting is that the value of ξ_{n+1} at time $n+1$ depends on the past only through the value at time n. This is the Markov property characterizing Markov chains. There are numerous examples of Markov chains, with a multitude of applications.

From the mathematical point of view, Markov chains are both simple and difficult. Their definition and basic properties do not involve any complicated notions or sophisticated mathematics. Yet, any deeper understanding of Markov chains requires quite advanced tools. For example, this is so for problems related to the long-time behaviour of Markov processes. In this chapter we shall try to maintain a balance between the accessibility of exposition and the depth of mathematical results. Various concepts will be introduced. In particular, we shall discuss the classification of states and its relevance to the asymptotic behaviour of transition probabilities. This will turn out to be closely linked to ergodicity and the existence and uniqueness of invariant measures.

CHAPTER 2

Markov Chains

1. First Examples and Definitions

1.1. Example 5.1. In some homes the use of the telephone can become quite a sensitive issue. Suppose that if the phone is free during some period of time, say the nth minute, then with probability p, where $0 < p < 1$, it will be busy during the next minute. If the phone has been busy during the nth minute, it will become free during the next minute with probability q, where $0 < q < 1$. Assume that the phone is free in the 0th minute. We would like to answer the following two questions.

(1) What is the probability x_n that the telephone will be free in the nth minute?
(2) What is $\lim_{n\to\infty} x_n$, if it exists?

Denote by A_n the event that the phone is free during the nth minute and let $B_n = \Omega \setminus A_n$ be its complement, i.e. the event that the phone is busy during the nth minute. The conditions of the example give us

$$P(B_{n+1}|A_n) = p, \tag{5.1}$$
$$P(A_{n+1}|B_n) = q. \tag{5.2}$$

We also assume that $P(A_0) = 1$, i.e. $x_0 = 1$. Using this notation, we have $x_n = P(A_n)$. Then the total probability formula, see Exercise 1.10, together with (5.1)-(5.2) imply that

$$\begin{aligned} x_{n+1} &= P(A_{n+1}) && (5.3)\\ &= P(A_{n+1}|A_n)P(A_n) + P(A_{n+1}|B_n)P(B_n) && (1)\\ &= (1-p)x_n + q(1-x_n) = q + (1-p-q)x_n. && (2) \end{aligned}$$

It's a bit tricky to find an explicit formula for x_n. To do so we suppose first that the sequence $\{x_n\}$ is convergent, i.e.

$$\lim_{n\to\infty} x_n = x. \tag{5.4}$$

The elementary properties of limits and equation (5.3), i.e. $x_{n+1} = q + (1-p-q)x_n$, yield

$$x = q + (1-p-q)x. \tag{5.5}$$

The unique solution to the last equation is

$$x = \frac{q}{q+p}. \tag{5.6}$$

In particular,

$$\frac{q}{q+p} = q + (1-p-q)\frac{q}{q+p}. \tag{5.7}$$

Subtracting (5.7) from (5.3), we infer that

$$x_{n+1} - \frac{q}{q+p} = (1-p-q)\left(x_n - \frac{q}{q+p}\right). \tag{5.8}$$

Thus, $\{x_n - \frac{q}{q+p}\}$ is a geometric sequence and therefore, for all $n \in \mathbb{N}$,

$$x_n - \frac{q}{q+p} = (1-p-q)^n\left(x_0 - \frac{q}{q+p}\right). \tag{3}$$

Hence, by taking into account the initial condition $x_0 = 1$, we have

$$\begin{aligned} x_n &= \frac{q}{q+p} + \left(x_0 - \frac{q}{q+p}\right)(1-p-q)^n && (5.9)\\ &= \frac{q}{q+p} + \frac{p}{q+p}(1-p-q)^n. && (4) \end{aligned}$$

Let us point out that although we have used the assumption (5.4) to derive (5.8), the proof of the latter is now complete. Indeed, having proven (5.9), we can show that the assumption (5.4) is indeed satisfied. This is because the conditions $0 < p, q < 1$ imply that $|1-p-q| < 1$, and so $(1-p-q)^n \to 0$ as $n \to \infty$. Thus, (5.4) holds. This provides an answer to the second part of the example, i.e. $\lim_{n\to\infty} x_n = \frac{q}{p+q}$.

Exercise 2.1. *In the framework of Example 1.1, let y_n denote the probability that the telephone is busy in the nth minute. Supposing that $y_0 = 1$, find an explicit formula for y_n and, if it exists, $\lim_{n\to\infty} y_n$.*

Remark 2.1. ***Hint:*** *This exercise can be solved directly by repeating the above argument, or indirectly by using some of the results in Example 1.1.*

Remark 2.2. *The formulae (5.3) and (5.9) can be written collectively in a compact form by using vector and matrix notation. First of all, since $x_n + y_n = 1$, we get*

$$x_{n+1} = (1-p)x_n + qy_n, \tag{5}$$

$$y_{n+1} = px_n + (1-q)y_n. \tag{6}$$

Hence, the matrix version takes the form

$$\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}\begin{bmatrix} x_n \\ y_n \end{bmatrix}. \tag{7}$$

The situation described in Example 1.1 is quite typical. Often the probability of a certain event at time $n+1$ depends only on what happens at time n, but not further into the past.

Example 1.1 provides us with a simple case of a Markov chain. See also the following definition and exercises.

DEFINITION 2.1 (Markov chain). *Suppose that S is a finite or a countable set. Suppose also that a probability space $(\Omega, \mathcal{F}, P)$ is given. An S-valued sequence of random variables ξ_n, $n \in \mathbb{N}$, is called an S-valued Markov chain or a Markov chain on S if for all $n \in \mathbb{N}$ and all $s \in S$*

$$P(\xi_{n+1} = s \mid \xi_0, \ldots, \xi_n) = P(\xi_{n+1} = s \mid \xi_n). \tag{5.10}$$

Here $P(\xi_{n+1} = s|\xi_n)$ is the conditional probability of the event $\{\xi_{n+1} = s\}$ with respect to random variable ξ_n, or equivalently, with respect to the σ-field $\sigma(\xi_n)$ generated by ξ_n. Similarly, $P(\xi_{n+1} = s|\xi_0, \ldots, \xi_n)$ is the conditional probability of $\{\xi_{n+1} = s\}$ with respect to the σ-field $\sigma(\xi_0, \cdots, \xi_n)$ generated by the random variables $\xi_0, \cdots, \xi_n$.

Property (5.10) will usually be referred to as the Markov property of the Markov chain $\xi_n, n \in \mathbb{N}$. The set S is called the state space and the elements of S are called states.

Proposition 2.2. *The model in Example 1.1 and Exercise 2.1 is a Markov chain.*

PROOF. *Let $S = \{0, 1\}$, where 0 and 1 represent the states of the phone being free or busy. First we need to construct an appropriate probability space. Let Ω be the set of all sequences $\omega_0, \omega_1, \ldots$ with values in S. Let μ_0 be any probability measure on S. For example, $\mu = \delta_0$ corresponds to the case when the phone is free at time 0. We shall define P by induction. For any S-valued sequence $s_0, s_1, \ldots$ we put*

$$P(\{\omega \in \Omega : \omega_0 = s_0\}) = \mu_0(\{s_0\}) \tag{5.11}$$

and

$$\begin{aligned} &P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \ldots, n+1\}) && (5.12)\\ &= p(s_{n+1} \mid s_n) P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \ldots, n\}), && (8) \end{aligned}$$

where $p(s \mid r)$ are the entries of the 2×2 matrix

$$\begin{bmatrix} p(0 \mid 0) & p(0 \mid 1) \\ p(1 \mid 0) & p(1 \mid 1) \end{bmatrix} = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}. \tag{9}$$

It seems reasonable to expect P to be a probability measure (with respect to the trivial σ-field of all subsets of Ω). Take this for granted, check only that $P(\Omega) = 1$.

How would you define the process $\xi_n, n \in \mathbb{N}$? We shall do it in the standard way, i.e.

$$\xi_n(\omega) = \omega_n, \omega \in \Omega. \tag{5.13}$$

First we shall show that the transition probabilities of ξ_n are what they should be, i.e.

$$P(\xi_{n+1} = 1 \mid \xi_n = 0) = p, \tag{5.14}$$

$$P(\xi_{n+1} = 0 \mid \xi_n = 1) = q. \tag{5.15}$$

The definition of conditional probability yields

$$P(\xi_{n+1}=1 \mid \xi_n=0)=\frac{P(\xi_{n+1}=1,\xi_n=0)}{P(\xi_n=0)}. \tag{10}$$

Next, the definition of P gives

$$\begin{aligned} P(\xi_{n+1}=1,\xi_n=0) &= P(\{\omega\in\Omega:\omega_n=0,\omega_{n+1}=1\}) \\ &= \sum_{s_0,\dots,s_{n-1}\in S} P(\{\omega\in\Omega:\omega_i=s_i, i=0,\dots,n-1,\omega_n=0,\omega_{n+1}=1\}) \\ &= \sum_{s_0,\dots,s_{n-1}\in S} pP(\{\omega\in\Omega:\omega_i=s_i, i=0,\dots,n-1,\omega_n=0\}) \\ &= pP(\xi_n=0), \end{aligned} \tag{11}$$

by (5.12). We have proven (5.14). Moreover, (5.15) follows by the same argument. A similar line of reasoning shows that ξ_n is indeed a Markov chain. For this we need to verify that for any $n\in\mathbb{N}$ and any $s_0,s_1,\cdots,s_{n+1}\in S$

$$P(\xi_{n+1}=s_{n+1}\mid \xi_0=s_0,\dots,\xi_n=s_n)=P(\xi_{n+1}=s_{n+1}\mid\xi_n=s_n). \tag{12}$$

We have

$$\begin{aligned} &P(\xi_0=s_0,\dots,\xi_n=s_n,\xi_{n+1}=s_{n+1}) \\ &=P(\{\omega\in\Omega:\omega_i=s_i, i=0,\dots,n+1\}) \\ &=p(s_{n+1}\mid s_n)P(\{\omega\in\Omega:\omega_i=s_i,i=0,\dots,n\}) \\ &=p(s_{n+1}\mid s_n)P(\xi_0=s_0,\dots,\xi_n=s_n) \end{aligned} \tag{13}$$

which, in view of the definition of conditional probability, gives

$$P(\xi_{n+1}=s_{n+1}\mid\xi_0=s_0,\dots,\xi_n=s_n)=p(s_{n+1}\mid s_n). \tag{14}$$

On the other hand, by an easy generalization of (5.14) and (5.15)

$$P(\xi_{n+1}=s_{n+1}\mid\xi_n=s_n)=p(s_{n+1}\mid s_n), \tag{15}$$

which proves (5.10). □

In Example 1.1 the transition probabilities from state i to state j do not depend on the time n. This is an important class of Markov chains.

Definition 2.3 (Time-homogeneous Markov chain). *An S-valued Markov chain $\xi_n, n\in\mathbb{N}$, is called time-homogeneous or homogeneous if for all $n\in\mathbb{N}$ and all $i,j\in S$*

$$P(\xi_{n+1}=j\mid\xi_n=i)=P(\xi_1=j\mid\xi_0=i). \tag{5.16}$$

The number $P(\xi_1 = j \mid \xi_0 = i)$ is denoted by $p(j \mid i)$ and called the transition probability from state i to state j. The matrix $P = [p(j \mid i)]_{j,i\in S}$ is called the transition matrix of the chain ξ_n.

Exercise 2.2. *In the discussion so far we have seen an example of a transition matrix, $P = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}$. Obviously the sum of the entries in each column of P is equal to 1. Prove that this is true in general.*

Remark 2.3. ***Hint:*** *Remember that $P(\Omega \mid A) = 1$ for any event A.*

Definition 2.4 (Stochastic matrix). *$A = [a_{ji}]_{i,j\in S}$ is called a stochastic matrix if*

(1) *$a_{ji} \geq 0$, for all $i, j \in S$;*
(2) *the sum of the entries in each column is 1, i.e. $\sum_{j\in S} a_{ji} = 1$ for any $i \in S$.*

A is called a double stochastic matrix if both A and its transpose A^t are stochastic matrices.

Proposition 2.5. *Show that a stochastic matrix is doubly stochastic if and only if the sum of the entries in each row is 1, i.e. $\sum_{i\in S} a_{ji} = 1$ for any $j \in S$.*

Exercise 2.3. *Show that if $P = [p_{ji}]_{j,i\in S}$ is a stochastic matrix, then any natural power P^n of P is a stochastic matrix. Is the corresponding result true for a double stochastic matrix?*

Remark 2.4. ***Hint:*** *Show that if A and B are two stochastic matrices, then so is BA. For the second problem, recall that $(BA)^t = A^t B^t$.*

Exercise 2.4. *Let $P = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}$. Show that*

$$P^2 = \begin{bmatrix} 1+p^2-2p+pq & 2q-pq-q^2 \\ 2p-pq-p^2 & 1+q^2-2q+pq \end{bmatrix}. \tag{16}$$

Remark 2.5. ***Hint:*** *This is just simple matrix multiplication.*

We see that there is a problem with finding higher powers of the matrix P. When multiplying P^2 by P, P^3, and so on, we obtain more and more complicated expressions.

Definition 2.6 (n-step transition matrix). *The n-step transition matrix of a Markov chain ξ_n with transition probabilities $p(j \mid i)$, $j, i \in S$ is the matrix P_n with entries*

$$p_n(j \mid i) = P(\xi_n = j \mid \xi_0 = i). \tag{5.17}$$

Exercise 2.5. *Find an exact formula for P_n for the matrix P from Exercise 2.4.*

Remark 2.6. ***Hint:*** *Put $x_n = P(\xi_n = 0 \mid \xi_0 = 0)$ and $y_n = P(\xi_n = 1 \mid \xi_0 = 1)$. Is it correct to suppose that $p_n(0 \mid 0) = x_n$ and $p_n(1 \mid 1) = y_n$? If yes, you may be able use Example 1.1 and Exercise 2.1.*

Exercise 2.6. *You may suspect that P_n equals P^n, the nth power of the matrix P. This holds for $n = 1$. Check if it is true for $n = 2$. If this is the case, try to prove that $P_n = P^n$ for all $n \in \mathbb{N}$.*

Remark 2.7. ***Hint:*** *Once again, this is an exercise in matrix multiplication.*

The following is a generalization of Exercise 2.6.

Proposition 2.7 (Chapman–Kolmogorov equation)**.** *Suppose that $\xi_n, n \in \mathbb{N}$, is an S-valued Markov chain with n-step transition probabilities $p_n(j \mid i)$. Then for all $k, n \in \mathbb{N}$*

$$p_{n+k}(j \mid i) = \sum_{s \in S} p_n(j \mid s) p_k(s \mid i), \quad i, j \in S. \tag{5.18}$$

Exercise 2.7. *Prove Proposition 2.7.*

Remark 2.8. ***Hint:*** *$p_{n+k}(j \mid i)$ are the entries of the matrix $P_{n+k} = P^{n+k}$.*

Exercise 2.8. *Suppose that $S = \mathbb{Z}$. Let $\eta_n, n \geq 1$ be a sequence of independent identically distributed random variables with $P(\eta_1 = 1) = p$ and $P(\eta_1 = -1) = q = 1 - p$. Define $\xi_n = \sum_{i=1}^n \eta_i$ for $n \geq 1$ and $\xi_0 = 0$. Show that ξ_n is a Markov chain with transition probabilities*

$$p(j \mid i) = \begin{cases} p, & \text{if } j = i+1, \\ q, & \text{if } j = i-1, \\ 0, & \text{otherwise.} \end{cases} \tag{17}$$

$\xi_n, n \geq 0$, is called a random walk starting at 0. Replacing $\xi_0 = 0$ with $\xi_0 = i$, we get a random walk starting at i.

Remark 2.9. ***Hint:*** *$\xi_{n+1} = \xi_n + \eta_{n+1}$. Are ξ_n and η_{n+1} independent?*

Exercise 2.9. *For the random walk ξ_n defined in Exercise 2.8 prove that*

$$P(\xi_n = j \mid \xi_0 = i) = \binom{n}{\frac{n+j-i}{2}} p^{\frac{n+j-i}{2}} q^{\frac{n-j+i}{2}} \tag{5.19}$$

if $n + j - i$ is an even non-negative integer, and $P(\xi_n = j \mid \xi_0 = i) = 0$ otherwise.

Remark 2.10. ***Hint:*** *Use induction. Note that $\binom{n}{\frac{n+j-i}{2}} p^{\frac{n+j-i}{2}}$ equals 0 if $|j - i| \geq n + 1$.*

Proposition 2.8. *For all $p \in (0, 1)$*

$$P(\xi_n = i \mid \xi_0 = i) \to 0, \text{ as } n \to \infty. \tag{5.20}$$

Proposition 2.9. *The probability that the random walk ξ_n ever returns to the starting point is*

$$1 - |p - q|. \tag{18}$$

Exercise 2.10. *Prove formula (5.24).*

Remark 2.11. ***Hint:*** *Use the Taylor formula to expand the right-hand side of (5.24) into a power series.*

EXERCISE 2.11 (Branching process). *On the island Elschenbieden there lives an almost extinct species called Vugiel. Vugiel's males can produce zero, one, two, $\cdots$ male offspring with probability $p_0, p_1, p_2, \cdots$ respectively, where $p_i \geq 0$ and $\sum_{i=0}^{\infty} p_i = 1$. A challenging problem would be to find the Vugiel's chances of survival assuming that each individual lives exactly one year. At this moment, we ask you only to rewrite the problem in the language of Markov chains.*

REMARK 2.12. ***Hint:*** *The number of descendants of each male has the same distribution.*

EXERCISE 2.12. *Consider the following two cases:*

(1) *In Exercise 2.11 suppose that*

$$p_m = \binom{N}{m} p^m (1-p)^{N-m}. \tag{19}$$

for some $p \in (0,1)$ and $N \in \mathbb{N}^*$, where $\mathbb{N}^* = \{1, 2, 3, \cdots\}$. (Note that $p_m = 0$ if $m > N$.) Show that

$$p(j \mid i) = \binom{Ni}{j} p^j (1-p)^{Ni-j}. \tag{5.26}$$

Deduce that, in particular, $p(j \mid i) = 0$ if $j > Ni$.

(2) *Suppose that*

$$p_m = \frac{\lambda^m}{m!} e^{-\lambda}, m \in \mathbb{N}, \tag{20}$$

for some $\lambda > 0$. In other words, assume that each X_j has the Poisson distribution with mean λ. Show that

$$p(j \mid i) = \frac{(\lambda i)^j}{j!} e^{-\lambda i}, j, i \geq 0. \tag{5.27}$$

REMARK 2.13. ***Hint:*** *If X_1 has the binomial distribution $P(X_1 = m) = \binom{N}{m} p^m (1-p)^{N-m}, m \in \mathbb{N}$, then there exists a finite sequence $\eta_1^1, \cdots, \eta_N^1$ of independent identically distributed random variables such that $P(\eta_j^1 = 1) = p$, $P(\eta_j^1 = 0) = 1 - p$ and $X_j = \eta_1^j + \cdots + \eta_N^j = m$. Hence, $X_1 + \cdots + X_i = \sum_{r=1}^{i} \sum_{s=1}^{N} \eta_s^r$, i.e. the sum of Ni independent identically distributed random variables with distribution as above. Hence we infer (5.26).*

Proposition 2.10. *The probability of survival in Exercise 2.12, part 2) equals 0 if $\lambda \leq 1$, and $1 - \hat{r}^k$ if $\lambda > 1$, where k is the initial Vugiel population and $\hat{r} \in (0,1)$ is a solution to*

$$r = e^{(r-1)\lambda}. \tag{5.28}$$

REMARK 2.14. *The method presented in the last solution works for any distribution of the variables X_j. It turns out that the mean value λ of X_1 plays the same role as above. One can show that if $\lambda \leq 1$, then the population will become extinct with probability 1, while for $\lambda > 1$ the probability of extinction is larger than 0 and smaller than 1.*

Exercise 2.13. *On the, now familiar, island of Elschenbieden the question of survival of the Vugiel is a hot political issue. The (human) population of the island is N. Those who believe that action should be taken in order to help the animals preach their conviction quite convincingly. For if a supporter discusses the issue with a non-supporter, the latter will change his mind with probability one. However, they do so only in face-to-face encounters. Suppose that the probability of an encounter of exactly one pair of humans during one day is p and that with probability q this pair is a supporter-non-supporter one. Write down a Markov chain model of this situation. Neglect the probability of two or more encounters during one day.*

Remark 2.15. ***Hint:*** *On each day the number of supporters can either increase by 1 or remain unchanged. What is the probability of the former?*

Exercise 2.14 (Queuing model). *A car wash machine can serve at most one customer at a time. With probability p, $0 < p < 1$, the machine can finish serving a customer in a unit time. If this happens, the next waiting car (if any) can be served at the beginning of the next unit of time. During the time interval between the nth and $(n+1)$st unit of time the number of cars arriving has the Poisson distribution with parameter $\lambda > 0$. Let ξ_n denote the number of cars being served or waiting to be served at the beginning of unit n. Show that $\xi_n, n \in \mathbb{N}$, is a Markov chain and find its transition probabilities.*

Remark 2.16. ***Hint:*** *Let $Z_n, n = 0, 1, 2, \dots$ be a sequence of independent identically distributed random variables, each having the Poisson distribution with parameter λ. Then $\xi_{n+1} - \xi_n - Z_n$ equals -1 or 0.*

Remark 2.17. *In the last model we are interested in the behaviour of ξ_n for large values of n. In particular, it is interesting to determine whether the limit of ξ_n or that of $\mathbb{E}\xi_n$ (as $n \to \infty$) exists. In Exercise 2.31 we shall find conditions which guarantee the existence of a unique invariant measure and imply that the Markov chain in question is ergodic.*

2. Classification of States

In what follows we fix an S-valued Markov chain with transition matrix $P = [p(j \mid i)]_{j,i \in S}$, where S is a non-empty and at most countable set.

Definition 2.11 (Recurrent and transient states). *A state i is called recurrent if the process ξ_n will eventually return to i given that it starts at i, i.e.*

$$P(\xi_n = i \text{ for some } n \geq 1 \mid \xi_0 = i) = 1. \tag{5.36}$$

If the condition (5.36) is not satisfied, then the state i is called transient.

Theorem 2.12. *Show that for a random walk on $\mathbb{Z}$ with parameter $p \in (0,1)$, the state 0 is recurrent if and only if $p = 1/2$. Show that the same holds if 0 is replaced by any other state $i \in \mathbb{Z}$.*

Definition 2.13 (Communicating states). *We say that a state i communicates with a state j if with positive probability the chain will visit the state j having started at i, i.e.*

$$P(\xi_n = j \text{ for some } n \geq 0 \mid \xi_0 = i) > 0. \tag{5.37}$$

If i communicates with j, then we shall write $i \to j$. We say that the state i intercommunicates with a state j, and write $i \leftrightarrow j$, if $i \to j$ and $j \to i$.

EXERCISE 2.15. *Show that $i \to j$ if and only if $p_k(j \mid i) > 0$ for some $k \geq 1$.*

REMARK 2.18. ***Hint:*** *Recall that $p_k(j \mid i) = P(\xi_k = j \mid \xi_0 = i)$.*

EXERCISE 2.16. *Show that*

(1) *$i \leftrightarrow i$,*
(2) *if $i \leftrightarrow j$ then $j \leftrightarrow i$, and*
(3) *if $i \leftrightarrow j, j \leftrightarrow k$ then $i \leftrightarrow k$.*

In other words, show that $\leftrightarrow$ is an equivalence relation on S.

REMARK 2.19. ***Hint:*** *1) and 2) are obvious. For 3) use the Chapman-Kolmogorov equations.*

EXERCISE 2.17. *For $|x| < 1$ and $j, i \in S$ define*

$$P_{ji}(x) = \sum_{n=0}^{\infty} p_n(j \mid i)x^n, \tag{5.38}$$

$$F_{ji}(x) = \sum_{n=1}^{\infty} f_n(j \mid i)x^n, \tag{5.39}$$

where $f_n(j \mid i) = P(\xi_n = j, \xi_k \neq j, k = 1, \cdots, n-1 \mid \xi_0 = i)$. Show that the power series in (5.38)-(5.39) are absolutely convergent for $|x| < 1$ and that

$$P_{ji}(x) = F_{ji}(x)P_{jj}(x), \text{ if } j \neq i, \tag{5.40}$$

$$P_{ii}(x) = 1 + F_{ii}(x)P_{ii}(x). \tag{5.41}$$

REMARK 2.20. ***Hint:*** *Note that $|p_n(j \mid i)| \leq 1$, so the radius of convergence of the power series (5.38) is ≥ 1.*

EXERCISE 2.18. *Show that $\lim_{x\to 1} P_{jj}(x) = \sum_{n=0}^{\infty} p_n(j \mid j)$ and $\lim_{x\to 1} F_{jj}(x) = \sum_{n=0}^{\infty} f_n(j \mid j)$.*

REMARK 2.21. ***Hint:*** *Apply Abel's lemma: If $a_k \geq 0$ for all $k \geq 0$ and $\limsup_{k\to\infty} \sqrt[k]{|a_k|} \leq 1$, then $\lim_{x\to 1} \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} a_k$, no matter whether this sum is finite or infinite.*

EXERCISE 2.19. *Show that a state j is recurrent if and only if $\sum_n p_n(j \mid j) = \infty$. Deduce that the state j is transient if and only if*

$$\sum_n p_n(j \mid j) < \infty. \tag{5.42}$$

Show that if j is transient, then for each $i \in S$

$$\sum_n p_n(j \mid i) < \infty. \tag{5.43}$$

Remark 2.22. ***Hint:*** *If j is recurrent, then $F_{jj}(x) \to \sum_n f_n(j \mid j) = 1$ as $x \nearrow 1$. Use (5.41) in conjunction with Abel's Lemma.*

Exercise 2.20. *For a Markov chain ξ_n with transition matrix $P = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}$ show that both states are recurrent.*

Remark 2.23. ***Hint:*** *Use Exercise 2.19 and Exercise 2.5.*

One may suspect that if the state space S is finite, then there must exist at least one recurrent state. For otherwise, if all states were transient and $S = \{1, 2, \cdots, N\}$, then with positive probability a chain starting from 1 would visit 1 only a finite number of times. Thus, after visiting that state for the last time, the chain would move to a different state, say i_2, in which it would stay for a finite time only with positive probability. Thus, in finite time, with positive probability, the chain will never return to states 1 and i_2. By induction, in finite time, with positive probability, the chain will never return to any of the states. This is impossible. The following exercise will give precision to this argument.

Exercise 2.21. *Show that if ξ_n is a Markov chain with finite state space S, then there exists at least one recurrent state $i \in S$.*

Remark 2.24. ***Hint:*** *Argue by contradiction and use (5.43).*

The following result is quoted here for reference. The proof is surprisingly difficult and falls beyond the scope of this book.

Theorem 2.14. *A state $j \in S$ is recurrent if and only if*

$$P(\xi_n = j \text{ for infinitely many } n \mid \xi_0 = j) = 1, \tag{21}$$

and is transient if and only if

$$P(\xi_n = j \text{ for infinitely many } n \mid \xi_0 = j) = 0. \tag{22}$$

Definition 2.15 (Null-recurrent and positive-recurrent states). *For an S-valued Markov chain $\xi_n, n \in \mathbb{N}$, a state $i \in S$ is called null-recurrent if it is recurrent and its mean recurrence time m_i defined by*

$$m_i := \sum_{n=0}^{\infty} n f_n(i \mid i) \tag{5.44}$$

equals ∞. A state $i \in S$ is called positive-recurrent if it is recurrent and its mean recurrence time m_i is finite.

Remark 2.25. *One can show that a recurrent state i is null-recurrent if and only if $p_n(i \mid i) \to 0$.*

We already know that for a random walk on $\mathbb{Z}$ the state 0 is recurrent if and only if $p = 1/2$, i.e. if and only if the random walk is symmetric. In the following problem we shall try to answer if 0 is a null-recurrent or positive-recurrent state (when $p = 1/2$).

Exercise 2.22. *Consider a symmetric random walk on $\mathbb{Z}$. Show that 0 is a null-recurrent state. Can you deduce whether other states are positive-recurrent or null-recurrent?*

REMARK 2.26. ***Hint:*** *State 0 is null-recurrent if and only if* $\sum_n nf_n(0 \mid 0) = \infty$. *As in Exercise 2.18,* $\sum_n nf_n(0 \mid 0) = \lim_{x\to 1} F'_{00}(x)$, *where* F_{00} *is defined by (5.39).*

EXERCISE 2.23. *For the Markov chain* ξ_n *from Exercise 2.20 show that not only are all states recurrent, but they are positive-recurrent.*

REMARK 2.27. ***Hint:*** *Calculate* $f_n(0 \mid 0)$ *directly.*

The last two exercises suggest that the type of a state $i \in S$, i.e. whether it is transient, null-recurrent or positive-recurrent is invariant under the equivalence $\leftrightarrow$. We shall investigate this question in more detail below, but even before doing so we need one more notion: that of a periodic state.

DEFINITION 2.16 (Periodic and aperiodic states). *Suppose that* $\xi_n, n \in \mathbb{N}$, *is a Markov chain on a state space* S. *Let* $i \in S$. *We say that* i *is a periodic state if and only if the greatest common divisor (gcd) of all* $n \in \mathbb{N}^*$, *where* $\mathbb{N}^* = \{1, 2, 3, \cdots\}$, *such that* $p_n(i \mid i) > 0$ *is* ≥ 2. *Otherwise, the state* i *is called aperiodic. In both cases, the gcd is denoted by* $d(i)$ *and is called the period of the state* i. *Thus,* i *is periodic if and only if* $d(i) \geq 2$. *A state* i *which is positive recurrent and aperiodic is called ergodic.*

EXERCISE 2.24. *Is this claim that* $p_{d(i)}(i \mid i) > 0$ *true or not?*

REMARK 2.28. ***Hint:*** *Think of a Markov chain in which it is possible to return to the starting point by two different routes. One route with four steps, the other one with six steps.*

One of the by-products of the following exercise is another example of the type asked for in Exercise 2.24.

EXERCISE 2.25. *Consider a Markov chain on* $S = \{1, 2\}$ *with transition probability matrix* $P = \begin{bmatrix} 0 & 1/2 \\ 1 & 1/2 \end{bmatrix}$. *This chain can also be described by the graph in Figure* **??**. *Find* $d(1)$ *and* $d(2)$.

REMARK 2.29. ***Hint:*** *Calculate* P^2 *and* P^3. *This can be done in two different ways: either algebraically or, probabilistically.*

Proposition 2.17. *Suppose that* $i, j \in S$ *and* $i \leftrightarrow j$. *Show that*

(1) i *is transient if and only if* j *is;*
(2) i *is recurrent if and only if* j *is;*
(3) i *is null-recurrent if and only if* j *is;*
(4) i *is positive-recurrent if and only if* j *is;*
(5) i *is periodic if and only if* j *is, in which case* $d(i) = d(j)$;
(6) i *is ergodic if and only if* j *is.*

EXERCISE 2.26. *Show that the following modification of 2) above is true: If* i *is recurrent and* $i \to j$, *then* $j \to i$. *Deduce that if* i *is recurrent and* $i \to j$, *then* j *is recurrent and* $j \leftrightarrow i$.

REMARK 2.30. ***Hint:*** *Is it possible for a chain starting from* i *to visit* j *and then never return to* i? *Is such a situation possible when* i *is a recurrent state?*

The following result describes how the state space S can be partitioned into a countable sum of classes. One of these classes consists of all transient states. Each of the other classes consists of interconnecting recurrent states. If the chain enters one of the classes of second type, it will never leave it. However, if the chain enters the class of transient states, it will eventually leave it (and so never return to it). We begin with a definition.

DEFINITION 2.18 (Closed and irreducible sets). *Suppose that $\xi_n, n \in \mathbb{N}$, is a Markov chain on a countable state space S.*

(1) *A set $C \subset S$ is called closed if once the chain enters C it will never leave it, i.e.*

$$P(\xi_k \in S \setminus C \text{ for some } k \geq n \mid \xi_n \in C) = 0. \tag{5.45}$$

(2) *A set $C \subset S$ is called irreducible if any two elements i, j of C intercommunicate, i.e. for all $i, j \in C$ there exists an $n \in \mathbb{N}$ such that $p_n(j \mid i) > 0$.*

Theorem 2.19. *Suppose that $\xi_n, n \in \mathbb{N}$, is a Markov chain on a countable state space S. Then*

$$S = T \cup \bigcup_{j=1}^{N} C_j, \text{ (disjoint sum)}, \tag{5.46}$$

where T is the set of all transient states in S and each C_j is a closed irreducible set of recurrent states.

EXERCISE 2.27. *Suppose that $\xi_n, n \in \mathbb{N}$, is a Markov chain on a countable state space S. Show that a set $C \subset S$ is closed if and only if $p(j \mid i) = 0$ for all $i \in C$ and $j \in S \setminus C$.*

REMARK 2.31. ***Hint:*** *One implication is trivial. For the other one use the countable additivity of the measure P.*

3. Long-Time Behaviour of Markov Chains: General Case

For convenience we shall denote the countable state space S by $\{1, 2, 3, \cdots\}$ when S is an infinite set and by $\{1, 2, \cdots, n\}$ when S is finite.

Proposition 2.20. *Let $P = [p(j \mid i)]$ be the transition matrix of a Markov chain with state space S. Suppose that for all $i, j \in S$*

$$\lim_{n \to \infty} p_n(j \mid i) =: \pi_j. \tag{5.47}$$

(In particular, the limit is independent of i.) Then

(1) $\sum_j \pi_j \leq 1$;
(2) $\sum_i p(j \mid i)\pi_i = \pi_j$;
(3) *either* $\sum_j \pi_j = 1$, *or* $\pi_j = 0$ *for all* $j \in S$.

Proposition 2.21. *Let $P = [p(j \mid i)]$ be the transition matrix of a Markov chain with state space S. Suppose that for all $i, j \in S$*

$$\lim_{n \to \infty} p_n(j \mid i) =: \pi_j. \tag{5.47}$$

Then the vector $\pi = (\pi_j)_{j \in S}$ satisfies $\pi = P^T \pi$, i.e.,

$$\sum_{i \in S} p(j \mid i)\pi_i = \pi_j \quad \textit{for all } j \in S. \tag{5.65}$$

Theorem 2.22. *Suppose that $a \geq 1$ is a natural number. Consider a random walk on $S = \{0, 1, \cdots, a\}$ with absorbing barriers at 0 and a, and with probability p of moving to the right and probability $q = 1 - p$ of moving to the left from any of the states $1, \ldots, a-1$. Hence, our random walk is a Markov chain with transition probabilities*

$$p(j \mid i) = \begin{cases} p, & \textit{if } 1 \leq i \leq a-1, j = i+1, \\ q, & \textit{if } 1 \leq i \leq a-1, j = i-1, \\ 1, & \textit{if } i = j = 0 \textit{ or } i = j = a, \\ 0, & \textit{otherwise.} \end{cases} \tag{23}$$

Find, (a) all invariant measures (there may be just one), (b) the probability of hitting the right-hand barrier prior to hitting the left-hand one.

Remark 2.32. ***Hint:*** *For (a) recall Exercise 2.33 and for (b) Exercise 2.12.*

Exercise 2.28. *Ian plays a fair game of dice. After the nth roll of a die he writes down the maximum outcome ξ_n obtained so far. Show that ξ_n is a Markov chain and find its transition probabilities.*

Remark 2.33. ***Hint:*** $\xi_{n+1} = \max\{\xi_n, X_{n+1}\}$, *where X_k is the outcome of the kth roll.*

Exercise 2.29. *Analyse the Markov chain described in Exercise 2.28, but with fair die replaced by a fair pyramid.*

Remark 2.34. ***Hint:*** *A pyramid has four faces only.*

Exercise 2.30. *Consider a random walk on $S = \{0, 1, \ldots, a\}$ with absorbing barriers at 0 and a, and with probability p of moving to the right and probability $q = 1 - p$ of moving to the left from any of the states $1, \ldots, a-1$. Find the probability of hitting the right-hand barrier before the left-hand one.*

Remark 2.35. ***Hint:*** *Use the result from Proposition 2.23 or solve directly using recurrence relations.*

Proposition 2.23. *For the random walk with absorbing barriers described above, the probability of hitting state a before state 0 when starting from i ($1 \leq i \leq a-1$) is given by*

$$P(\textit{hit } a \textit{ before } 0 \mid \xi_0 = i) = \frac{1 - (q/p)^i}{1 - (q/p)^a} \quad \textit{if } p \neq q, \tag{24}$$

and

$$P(\textit{hit } a \textit{ before } 0 \mid \xi_0 = i) = \frac{i}{a} \quad \textit{if } p = q = 1/2. \tag{25}$$

Proposition 2.24. *The stationary distribution π for the random walk with absorbing barriers (when considered on the non-absorbing states) satisfies*

$$\pi_i = \frac{1}{Z}\left(\frac{p}{q}\right)^i, \quad i = 0, 1, \ldots, a, \tag{26}$$

where $Z = \sum_{i=0}^{a} (p/q)^i$ *is the normalizing constant.*

Theorem 2.25. *For the Poisson queueing model in Exercise 2.14, the stationary distribution exists if and only if* $\lambda' < 1$*, where* λ' *is the mean arrival rate. When it exists, the stationary distribution is Poisson with parameter* λ'*, i.e.,*

$$\pi_j = e^{-\lambda'} \frac{(\lambda')^j}{j!}, \quad j = 0, 1, 2, \ldots \tag{27}$$

Exercise 2.31. *Prove that if there exists an invariant measure for the Markov chain in Exercise 2.14, then* $\lambda' := \sum_{j=0}^{\infty} j q'_j < 1$*. Assuming that the converse is also true, conclude that the chain is ergodic if and only if* $\lambda' < 1$*. Show that if such an invariant measure exists, then it is unique.*

Remark 2.36. ***Hint:*** *Suppose that* $\pi = \sum_{j=0}^{\infty} \pi_j \delta_j$ *is an invariant measure. Write down an infinite system of linear equations for* π_j*. If you don't know how to follow, look at the solution.*

4. Long-Time Behaviour of Markov Chains with Finite State Space

As we have seen above, the existence of the π_j plays a very important role in the study of invariant measures. In what follows we shall investigate this question in the case when the state space S is finite.

Theorem 2.26. *Suppose that* S *is finite and the transition matrix* $P = [p(j \mid i)]$ *of a Markov chain on* S *satisfies the condition*

$$\exists n_0 \in \mathbb{N} \exists \varepsilon > 0 : p_{n_0}(j \mid i) \geq \varepsilon, \quad i, j \in S.$$

Then, the following limit exists for all $i, j \in S$ *and is independent of* i*:*

$$\lim_{n \to \infty} p_n(j \mid i) = \pi_j. \tag{5.55}$$

The numbers π_j *satisfy*

$$\pi_j > 0, j \in S \text{ and } \sum_{j \in S} \pi_j = 1. \tag{5.56}$$

Conversely, if a sequence of numbers $\pi_j, j \in S$ *satisfies conditions (5.55)-(5.56), then assumption (5.54) is also satisfied.*

Exercise 2.32. *Show that* $p_n(j \mid i) \to \pi_j$ *at an exponential rate.*

Remark 2.37. ***Hint:*** *Recall that* $m_n(j) \leq \pi_j \leq M_n(j)$ *and use (5.62).*

The above proves the following important result.

Theorem 2.27. *Suppose that the transition matrix $P = [p(j \mid i)]$ of a Markov chain $\xi_n, n \in \mathbb{N}$, satisfies assumption (5.54). Show that there exists a unique invariant measure μ. Moreover, for some $A > 0$, and $\alpha < 1$*

$$|p_n(j \mid i) - \pi_j| \leq A\alpha^n, \quad i, j \in S, n \in \mathbb{N}. \tag{5.63}$$

Exercise 2.33. *Investigate the existence and uniqueness of an invariant measure for the Markov chain in Proposition 2.2.*

Remark 2.38. ***Hint:*** *Are the assumptions of Theorem 2.26 satisfied?*

Remark 2.39. *The solution to Exercise 2.33 allows us to find the unique invariant measure by direct methods, i.e. by solving the linear equations (5.79)-(5.80).*

Exercise 2.34. *Find the invariant measure from Exercise 2.33 by calculating the limits (5.55).*

Remark 2.40. ***Hint:*** *Refer to Solution 5.5.*

Exercise 2.35. *For each of the following transition matrices, determine: (a) whether the chain is irreducible, (b) whether it is aperiodic, (c) the invariant distribution (if it exists), (d) whether the chain is ergodic.*

(1) $P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$

(2) $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(3) $P = \begin{bmatrix} 0.5 & 0.25 & 0.25 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}$

Remark 2.41. ***Hint:*** *Check the definitions of irreducibility, aperiodicity, and use Theorem 2.27.*

CHAPTER 3

Stochastic Processes in Continuous Time

1. General Notions

The following definitions are straightforward extensions of those introduced earlier for sequences of random variables, the underlying idea being that of a family of random variables depending on time.

DEFINITION 3.1 (Stochastic Process). *A stochastic process is a family of random variables $\xi(t)$ parametrized by $t \in T$, where $T \subset \mathbb{R}$. When $T = \{1, 2, \ldots\}$, we shall say that $\xi(t)$ is a stochastic process in discrete time (i.e. a sequence of random variables). When T is an interval in $\mathbb{R}$ (typically $T = [0, \infty)$), we shall say that $\xi(t)$ is a stochastic process in continuous time.*

For every $\omega \in \Omega$ the function

$$T \ni t \mapsto \xi(t, \omega)$$

is called a path (or sample path) of $\xi(t)$.

DEFINITION 3.2 (Filtration). *A family $\mathcal{F}_t$ of σ-fields on Ω parametrized by $t \in T$, where $T \subset \mathbb{R}$, is called a filtration if*

$$\mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$$

for any $s, t \in T$ such that $s \leq t$.

DEFINITION 3.3 (Martingale). *A stochastic process $\xi(t)$ parametrized by $t \in T$ is called a martingale (submartingale, supermartingale) with respect to a filtration $\mathcal{F}_t$ if*

(1) *$\xi(t)$ is integrable for each $t \in T$;*
(2) *$\xi(t)$ is $\mathcal{F}_t$-measurable for each $t \in T$ (in which case we say that $\xi(t)$ is adapted to $\mathcal{F}_t$);*
(3) *$\xi(s) = E(\xi(t) \mid \mathcal{F}_s)$ (respectively, $\leq$ or $\geq$) for every $s, t \in T$ such that $s \leq t$.*

In earlier chapters we have seen various stochastic processes, in particular, martingales in discrete time such as the symmetric random walk, for example. In what follows we shall study in some detail two processes in continuous time, namely, the Poisson process and Brownian motion.

2. Poisson Process

2.1. Exponential Distribution and Lack of Memory.

DEFINITION 3.4 (Exponential Distribution). *We say that a random variable η has the exponential distribution of rate $\lambda > 0$ if*

$$P\{\eta > t\} = e^{-\lambda t} \tag{28}$$

for all $t \geq 0$.

For example, the emissions of particles by a sample of radioactive material (or calls made at a telephone exchange) occur at random times. The probability that no particle is emitted (no call is made) up to time t is known to decay exponentially as t increases. That is to say, the time η of the first emission has the exponential distribution, $P\{\eta > t\} = e^{-\lambda t}$.

EXERCISE 3.1. *What is the distribution function of a random variable η with exponential distribution? Does it have a density? If so, find the density.*
Hint: *What is the probability that $\eta > 0$? What is the probability that $\eta > t$ for any given $t < 0$? Can you express the distribution function in terms of $P\{\eta > t\}$? Is the distribution function differentiable?*

EXERCISE 3.2. *Compute the expectation and variance of a random variable having the exponential distribution.*
Hint: *Use the density found in Exercise 3.1.*

EXERCISE 3.3. *Show that a random variable η with exponential distribution satisfies*

$$P\{\eta > t + s\} = P\{\eta > t\}P\{\eta > s\} \tag{6.1}$$

for any $s, t \geq 0$.
Hint: *When the probabilities are replaced by exponents, the equality should become obvious.*

EXERCISE 3.4. *Show that the equality in Exercise 3.3 is equivalent to*

$$P\{\eta > t + s \mid \eta > s\} = P\{\eta > t\} \tag{6.2}$$

for any $s, t \geq 0$.
Hint: *Recall how to compute conditional probability. Observe that $\eta > s + t$ implies $\eta > s$.*

The equality **??** (or, equivalently, **??**) is known as the **lack of memory property**. The odds that no particle will be emitted (no call will be made) in the next time interval of length t are not affected by the length of time s it has already taken to wait, given that no emission (no call) has occurred yet.

EXERCISE 3.5. *Show that the exponential distribution is the only probability distribution satisfying the lack of memory property.*
Hint: *The lack of memory property means that $g(t) = P\{\eta > t\}$ satisfies the functional equation*

$$g(t + s) = g(t)g(s)$$

for any $s, t > 0$. Find all non-negative non-increasing solutions of this functional equation.

2.2. Construction of the Poisson Process. Let $\eta_1, \eta_2, \dots$ be a sequence of independent random variables, each having the same exponential distribution of rate λ. For example, the times between the emissions of radioactive particles (or between calls made at a telephone exchange) have this property. We put

$$\xi_n = \eta_1 + \cdots + \eta_n,$$

which can be thought of as the time of the n-th emission (the n-th call). We also put $\xi_0 = 0$ for convenience. The number of emissions (calls) up to time $t \geq 0$ is an n such that $\xi_{n+1} > t \geq \xi_n$. In other words, the number of emissions (calls) up to time $t \geq 0$ is equal to $\max\{n : t \geq \xi_n\}$.

Definition 3.5 (Poisson Process). *We say that $N(t)$, where $t \geq 0$, is a Poisson process if*

$$N(t) = \max\{n : t \geq \xi_n\}. \tag{29}$$

Thus, $N(t)$ can be regarded as the number of particles emitted (calls made) up to time t. It is an example of a stochastic process in continuous time. A typical path of $N(t)$ is shown in fig. 1. It begins at the origin, $N(0) = 0$ (no particles emitted at time 0), and is right-continuous, non-decreasing and piecewise constant with jumps of size 1 at the times ξ_n.

Figure 1. A typical path of $N(t)$ and the jump times ξ_n

What is the distribution of $N(t)$? To answer this question we need to recall the definition of the Poisson distribution.

Definition 3.6 (Poisson Distribution). *A random variable ν has the Poisson distribution with parameter $\alpha > 0$ if*

$$P\{\nu = n\} = e^{-\alpha}\frac{\alpha^n}{n!} \tag{30}$$

for any $n = 0, 1, 2, \ldots$

The probabilities $P\{\nu = n\}$ for various values of α are shown in fig. 2.

Figure 2. Poisson distribution with parameter α

Proposition 3.7 (Poisson Process Distribution)**.** *$N(t)$ has the Poisson distribution with parameter λt,*

$$P\{N(t) = n\} = e^{-\lambda t}\frac{(\lambda t)^n}{n!}. \tag{31}$$

Proof. First of all, observe that

$$\{N(t) < n\} = \{\xi_n > t\}. \tag{32}$$

It suffices to compute the probability of this event for any n because

$$\begin{aligned} P\{N(t) = n\} &= P\{N(t) < n + 1\} - P\{N(t) < n\} \\ &= P\{\xi_{n+1} > t\} - P\{\xi_n > t\}. \end{aligned} \tag{33}$$

We shall prove by induction on n that

$$P\{\xi_n > t\} = e^{-\lambda t}\sum_{k=0}^{n-1}\frac{(\lambda t)^k}{k!}. \tag{6.4}$$

For $n = 1$:

$$P\{\xi_1 > t\} = P\{\eta_1 > t\} = e^{-\lambda t}.$$

Next, suppose that **??** holds for some n. Then, expressing ξ_{n+1} as the sum of the independent random variables ξ_n and η_{n+1}, we compute

$$
\begin{aligned}
P\{\xi_{n+1} > t\} &= P\{\xi_n + \eta_{n+1} > t\} \\
&= P\{\eta_{n+1} > t\} + P\{\xi_n > t - \eta_{n+1}, t \geq \eta_{n+1} > 0\} \\
&= e^{-\lambda t} + \int_0^t P\{\xi_n > t - s\} f_{\eta_{n+1}}(s)\, ds \\
&= e^{-\lambda t} + \int_0^t e^{-\lambda(t-s)} \sum_{k=0}^{n-1} \frac{(\lambda(t-s))^k}{k!} \lambda e^{-\lambda s}\, ds \qquad (34) \\
&= e^{-\lambda t} + e^{-\lambda t} \sum_{k=0}^{n-1} \frac{\lambda^{k+1}}{k!} \int_0^t (t-s)^k\, ds \\
&= e^{-\lambda t} \sum_{k=0}^{n} \frac{(\lambda t)^k}{k!},
\end{aligned}
$$

where $f_{\eta_{n+1}}(s)$ is the density of η_{n+1}. By induction, **??** holds for any n. Now apply **??** to complete the proof.

Exercise 3.6. *What is the expectation of $N(t)$?*
Hint: *What are the possible values of $N(t)$? What are the corresponding probabilities? Can you compute the expectation from these? To simplify the result use the Taylor expansion of e^x.*

Exercise 3.7. *Compute $P\{N(s) = 1, N(t) = 2\}$ for any $0 \leq s < t$.*
Hint: *Express $\{N(s) = 1, N(t) = 2\}$ as $\{\eta_1 \leq s < \eta_1 + \eta_2 \leq t < \eta_1 + \eta_2 + \eta_3\}$. You can compute the probability of the latter, since η_1, η_2, η_3 are exponentially distributed and independent.*

2.3. Poisson Process Starts from Scratch at Time t. *Imagine that you are to take part in an experiment to count the emissions of a radioactive particle. Unfortunately, in the excitement of proving the lack of memory property you forget about the engagement and arrive late to find that the experiment has already been running for time t and you have missed the first $N(t)$ emissions. Determined to make the best of it, you start counting right away, so at time $t + s$ you will have registered $N(t + s) - N(t)$ emissions. It will now be necessary to discuss $N(t + s) - N(t)$ instead of $N(s)$ in your report.*

What are the properties of $N(t + s) - N(t)$? Perhaps you can guess something from the physical picture? After all, a sample of radioactive material will keep emitting particles no matter whether anyone cares to count them or not. So the moment when someone starts counting does not seem important. You can expect $N(t+s)-N(t)$ to behave in a similar way as $N(s)$. And because radioactive emissions have no memory of the past, $N(t + s) - N(t)$ should be independent of $N(t)$.

To study this conjecture recall the construction of a Poisson process $N(t)$ based on a sequence of independent random variables $\eta_1, \eta_2, \ldots$, all having the same exponential distribution. We shall try to represent $N(t + s) - N(t)$ in a similar way.

Let us put

$$
\eta_1^t := \xi_{N(t)+1} - t, \quad \eta_n^t := \eta_{N(t)+n}, \quad n = 2, 3, \ldots, \qquad (35)
$$

see fig. 3. These are the times between the jumps of $N(t+s)-N(t)$. Then we define

$$\xi_n^t = \eta_1^t + \cdots + \eta_n^t, \tag{36}$$

$$N^t(s) = \max\{n : \xi_n^t \le s\}. \tag{37}$$

FIGURE 3. The random variables $\eta_1^t, \eta_2^t, \eta_3^t, \ldots$

EXERCISE 3.8. *Show that*

$$N^t(s) = N(t+s) - N(t). \tag{38}$$

Hint: *First show that $\xi_n^t = \xi_{N(t)+n} - t$.*

If we can show that $\eta_1^t, \eta_2^t, \ldots$ are independent random variables having the same exponential distribution as $\eta_1, \eta_2, \ldots$, it will mean that $N(t+s)-N(t)$ is a Poisson process with the same probability distribution as $N(s)$. Moreover, if the η_n^t turn out to be independent of $N(t)$, it will imply that $N(t+s)-N(t)$ is also independent of $N(t)$.

Before setting about this task beware of one common mistake. It is sometimes claimed that the times $\eta_1^t, \eta_2^t, \eta_3^t, \ldots$ between the jumps of $N(t+s)-N(t)$ are equal to $\xi_{n+1} - t, \eta_{n+2}, \eta_{n+3}, \ldots$ for some n. Hence $\eta_1^t, \eta_2^t, \eta_3^t, \ldots$ are independent because the random variables $\xi_{n+1}, \eta_{n+2}, \eta_{n+3}, \ldots$ are. The flaw in this is that, in fact, $\eta_1^t, \eta_2^t, \eta_3^t, \ldots$ are equal to $\xi_{n+1} - t, \eta_{n+2}, \eta_{n+3}, \ldots$ only on the set $\{N(t) = n\}$. However, the argument can be saved by conditioning with respect to $N(t)$. Our task becomes an exercise in computing conditional probabilities.

EXERCISE 3.9. *Show that*

$$P\{\eta_1^t > s \mid N(t)\} = P\{\eta_1 > s\}. \tag{39}$$

Hint: *It suffices (why?) to verify that*

$$P\{\eta_1^t > s, N(t) = n\} = P\{\eta_1 > s\}P\{N(t) = n\}$$

for any n. To this end, write the sets $\{N(t) = n\}$ and $\{\eta_1^t > s, N(t) = n\}$ in terms of ξ_n and η_{n+1}, which are independent random variables, and use the lack of memory property for η_{n+1}.

EXERCISE 3.10. *Show that*

$$P\{\eta_1^t > s_1, \ldots, \eta_k^t > s_k \mid N(t)\} = P\{\eta_1 > s_1\} \cdots P\{\eta_k > s_k\}. \tag{40}$$

Hint: *Verify that*

$$P\{\eta_1^t > s_1, \eta_2^t > s_2, \ldots, \eta_k^t > s_k, N(t) = n\} = P\{\eta_1 > s_1\} \cdots P\{\eta_k > s_k\}P\{N(t) = n\}$$

for any n. This is done in Exercise 3.9 for $k = 1$.

EXERCISE 3.11. *From the formula in Exercise 3.10 deduce that the random variables η_n^t and $N(t)$ are independent, and the η_n^t have the same probability distribution as η_n, i.e. exponential with parameter λ.*

Hint: *What do you get if you take the expectation on both sides of the equality in Exercise 3.10? Can you deduce the probability distribution of η_n^t? Can you see that the η_n^t are independent?*

To prove that the η_n^t are independent of $N(t)$ you need to be a little more careful and integrate over $\{N(t) = n\}$ instead of taking the expectation.

Because $N(t+s)-N(t)$ can be defined in terms of $\eta_1^t, \eta_2^t, \ldots$ in the same way the original Poisson process $N(t)$ is defined in terms of $\eta_1, \eta_2, \ldots$, the result in Exercise 3.11 proves the theorem below.

Theorem 3.8 (Poisson Process Stationarity)**.** *For any fixed $t \geq 0$*

$$N^t(s) = N(t+s) - N(t), \quad s \geq 0$$

is a Poisson process independent of $N(t)$ with the same probability law as $N(s)$.

That is to say, for any $s, t \geq 0$ the increment $N(t+s) - N(t)$ is independent of $N(t)$ and has the same probability distribution as $N(s)$. The assertion can be generalized to several increments, resulting in the following important theorem.

Theorem 3.9 (Poisson Process Independent Increments)**.** *For any $0 \leq t_1 \leq t_2 \leq \cdots \leq t_n$ the increments*

$$N(t_1), N(t_2) - N(t_1), N(t_3) - N(t_2), \ldots, N(t_n) - N(t_{n-1})$$

are independent and have the same probability distribution as

$$N(t_1), N(t_2 - t_1), N(t_3 - t_2), \ldots, N(t_n - t_{n-1}).$$

Proof. *From definition 3.8 it follows immediately that each increment $N(t_i) - N(t_{i-1})$ has the same distribution as $N(t_i - t_{i-1})$ for $i = 1, \ldots, n$.*

It remains to prove independence. This can be done by induction. The case when $n = 2$ is covered by definition 3.8. Now suppose that independence holds for n increments of a Poisson process for some $n \geq 2$. Take any sequence

$$0 \leq t_1 \leq t_2 \leq \cdots \leq t_n \leq t_{n+1}.$$

By the induction hypothesis

$$N(t_{n+1}) - N(t_n), \ldots, N(t_2) - N(t_1)$$

are independent, since they can be regarded as increments of

$$N^{t_1}(s) = N(t_1 + s) - N(t_1),$$

which is a Poisson process by definition 3.8. By the same theorem these increments are independent of $N(t_1)$. It follows that the $n+1$ random variables

$$N(t_{n+1}) - N(t_n), \ldots, N(t_2) - N(t_1), N(t_1)$$

are independent, completing the proof.

Definition 3.10 (Independent Increments). *We say that a stochastic process $\xi(t)$, where $t \in T$, has independent increments if*

$$\xi(t_1) - \xi(t_0), \ldots, \xi(t_n) - \xi(t_{n-1})$$

are independent for any $t_0 < t_1 < \cdots < t_n$ such that $t_0, t_1, \ldots, t_n \in T$.

Definition 3.11 (Stationary Increments). *A stochastic process $\xi(t)$, where $t \in T$, is said to have stationary increments if for any $s, t \in T$ the probability distribution of $\xi(t+h) - \xi(s+h)$ is the same for each h such that $s+h, t+h \in T$.*

definition 3.9 implies that the Poisson process has stationary independent increments. The result in the next exercise is also a consequence of definition 3.9.

EXERCISE 3.12. *Show that $N(t) - \lambda t$ is a martingale with respect to the filtration $\mathcal{F}_t$ generated by the family of random variables $\{N(s) : s \in [0,t]\}$.*
Hint: *Observe that $N(t) - N(s)$ is independent of $\mathcal{F}_s$ by definition 3.9.*

2.4. Various Exercises on the Poisson Process.

EXERCISE 3.13. *Show that $\xi_0 < \xi_1 < \xi_2 < \cdots$ a.s.*
Hint: *What is the probability of the event $\{\xi_{n-1} < \xi_n\} = \{\eta_n > 0\}$? What is the probability of the intersection of all such events?*

EXERCISE 3.14. *Show that $\lim_{n\to\infty} \xi_n = \infty$ a.s.*
Hint: *If $\lim_{n\to\infty} \xi_n < \infty$, then the sequence $\eta_1, \eta_2, \dots$ of independent random variables, all having the same exponential distribution, must be bounded. What is the probability that such a sequence is bounded? Begin with computing the probability $P\{\eta_1 \le m, \dots, \eta_n \le m\}$ for any fixed $m > 0$.*

Although it is instructive to estimate $P\{\lim_{n\to\infty} \xi_n < \infty\}$ in this way, there is a more elegant argument based on the strong law of large numbers. What does the law of large numbers tell us about the limit of $\frac{\xi_n}{n}$ as $n \to \infty$?

EXERCISE 3.15. *Show that ξ_n has absolutely continuous distribution with density*

$$f_n(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!} \tag{41}$$

with parameters n and λ. The density $f_n(t)$ of the gamma distribution is shown in fig. 4 for $n = 2, 4$ and $\lambda = 1$.

FIGURE 4. Density $f_n(t)$ of the gamma distribution with parameters $n = 2, \lambda = 1$ and $n = 4, \lambda = 1$

Hint: *Use the formula for $P\{\xi_n > t\}$ in the proof of definition 3.7 to find the distribution function of ξ_n. Is this function differentiable? What is the derivative?*

EXERCISE 3.16. *Prove that*

$$\lim_{t\to\infty} N(t) = \infty \quad a.s.$$

Hint: *What is the limit of $P\{N(k) \ge n\}$ as $k \to \infty$? Can you express $\{\lim_{t\to\infty} N(t) = \infty\}$ in terms of the sets $\{N(k) \ge n\}$?*

EXERCISE 3.17. *Verify that*

$$P\{N(t) \text{ is odd}\} = e^{-\lambda t} \sinh(\lambda t),$$
$$P\{N(t) \text{ is even}\} = e^{-\lambda t} \cosh(\lambda t).$$

Hint: *What is the probability that $N(t) = 2n + 1$? Compare this to the n-th term of the Taylor expansion of $\sinh(\lambda t)$.*

EXERCISE 3.18. *Show that*

$$\lim_{t\to\infty} \frac{N(t)}{t} = \lambda \quad a.s.$$

if $N(t)$ is a Poisson process with parameter λ.
Hint: *$N(n)$ is the sum of independent identically distributed random variables $N(1), N(2) - N(1), \dots, N(n) - N(n-1)$, so the strong law of large numbers can be applied to obtain the*

limit of $N(n)/n$ as $n \to \infty$. Because $N(t)$ is non-decreasing, the limit will not be affected if n is replaced by a continuous parameter $t \geq 0$.

3. Brownian Motion

Imagine a cloud of smoke in completely still air. In time, the cloud will spread over a large volume, the concentration of smoke varying in a smooth manner. However, if a single smoke particle is observed, its path turns out to be extremely rough due to frequent collisions with other particles. This exemplifies two aspects of the same phenomenon called diffusion: erratic particle trajectories at the microscopic level, giving rise to a very smooth behaviour of the density of the whole ensemble of particles. The Wiener process $W(t)$ defined below is a mathematical device designed as a model of the motion of individual diffusing particles. In particular, its paths exhibit similar erratic behaviour to the trajectories of real smoke particles. Meanwhile, the density $f_{W(t)}$ of the random variable $W(t)$ is very smooth, given by the exponential function

$$f_{W(t)}(x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}, \tag{42}$$

which is a solution of the diffusion equation

$$\frac{\partial f}{\partial t} = \frac{1}{2}\frac{\partial^2 f}{\partial x^2} \tag{43}$$

and can be interpreted as the density at time t of a cloud of smoke issuing from a single point source at time 0. The Wiener process $W(t)$ is also associated with the name of the British botanist Robert Brown, who around 1827 observed the random movement of pollen particles in water. We shall study mainly the one-dimensional Wiener process, which can be thought of as the projection of the position of a smoke particle onto one of the axes of a coordinate system.

Apart from describing the motion of diffusing particles, the Wiener process is widely applied in mathematical models involving various noisy systems, for example, the behaviour of asset prices at the stock exchange. If the noise in the system is due to a multitude of independent random changes, then the Central Limit Theorem predicts that the net result will have the normal distribution, a property shared by the increments $W(t) - W(s)$ of the Wiener process. This is one of the main reasons of the widespread use of $W(t)$ in mathematical models.

3.1. Definition and Basic Properties.

Definition 3.12 (Wiener Process / Brownian Motion). *The Wiener process (or Brownian motion) is a stochastic process $W(t)$ with values in $\mathbb{R}$ defined for $t \in [0, \infty)$ such that*

1. *$W(0) = 0$ a.s.;*
2. *the sample paths $t \mapsto W(t)$ are a.s. continuous;*
3. *for any finite sequence of times $0 < t_1 < \cdots < t_n$ and Borel sets $A_1, \ldots, A_n \subset \mathbb{R}$*

$$\begin{aligned} &P\{W(t_1) \in A_1, \ldots, W(t_n) \in A_n\} \\ &= \int_{A_1} \cdots \int_{A_n} p(t_1, 0, x_1) p(t_2 - t_1, x_1, x_2) \cdots \\ &\qquad \cdots p(t_n - t_{n-1}, x_{n-1}, x_n)\, dx_1 \cdots dx_n, \end{aligned}$$

where

$$p(t,x,y) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}} \tag{6.5}$$

defined for any $x, y \in \mathbb{R}$ and $t > 0$ is called the transition density.

A typical sample path of the Wiener process is shown in fig. 5.

FIGURE 5. A typical path of $W(t)$

EXERCISE 3.19. *Show that*

$$f_{W(t)}(x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}$$

is the probability density of $W(t)$ and find the expectation and variance of $W(t)$.
Hint: *The density of $W(t)$ can be obtained from condition 3) of definition 3.12 written for a single time t and a single Borel set. You will need the formula*

$$\int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}}\, dx = \sqrt{2\pi}$$

to compute the integrals in the expressions for the expectation and variance.

The results of Exercise 3.19 mean that $W(t)$ has the normal distribution with mean 0 and variance t.

EXERCISE 3.20. *Show that*

$$E(W(s)W(t)) = \min\{s,t\}.$$

Hint: *The joint density of $W(s)$ and $W(t)$ will be needed. It can be found from condition 3) of definition 3.12 written for two times s and t and two Borel sets.*

EXERCISE 3.21. *Show that*

$$E\left(|W(t) - W(s)|^2\right) = |t-s|.$$

Hint: *Expand the square and use the formula in Exercise 3.20.*

EXERCISE 3.22. *Compute the characteristic function $E(\exp(i\lambda W(t)))$ for any $\lambda \in \mathbb{R}$.*
Hint: *Use the density of $W(t)$ found in Exercise 3.19.*

EXERCISE 3.23. *Find $E(W(t)^4)$.*
Hint: *This can be done, for example, by expressing the expectation in terms of the density of $W(t)$ and computing the resulting integral, or by computing the fourth derivative of the characteristic function of $W(t)$ at 0. The second method is more efficient.*

DEFINITION 3.13 (n-dimensional Wiener Process). *We call $W(t) = (W^1(t), \ldots, W^n(t))$ an n-dimensional Wiener process if $W^1(t), \ldots, W^n(t)$ are independent $\mathbb{R}$-valued Wiener processes.*

EXERCISE 3.24. *For a two-dimensional Wiener process $W(t) = (W^1(t), W^2(t))$ find the probability that $|W(t)| < R$, where $R > 0$ and $|x|$ is the Euclidean norm of $x = (x^1, x^2)$ in $\mathbb{R}^2$, i.e. $|x|^2 = (x^1)^2 + (x^2)^2$.*

Hint: *Express the probability in terms of the joint density of $W^1(t)$ and $W^2(t)$. Independence means that the joint density of $W^1(t)$ and $W^2(t)$ is the product of their respective densities, which are known from Exercise 3.19. It is convenient to use polar coordinates to compute the resulting integral over a disc.*

3.2. Increments of Brownian Motion.

Proposition 3.14 (Increments are Normal)**.** *For any $0 \leq s < t$ the increment $W(t) - W(s)$ has the normal distribution with mean 0 and variance $t - s$.*

PROOF. *By condition 3) of definition 3.12 the joint density of $W(s), W(t)$ is*

$$f_{W(s),W(t)}(x, y) = p(s, 0, x)p(t - s, x, y).$$

Hence, for any Borel set A

$$\begin{aligned} P\{W(t) - W(s) \in A\} &= \int_{\{(x,y):y-x\in A\}} p(s, 0, x)p(t - s, x, y)\,dx\,dy \\ &= \int_{-\infty}^{+\infty} p(s, 0, x) \left(\int_{\{y:y-x\in A\}} p(t - s, x, y)\,dy \right) dx \\ &= \int_{-\infty}^{+\infty} p(s, 0, x) \left(\int_A p(t - s, x, x + u)\,du \right) dx \\ &= \int_{-\infty}^{+\infty} p(s, 0, x) \left(\int_A p(t - s, 0, u)\,du \right) dx \\ &= \int_A p(t - s, 0, u)\,du \int_{-\infty}^{+\infty} p(s, 0, x)\,dx \\ &= \int_A p(t - s, 0, u)\,du. \end{aligned} \tag{44}$$

But $f(u) = p(t-s, 0, u)$ is the density of the normal distribution with mean 0 and variance $t - s$, which proves the claim.

Corollary 3.15. *definition 3.14 implies that $W(t)$ has stationary increments.*

Proposition 3.16 (Independent Increments)**.** *For any $0 = t_0 \leq t_1 \leq \cdots \leq t_n$ the increments*

$$W(t_1) - W(t_0), \ldots, W(t_n) - W(t_{n-1})$$

are independent.

PROOF. *From definition 3.14 we know that the increments of $W(t)$ have the normal distribution. Because normally distributed random variables are independent if and only if they are uncorrelated, it suffices to verify that*

$$E[(W(u) - W(t))(W(s) - W(r))] = 0$$

for any $0 \leq r \leq s \leq t \leq u$. But this follows immediately from Exercise 3.20:

$$\begin{aligned} E[(W(u) - W(t))(W(s) - W(r))] &= E(W(u)W(s)) - E(W(u)W(r)) \\ &\quad - E(W(t)W(s)) + E(W(t)W(r)) \\ &= s - r - s + r \\ &= 0, \end{aligned} \tag{45}$$

as required.

Corollary 3.17. *For any $0 \leq s < t$ the increment $W(t) - W(s)$ is independent of the σ-field*

$$\mathcal{F}_s = \sigma\{W(r) : 0 \leq r \leq s\}.$$

Proof. *By definition 3.16 the random variables $W(t) - W(s)$ and $W(r) - W(0) = W(r)$ are independent if $0 \leq r \leq s \leq t$. Because the σ-field $\mathcal{F}_s$ is generated by such $W(r)$, it follows that $W(t) - W(s)$ is independent of $\mathcal{F}_s$.*

Exercise 3.25. *Show that $W(t)$ is a martingale with respect to the filtration $\mathcal{F}_t$.*
Hint: *Take advantage of the fact that $W(t) - W(s)$ is independent of $\mathcal{F}_s$ if $s < t$.*

Exercise 3.26. *Show that $|W(t)|^2 - t$ is a martingale with respect to the filtration $\mathcal{F}_t$.*
Hint: *Once again, use the fact that $W(t) - W(s)$ is independent of $\mathcal{F}_s$ if $s < t$.*

Let us state without proof the following useful characterization of the Wiener process in terms of its increments.

Theorem 3.18 (Wiener Process Characterization)**.** *A stochastic process $W(t), t \geq 0$, is a Wiener process if and only if the following conditions hold:*

1. *$W(0) = 0$ a.s.;*
2. *the sample paths $t \mapsto W(t)$ are continuous a.s.;*
3. *$W(t)$ has stationary independent increments;*
4. *the increment $W(t) - W(s)$ has the normal distribution with mean 0 and variance $t - s$ for any $0 \leq s < t$.*

Exercise 3.27. *Show that for any $T > 0$*

$$V(t) = W(t + T) - W(T)$$

is a Wiener process if $W(t)$ is.
Hint: *Are the increments of $V(t)$ independent? What is their distribution? Does $V(t)$ have continuous paths? Is it true that $V(0) = 0$?*

The Wiener process can also be characterized by its martingale properties. The following theorem is also given without proof.

Theorem 3.19 (Lévy's Martingale Characterization)**.** *Let $W(t), t \geq 0$, be a stochastic process and let $\mathcal{F}_t = \sigma(W_s, s \leq t)$ be the filtration generated by it. Then $W(t)$ is a Wiener process if and only if the following conditions hold:*

1. *$W(0) = 0$ a.s.;*
2. *the sample paths $t \mapsto W(t)$ are continuous a.s.;*
3. *$W(t)$ is a martingale with respect to the filtration $\mathcal{F}_t$;*
4. *$|W(t)|^2 - t$ is a martingale with respect to $\mathcal{F}_t$.*

Exercise 3.28. *Let $c > 0$. Show that $V(t) = \frac{1}{c}W(c^2 t)$ is a Wiener process if $W(t)$ is.*
Hint: *Is $V(t)$ a martingale? With respect to which filtration? Is $|V(t)|^2 - t$ a martingale? Are the paths of $V(t)$ continuous? Is it true that $V(0) = 0$?*

3.3. Sample Paths. *Let*

$$0 = t_0^n < t_1^n < \cdots < t_n^n = T,$$

where

$$t_i^n = \frac{iT}{n},$$

be a partition of the interval $[0,T]$ into n equal parts. We denote by

$$\Delta_i^n W = W(t_{i+1}^n) - W(t_i^n)$$

the corresponding increments of the Wiener process $W(t)$.

EXERCISE 3.29. *Show that*

$$\lim_{n\to\infty} \sum_{i=0}^{n-1} (\Delta_i^n W)^2 = T \quad \text{in } L^2.$$

Hint: *You need to show that*

$$\lim_{n\to\infty} E\left(\left[\sum_{i=0}^{n-1} (\Delta_i^n W)^2 - T\right]^2\right) = 0.$$

Use the independence of increments to simplify the expectation. What are the expectations of $\Delta_i^n W$, $(\Delta_i^n W)^2$ and $(\Delta_i^n W)^4$?

The next theorem on the variation of the paths of $W(t)$ is a consequence of the result in Exercise 3.29. First, let us recall that the variation of a function is defined as follows.

DEFINITION 3.20 (Variation). *The variation of a function $f : [0,T] \to \mathbb{R}$ is defined to be*

$$\limsup_{\Delta t\to 0} \sum_{i=0}^{n-1} |f(t_{i+1}) - f(t_i)|,$$

where $t = (t_0, t_1, \dots, t_n)$ is a partition of $[0,T]$, i.e. $0 = t_0 < t_1 < \cdots < t_n = T$, and where

$$\Delta t = \max_{i=0,\dots,n-1} |t_{i+1} - t_i|.$$

Theorem 3.21 (Infinite Variation)**.** *The variation of the paths of $W(t)$ is infinite a.s.*

PROOF. *Consider the sequence of partitions $t^n = (t_0^n, t_1^n, \dots, t_n^n)$ of $[0,T]$ into n equal parts. Then*

$$\sum_{i=0}^{n-1} |\Delta_i^n W|^2 \le \left(\max_{i=0,\dots,n-1} |\Delta_i^n W|\right) \sum_{i=0}^{n-1} |\Delta_i^n W|.$$

Since the paths of $W(t)$ are a.s. continuous on $[0,T]$,

$$\lim_{n\to\infty} \left(\max_{i=0,\dots,n-1} |\Delta_i^n W|\right) = 0 \quad a.s.$$

By Exercise 3.29 there is a subsequence $t^{n_k} = (t_0^{n_k}, t_1^{n_k}, \dots, t_{n_k}^{n_k})$ of partitions such that

$$\lim_{k\to\infty} \sum_{i=0}^{n_k-1} |\Delta_i^{n_k} W|^2 = T \quad a.s.$$

This is because every sequence of random variables convergent in L^2 has a subsequence convergent a.s. It follows that

$$\lim_{k\to\infty} \sum_{i=0}^{n_k-1} |\Delta_i^{n_k} W| = \infty \quad a.s.,$$

while

$$\lim_{k\to\infty} \Delta t^{n_k} = \lim_{k\to\infty} \frac{T}{n_k} = 0,$$

which proves the theorem.

definition 3.21 has important consequences for the theory of stochastic integrals presented in the next chapter. This is because an integral of the form

$$\int_0^T f(t)\,dW(t)$$

cannot be defined pathwise (that is, separately for each $\omega \in \Omega$) as the Riemann-Stieltjes integral if the paths have infinite variation. It turns out that an intrinsically stochastic approach will be needed to tackle such integrals, see **??**.

EXERCISE 3.30. *Show that $W(t)$ is a.s. non-differentiable at $t = 0$.*
Hint: *By Exercise 3.28 $V_c(t) = \frac{1}{c}W(c^2 t)$ is a Wiener process for any $c > 0$. Deduce that the probability*

$$P\left\{\frac{|W(t)|}{t} > cM \text{ for some } t \in [0, \frac{1}{c^2}]\right\}$$

is the same for each $c > 0$. What is the probability that the limit of $\frac{|W(t)|}{t}$ exists as $t \searrow 0$, then?

EXERCISE 3.31. *Show that for any $t \geq 0$ the Wiener process $W(t)$ is a.s. non-differentiable at t.*
Hint: *$V_t(s) = W(s + t) - W(t)$ is a Wiener process for any $t \geq 0$.*

A weak point in the assertion in Exercise 3.31 is that for each t the event of measure 1 in which $W(t)$ is non-differentiable at t may turn out to be different for each $t \geq 0$. The theorem below, which is presented without proof, shows that in fact the same event of measure 1 can be chosen for each $t \geq 0$. This is not a trivial conclusion because the set of $t \geq 0$ is uncountable.

Theorem 3.22. *With probability 1 the Wiener process $W(t)$ is non-differentiable at any $t \geq 0$.*

3.4. Doob's Maximal L^2 Inequality for Brownian Motion. *The inequality proved in this section is necessary to study the properties of stochastic integrals in the next chapter. It can be viewed as an extension of Doob's maximal L^2 inequality in thm:doob-maximal-inequality-discrete to the case of continuous time. In fact, in the result below the Wiener process can be replaced by any square integrable martingale $\xi(t), t \geq 0$ with a.s. continuous paths.*

Theorem 3.23 (Doob's Maximal L^2 Inequality)**.** *For any $t > 0$*

$$E\left(\max_{s\leq t} |W(s)|^2\right) \leq 4E|W(t)|^2. \tag{6.6}$$

PROOF. *For $t > 0$ and $n \in \mathbb{N}$ we define*

$$M_k^n = \left|W\left(\frac{kt}{2^n}\right)\right|, \quad 0 \leq k \leq 2^n. \tag{6.7}$$

Then, by Jensen's inequality, $M_k^n, k = 0, \dots, 2^n$ is a non-negative square integrable submartingale with respect to the filtration $\mathcal{F}_k^n = \mathcal{F}_{\frac{k+1}{2^n}}$, so by thm:doob-maximal-inequality-discrete

$$E\left(\max_{k\le 2^n}|M_k^n|^2\right) \le 4E|M_{2^n}^n|^2 = 4E|W(t)|^2.$$

Since $W(t)$ has a.s. continuous paths,

$$\lim_{n\to\infty}\max_{k\le 2^n}|M_k^n|^2 = \max_{s\le t}|W(s)|^2 \quad a.s.$$

Moreover, since $M_k^n = M_{2k}^{n+1}$, the sequence $\sup_{k\le 2^n}|M_k^n|^2, n\in\mathbb{N}$, is increasing. Hence by the Lebesgue monotone convergence theorem $\max_{s\le t}|W(s)|^2$ is an integrable function and

$$E\left(\max_{s\le t}|W(s)|^2\right) = \lim_{n\to\infty} E\left(\max_{k\le 2^n}|M_k^n|^2\right) \le 4E|W(t)|^2,$$

completing the proof.

3.5. Various Exercises on Brownian Motion.

Exercise 3.32. *Verify that the transition density $p(t,x,y)$ satisfies the diffusion equation*

$$\frac{\partial p}{\partial t} = \frac{1}{2}\frac{\partial^2 p}{\partial y^2}.$$

Hint: *Simply differentiate the expression* **??** *for the transition density.*

Exercise 3.33. *Show that $Z(t) = -W(t)$ is a Wiener process if $W(t)$ is.*
Hint: *Are the increments of $Z(t)$ independent? How are they distributed? Are the paths of $Z(t)$ continuous? Is it true that $Z(0) = 0$?*

Exercise 3.34. *Show that for any $0 \le s < t$*

$$P\{W(t)\in A \mid W(s)\} = \int_A p(t-s, W(s), y)\,dy.$$

Hint: *Write the conditional probability as the conditional expectation of $1_A(W(t))$ given $W(s)$. Compute the conditional expectation by transforming the integral of $1_A(W(t))$ over any event in the σ-field generated by $W(s)$. This can be done using the joint density of $W(s)$ and $W(t)$. Refer to the chapter on conditional expectation if necessary.*

Exercise 3.35. *Show that $e^{W(t)}e^{-\frac{t}{2}}$ is a martingale. (It is called the exponential martingale.)*
Hint: *What is the expectation of $e^{W(t)-W(s)}$ for $s < t$? By independence it is equal to the conditional expectation of $e^{W(t)-W(s)}$ given $\mathcal{F}_s$. This will give the martingale condition.*

Exercise 3.36. *Compute $E(W(s) \mid W(t))$ for $0 \le s < t$.*
Hint: *You want to find a Borel function F such that $E(W(s) \mid W(t)) = F(W(t))$, i.e.*

$$\int_{\{W(t)\in A\}} W(s)\,dP = \int_{\{W(t)\in A\}} F(W(t))\,dP.$$

Either side of this equality can be transformed using the joint density of $W(t)$ and $W(s)$.

4. Solutions

4.1. Solution to Exercise 3.1. *Suppose that η is a random variable with exponential distribution of rate λ. The distribution function of η is*

$$F(t) = P\{\eta \le t\} = 1 - P\{\eta > t\} = \begin{cases} 0 & \text{if } t < 0, \\ 1 - e^{-\lambda t} & \text{if } t \ge 0. \end{cases} \tag{46}$$

Therefore η has density

$$f(t) = \frac{d}{dt} F(t) = \begin{cases} 0 & \text{if } t < 0, \\ \lambda e^{-\lambda t} & \text{if } t > 0. \end{cases} \tag{47}$$

The distribution function $F(t)$ and density $f(t)$ are shown in fig. 6.

Figure 6. The distribution function $F(t)$ and density $f(t)$ of a random variable with exponential distribution of rate $\lambda = 2$

4.2. Solution to Exercise 3.2. *Using the density $f(t) = \lambda e^{-\lambda t}$ found in Exercise 3.1 and integrating by parts, we obtain*

$$\begin{aligned} E(\eta) &= \int_{-\infty}^{\infty} t f(t)\, dt = \int_0^{\infty} t \lambda e^{-\lambda t}\, dt = -\int_0^{\infty} t \frac{d}{dt} e^{-\lambda t}\, dt \\ &= -t e^{-\lambda t}\Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda t}\, dt = 0 - \frac{1}{\lambda} e^{-\lambda t}\Big|_0^{\infty} = \frac{1}{\lambda}. \end{aligned} \tag{48}$$

In a similar way we compute

$$\begin{aligned} E(\eta^2) &= \int_{-\infty}^{\infty} t^2 f(t)\, dt = \int_0^{\infty} t^2 \lambda e^{-\lambda t}\, dt = -\int_0^{\infty} t^2 \frac{d}{dt} e^{-\lambda t}\, dt \\ &= -t^2 e^{-\lambda t}\Big|_0^{\infty} + 2\int_0^{\infty} t e^{-\lambda t}\, dt = 0 + \frac{2}{\lambda} \int_0^{\infty} t f(t)\, dt = \frac{2}{\lambda^2}. \end{aligned} \tag{49}$$

It follows that the variance is equal to

$$\operatorname{var}(\eta) = E(\eta^2) - (E(\eta))^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}. \tag{50}$$

4.3. Solution to Exercise 3.3. *By the definition of a random variable with exponential distribution*

$$P\{\eta > s + t\} = e^{-\lambda(s+t)} = e^{-\lambda s} e^{-\lambda t} = P\{\eta > s\} P\{\eta > t\}.$$

4.4. Solution to Exercise 3.4. *By the definition of conditional probability*

$$\begin{aligned} P\{\eta > t + s \mid \eta > s\} &= \frac{P\{\eta > t + s, \eta > s\}}{P\{\eta > s\}} \\ &= \frac{P\{\eta > t + s\}}{P\{\eta > s\}}, \end{aligned} \tag{51}$$

since $\{\eta > t + s, \eta > s\} = \{\eta > t + s\}$ (because $\eta > s + t$ implies that $\eta > s$). Now it is clear that **??** *is equivalent to* **??**.

4.5. Solution to Exercise 3.5. *The exponential distribution is the only probability distribution satisfying the lack of memory property because the only non-negative non-increasing solutions of the functional equation*

$$g(t+s) = g(t)g(s)$$

are of the form $g(t) = a^t$ *for some* $0 \leq a \leq 1$.

To verify this observe that $[g(m/n)]^n = g(m) = [g(1)]^m$ *for any integers* m *and* $n \neq 0$. *Let* $a := g(1)$. *It follows that*

$$g(q) = a^q \quad \text{for any } q \in \mathbb{Q}.$$

Since g *is non-increasing,* $0 \leq a \leq 1$ *and*

$$a^t = \inf_{t>q\in\mathbb{Q}} g(q) \geq g(t) \geq \sup_{t<q\in\mathbb{Q}} g(q) = a^t,$$

so indeed

$$g(t) = a^t \quad \text{for any } t \in \mathbb{R}.$$

As a result, $P\{\eta > t\} = a^t$ *for some* $0 \leq a \leq 1$. *But the distribution function of a random variable cannot be constant, so* $0 \neq a \neq 1$. *Hence* $a = e^{-\lambda}$ *for some* $\lambda > 0$, *completing the argument.*

4.6. Solution to Exercise 3.6. *Since* $N(t)$ *has the Poisson distribution with parameter* λt *we have* $E(N(t)) = \lambda t$. *Indeed*

$$\begin{aligned} E(N(t)) &= \sum_{n=0}^{\infty} nP\{N(t) = n\} = \sum_{n=0}^{\infty} ne^{-\lambda t}\frac{(\lambda t)^n}{n!} \\ &= \lambda te^{-\lambda t}\sum_{n=1}^{\infty}\frac{(\lambda t)^{n-1}}{(n-1)!} = \lambda te^{-\lambda t}e^{\lambda t} = \lambda t. \end{aligned} \tag{52}$$

4.7. Solution to Exercise 3.7. *Using the fact that* $\eta_1, \eta_2, \ldots$ *are independent and exponentially distributed, we obtain*

$$\begin{aligned} P\{N(s) = 1, N(t) = 2\} &= P\{\xi_1 \leq s < \xi_2 \leq t < \xi_3\} \\ &= P\{\eta_1 \leq s < \eta_1 + \eta_2 \leq t < \eta_1 + \eta_2 + \eta_3\} \\ &= \int_0^s P\{s < u + \eta_2 \leq t < u + \eta_2 + \eta_3\}\lambda e^{-\lambda u}\,du \\ &= \int_0^s \left(\int_{s-u}^{t-u} P\{t < u + v + \eta_3\}\lambda e^{-\lambda v}\,dv\right)\lambda e^{-\lambda u}\,du \\ &= \int_0^s \left(\int_{s-u}^{t-u} e^{-\lambda(t-u-v)}\lambda e^{-\lambda v}\,dv\right)\lambda e^{-\lambda u}\,du \\ &= \lambda^2 e^{-\lambda t}s(t-s). \end{aligned} \tag{53}$$

4.8. Solution to Exercise 3.8. *Since*

$$\begin{aligned} \xi_n^t &= \eta_1^t + \cdots + \eta_n^t \\ &= \xi_{N(t)+1} - t + \eta_{N(t)+2} + \cdots + \eta_{N(t)+n} \\ &= \xi_{N(t)+n} - t, \end{aligned} \tag{54}$$

it follows that

$$N^t(s) = \max\{n : \xi_n^t \le s\}$$
$$= \max\{n : \xi_{N(t)+n} \le t+s\}$$
$$= \max\{n : \xi_n \le t+s\} - N(t)$$
$$= N(t+s) - N(t). \tag{55}$$

4.9. Solution to Exercise 3.9. *It is easily verified that*

$$\{N(t) = n\} = \{\eta_{n+1} > t - \xi_n, t \ge \xi_n\}$$

$$\{\eta_1^t > s, N(t) = n\} = \{\eta_{n+1} > s + t - \xi_n, t \ge \xi_n\}$$

Since ξ_n, η_{n+1} are independent and η_{n+1} satisfies the lack of memory property from Exercise 3.3,

$$P\{\eta_1^t > s, N(t) = n\} = P\{\eta_{n+1} > s + t - \xi_n, t \ge \xi_n\}$$
$$= \int_{-\infty}^t P\{\eta_{n+1} > s + t - u\} P_{\xi_n}(du)$$
$$= P\{\eta_{n+1} > s\} \int_{-\infty}^t P\{\eta_{n+1} > t - u\} P_{\xi_n}(du) \tag{56}$$
$$= P\{\eta_{n+1} > s\} P\{\eta_{n+1} > t - \xi_n, t \ge \xi_n\}$$
$$= P\{\eta_{n+1} > s\} P\{N(t) = n\}$$
$$= P\{\eta_1 > s\} P\{N(t) = n\}.$$

The last equality holds because η_{n+1} has the same distribution as η_1. Now divide both sides by $P\{N(t) = n\}$ to get

$$P\{\eta_1^t > s \mid N(t) = n\} = P\{\eta_1 > s\}$$

for any $n = 0, 1, 2, \ldots$. As a result,

$$P\{\eta_1^t > s \mid N(t)\} = P\{\eta_1 > s\}$$

because $N(t)$ is a discrete random variable with values $0, 1, 2, \ldots$.

4.10. Solution to Exercise 3.10. *As in Solution 3.9,*

$$\{N(t) = n\} = \{\eta_{n+1} > t - \xi_n, t \ge \xi_n\},$$

$$\{\eta_1^t > s_1, N(t) = n\} = \{\eta_{n+1} > s_1 + t - \xi_n, t \ge \xi_n\},$$

and, more generally,

$$\{\eta_1^t > s_1, \eta_2^t > s_2, \ldots, \eta_k^t > s_k, N(t) = n\}$$
$$= \{\eta_{n+1} > s_1 + t - \xi_n, t \ge \xi_n\} \cap \{\eta_{n+2} > s_2\} \cap \cdots \cap \{\eta_{n+k} > s_k\}. \tag{57}$$

Since $\xi_n, \eta_{n+1}, \ldots, \eta_{n+k}$ are independent and $\eta_{n+2}, \ldots, \eta_{n+k}$ have the same distribution as $\eta_2, \ldots, \eta_k$, using Exercise 3.9 we find that

$$\begin{aligned}
&P\{\eta_1^t > s_1, \eta_2^t > s_2, \ldots, \eta_k^t > s_k, N(t) = n\} \\
&= P\{\eta_{n+1} > s_1 + t - \xi_n, \xi_n \le t\} P\{\eta_{n+2} > s_2\} \cdots P\{\eta_{n+k} > s_k\} \\
&= P\{\eta_1^t > s_1, N(t) = n\} P\{\eta_2 > s_2\} \cdots P\{\eta_k > s_k\} \qquad (58) \\
&= P\{\eta_1^t > s_1 \mid N(t) = n\} P\{\eta_2 > s_2\} \cdots P\{\eta_k > s_k\} P\{N(t) = n\} \\
&= P\{\eta_1 > s_1\} P\{\eta_2 > s_2\} \cdots P\{\eta_k > s_k\} P\{N(t) = n\}.
\end{aligned}$$

As in Solution 3.9, this implies the desired equality.

4.11. Solution to Exercise 3.11. *Take the expectation on both sides of the equality in Exercise 3.10 to find that*

$$P\{\eta_1^t > s_1, \ldots, \eta_k^t > s_k\} = P\{\eta_1 > s_1\} \cdots P\{\eta_k > s_k\}. \tag{59}$$

If all the numbers s_n except perhaps one are zero, it follows that

$$P\{\eta_n^t > s_n\} = P\{\eta_n > s_n\}, \quad n = 1, \ldots, k,$$

so the random variables η_n^t have the same distribution as η_n. Inserting this back into the first equality, we obtain

$$P\{\eta_1^t > s_1, \ldots, \eta_k^t > s_k\} = P\{\eta_1^t > s_1\} \cdots P\{\eta_k^t > s_k\},$$

so the η_n^t are independent.

To prove that the η_n^t are independent of $N(t)$ integrate the formula in Exercise 3.10 over $\{N(t) = n\}$ and multiply by $P\{N(t) = n\}$ to get

$$P\{\eta_1^t > s_1, \ldots, \eta_k^t > s_k, N(t) = n\} = P\{\eta_1 > s_1\} \cdots P\{\eta_k > s_k\} P\{N(t) = n\}.$$

But $P\{\eta_n^t > s_n\} = P\{\eta_n > s_n\}$, hence $N(t)$ and the η_n^t are independent.

4.12. Solution to Exercise 3.12. *We need to verify conditions 1), 2), 3) of definition 3.3. Clearly, $N(t) - \lambda t$ is $\mathcal{F}_t$-measurable. By Exercise 3.6*

$$E(|N(t)|) = E(N(t)) = \lambda t < \infty,$$

which means that $N(t)$ is integrable, and so is $N(t) - \lambda t$.

definition 3.9 implies that $N(t) - N(s)$ is independent of $\mathcal{F}_s$ for any $0 \le s \le t$, so

$$E(N(t) - N(s) \mid \mathcal{F}_s) = E(N(t) - N(s)) = E(N(t)) - E(N(s)) = \lambda t - \lambda s.$$

It follows that

$$E(N(t) - \lambda t \mid \mathcal{F}_s) = E(N(s) - \lambda s \mid \mathcal{F}_s) = N(s) - \lambda s,$$

completing the proof.

4.13. Solution to Exercise 3.13. *Since $\eta_n = \xi_n - \xi_{n-1}$ and $P\{\eta_n > 0\} = e^0 = 1$,*

$$P\{\xi_0 < \xi_1 < \xi_2 < \cdots\} = P\left(\bigcap_{n=1}^{\infty} \{\eta_n > 0\}\right) = 1.$$

Here we have used the property that if $P(A_n) = 1$ for all $n = 1, 2, \ldots$, then $P\left(\bigcap_{n=1}^{\infty} A_n\right) = 1$.

4.14. Solution to Exercise 3.14. *Since*

$$\lim_{n\to\infty} \xi_n = \sum_{n=1}^{\infty} \eta_n,$$

it follows that

$$\begin{aligned}\left\{\lim_{n\to\infty} \xi_n < \infty\right\} &\subset \{\eta_1, \eta_2, \ldots \text{ is a bounded sequence}\} \\ &= \bigcup_{m=1}^{\infty} \bigcap_{n=1}^{\infty} \{\eta_n \le m\}.\end{aligned} \tag{60}$$

Let us compute the probability of this event. Because $\bigcap_{n=1}^{N} \{\eta_n \le m\}, N = 1, 2, \ldots$ *is a contracting sequence of events,*

$$\begin{aligned}P\left(\bigcap_{n=1}^{\infty} \{\eta_n \le m\}\right) &= \lim_{N\to\infty} P\left(\bigcap_{n=1}^{N} \{\eta_n \le m\}\right) \\ &= \lim_{N\to\infty} \prod_{n=1}^{N} P\{\eta_n \le m\} \\ &= \lim_{N\to\infty} (1 - e^{-\lambda m})^N \\ &= 0.\end{aligned} \tag{61}$$

It follows that

$$\begin{aligned}P\left(\lim_{n\to\infty} \xi_n < \infty\right) &\le P\left(\bigcup_{m=1}^{\infty} \bigcap_{n=1}^{\infty} \{\eta_n \le m\}\right) \\ &\le \sum_{m=1}^{\infty} P\left(\bigcap_{n=1}^{\infty} \{\eta_n \le m\}\right) \\ &= 0,\end{aligned} \tag{62}$$

completing the proof.

While it is instructive to work through the above estimates, there exists a much more elegant argument. By the strong law of large numbers

$$\lim_{n\to\infty} \frac{\xi_n}{n} = \frac{1}{\lambda} \quad a.s.$$

Here $\frac{1}{\lambda}$ *is the expectation of each of the independent identically distributed random variables* η_n *(see Exercise 3.2). It follows that*

$$\lim_{n\to\infty} \xi_n = \infty \quad a.s.,$$

as required.

4.15. Solution to Exercise 3.15. *In the proof of definition 3.7 it was shown that*

$$P\{\xi_n > t\} = e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!}$$

for $t \geq 0$, see **??**. *Therefore the distribution function*

$$F_n(t) = P\{\xi_n \leq t\} = 1 - P\{\xi_n > t\} = e^{-\lambda t} \sum_{k=n}^{\infty} \frac{(\lambda t)^k}{k!}$$

of ξ_n is differentiable, the density f_n of ξ_n being

$$\begin{aligned} f_n(t) &= \frac{d}{dt} F_n(t) \\ &= -\lambda e^{-\lambda t} \sum_{k=n}^{\infty} \frac{(\lambda t)^k}{k!} + \lambda e^{-\lambda t} \sum_{k=n}^{\infty} \frac{(\lambda t)^{k-1}}{(k-1)!} \\ &= \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!} \end{aligned} \tag{63}$$

for $t > 0$, and clearly $f_n(t) = 0$ for $t \leq 0$.

4.16. Solution to Exercise 3.16. *Because $N(t)$ has non-decreasing trajectories*

$$\left\{ \lim_{t\to\infty} N(t) = \infty \right\} = \bigcap_{n=1}^{\infty} \bigcup_{k=1}^{\infty} \{N(k) \geq n\}.$$

Also, $\{N(k) \geq n\}, k = 1, 2, \ldots$ is an expanding sequence of events and

$$\begin{aligned} P\{N(k) \geq n\} &= e^{-\lambda k} \sum_{i=n}^{\infty} \frac{(\lambda k)^i}{i!} \\ &= 1 - e^{-\lambda k} \sum_{i=0}^{n-1} \frac{(\lambda k)^i}{i!} \to 1 \quad \text{as } k \to \infty. \end{aligned} \tag{64}$$

It follows that

$$P\left\{ \bigcup_{k=1}^{\infty} \{N(k) \geq n\} \right\} = 1,$$

so

$$P\left\{ \lim_{t\to\infty} N(t) = \infty \right\} = P\left\{ \bigcap_{n=1}^{\infty} \bigcup_{k=1}^{\infty} \{N(k) \geq n\} \right\} = 1.$$

4.17. Solution to Exercise 3.17. *Since*

$$\sinh(x) = \frac{e^x - e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!},$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!},$$

we have

(65)
$$\begin{aligned}P\{N(t) \text{ is odd}\} &= \sum_{n=0}^{\infty} P\{N(t) = 2n+1\} \\ &= \sum_{n=0}^{\infty} e^{-\lambda t}\frac{(\lambda t)^{2n+1}}{(2n+1)!} \\ &= e^{-\lambda t}\sinh(\lambda t),\end{aligned}$$

(66)
$$\begin{aligned}P\{N(t) \text{ is even}\} &= \sum_{n=0}^{\infty} P\{N(t) = 2n\} \\ &= \sum_{n=0}^{\infty} e^{-\lambda t}\frac{(\lambda t)^{2n}}{(2n)!} \\ &= e^{-\lambda t}\cosh(\lambda t).\end{aligned}$$

4.18. Solution to Exercise 3.18. *We can write*

$$N(n) = N(1) + (N(2) - N(1)) + \cdots + (N(n) - N(n-1)),$$

where $N(1), N(2)-N(1), N(3)-N(2), \ldots$ is a sequence of independent identically distributed random variables with expectation

$$E(N(1)) = E(N(2) - N(1)) = E(N(3) - N(2)) = \cdots = \lambda.$$

By the strong law of large numbers

(6.8)
$$\lim_{n\to\infty}\frac{N(n)}{n} = \lambda \quad a.s.$$

Now, if $n \le t \le n+1$, then $N(n) \le N(t) \le N(n+1)$ and

$$\frac{N(n)}{n+1} \le \frac{N(t)}{t} \le \frac{N(n+1)}{n}.$$

By **??** *both sides tend to λ as $n \to \infty$, implying that*

$$\lim_{t\to\infty}\frac{N(t)}{t} = \lambda \quad a.s.$$

4.19. Solution to Exercise 3.19. *Condition 3) of definition 3.12 implies that*

$$f_{W(t)}(x) = p(t, 0, x)$$

is the density of $W(t)$. Therefore, integrating by parts, we can compute the expectation

(67)
$$\begin{aligned}E(W(t)) &= \int_{-\infty}^{+\infty} xp(t,0,x)\,dx \\ &= \frac{1}{\sqrt{2\pi t}}\int_{-\infty}^{+\infty} xe^{-\frac{x^2}{2t}}\,dx \\ &= -\frac{t}{\sqrt{2\pi t}}\int_{-\infty}^{+\infty}\frac{d}{dx}e^{-\frac{x^2}{2t}}\,dx \\ &= -\frac{t}{\sqrt{2\pi t}}e^{-\frac{x^2}{2t}}\Big|_{-\infty}^{+\infty} = 0,\end{aligned}$$

and variance

$$
\begin{aligned}
E\left((W(t))^2\right) &= \int_{-\infty}^{+\infty} x^2 p(t,0,x)\,dx \\
&= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} x^2 e^{-\frac{x^2}{2t}}\,dx \\
&= -\frac{t}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} x \frac{d}{dx} e^{-\frac{x^2}{2t}}\,dx \\
&= -\frac{t}{\sqrt{2\pi t}} x e^{-\frac{x^2}{2t}}\Big|_{-\infty}^{+\infty} + \frac{t}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} e^{-\frac{x^2}{2t}}\,dx \\
&= 0 + \frac{t}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{u^2}{2}}\,du = t.
\end{aligned} \tag{68}
$$

We have used the substitution $u = \frac{x}{\sqrt{t}}$ and the formula stated in the hint.

4.20. Solution to Exercise 3.20. *Suppose that $s < t$. Condition 3) of definition 3.12 implies that the joint density of $W(s)$ and $W(t)$ is*

$$f_{W(s),W(t)}(x,y) = p(s,0,x)p(t-s,x,y).$$

It follows that

$$
\begin{aligned}
E(W(s)W(t)) &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} xyp(s,0,x)p(t-s,x,y)\,dx\,dy \\
&= \int_{-\infty}^{+\infty} xp(s,0,x)\left(\int_{-\infty}^{+\infty} yp(t-s,x,y)\,dy\right)dx \\
&= \int_{-\infty}^{+\infty} x^2 p(s,0,x)\,dx = s.
\end{aligned} \tag{69}
$$

This is because by the results of Exercise 3.19

$$
\begin{aligned}
\int_{-\infty}^{+\infty} yp(t-s,x,y)\,dy &= \int_{-\infty}^{+\infty} (x+u)p(t-s,x,x+u)\,du \\
&= \int_{-\infty}^{+\infty} (x+u)p(t-s,0,u)\,du \\
&= x\int_{-\infty}^{+\infty} p(t-s,0,u)\,du + \int_{-\infty}^{+\infty} up(t-s,0,u)\,du \\
&= x + 0 = x,
\end{aligned} \tag{70}
$$

and

$$\int_{-\infty}^{+\infty} x^2 p(s,0,x)\,dx = s.$$

It follows that for arbitrary $s,t \geq 0$

$$E(W(s)W(t)) = \min\{s,t\}.$$

4.21. Solution to Exercise 3.21. *Suppose that* $s \le t$. *Then by Exercise 3.20*

$$\begin{aligned} E\left(|W(t)-W(s)|^2\right) &= E(W(t)^2) - 2E(W(s)W(t)) + E(W(s)^2) \\ &= t - 2s + s = t - s. \end{aligned} \tag{71}$$

In general, for arbitrary $s, t \ge 0$

$$E\left(|W(t)-W(s)|^2\right) = |t-s|.$$

4.22. Solution to Exercise 3.22. *Using the density* $f_{W(t)}(x) = p(t,0,x)$ *of* $W(t)$, *we compute*

$$\begin{aligned} E(\exp(i\lambda W(t))) &= \int_{-\infty}^{+\infty} e^{i\lambda x} p(t,0,x)\,dx \\ &= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} e^{i\lambda x} e^{-\frac{x^2}{2t}}\,dx \\ &= \frac{1}{\sqrt{2\pi t}} e^{-\frac{\lambda^2 t}{2}} \int_{-\infty}^{+\infty} e^{-\frac{(x-i\lambda t)^2}{2t}}\,dx \\ &= e^{-\frac{\lambda^2 t}{2}}. \end{aligned} \tag{72}$$

4.23. Solution to Exercise 3.23. *Using the formula for the characteristic function of* $W(t)$ *found in Exercise 3.22, we compute*

$$\begin{aligned} E(W(t)^4) &= \left.\frac{d^4}{d\lambda^4}\right|_{\lambda=0} E(\exp(i\lambda W(t))) \\ &= \left.\frac{d^4}{d\lambda^4}\right|_{\lambda=0} e^{-\frac{1}{2}\lambda^2 t} \\ &= 3t^2. \end{aligned} \tag{73}$$

4.24. Solution to Exercise 3.24. *Since* $W^1(t), W^2(t)$ *are independent, their joint density is the product of the densities of* $W^1(t)$ *and* $W^2(t)$. *Therefore*

$$\begin{aligned} P\{|W(t)| < R\} &= \int_{\{|x|<R\}} p(t,0,x)p(t,0,y)\,dx\,dy \\ &= \frac{1}{2\pi t} \int_{\{|x|<R\}} e^{-\frac{x^2+y^2}{2t}}\,dx\,dy \\ &= \frac{1}{2\pi t} \int_0^R \int_0^{2\pi} r e^{-\frac{r^2}{2t}}\,d\varphi\,dr \\ &= -\int_0^R \frac{d}{dr} e^{-\frac{r^2}{2t}}\,dr \\ &= 1 - e^{-\frac{R^2}{2t}}. \end{aligned} \tag{74}$$

We have used the polar coordinates r, φ *to compute the integral.*

4.25. Solution to Exercise 3.25. *For any* $0 \leq s < t$

$$\begin{aligned} E(W(t) \mid \mathcal{F}_s) &= E(W(t) - W(s) \mid \mathcal{F}_s) + E(W(s) \mid \mathcal{F}_s) \\ &= E(W(t) - W(s)) + W(s) \\ &= W(s), \end{aligned} \tag{75}$$

since $W(t) - W(s)$ *is independent of* $\mathcal{F}_s$ *by definition 3.17,* $W(s)$ *is* $\mathcal{F}_s$*-measurable and* $E(W(t)) = E(W(s)) = 0$.

4.26. Solution to Exercise 3.26. *For any* $0 \leq s < t$

$$\begin{aligned} E(W(t)^2 \mid \mathcal{F}_s) &= E(|W(t) - W(s)|^2 \mid \mathcal{F}_s) + E(2W(t)W(s) \mid \mathcal{F}_s) \\ &\quad - E(W(s)^2 \mid \mathcal{F}_s) \\ &= E(|W(t) - W(s)|^2) + 2W(s)E(W(t) \mid \mathcal{F}_s) - W(s)^2 \\ &= t - s + 2W(s)^2 - W(s)^2 \\ &= t - s + W(s)^2, \end{aligned} \tag{76}$$

since $W(t) - W(s)$ *is independent of* $\mathcal{F}_s$ *and has the normal distribution with mean 0 and variance* $t - s$, $W(s)$ *is* $\mathcal{F}_s$*-measurable, and* $W(t)$ *is a martingale. It follows that*

$$E(W(t)^2 - t \mid \mathcal{F}_s) = W(s)^2 - s,$$

as required.

4.27. Solution to Exercise 3.27. *For any* $0 \leq t_0 < t_1 < \cdots < t_n$ *the increments*

$$V(t_n) - V(t_{n-1}), \ldots, V(t_1) - V(t_0)$$

of $V(t)$ *are independent, since the increments*

$$W(t_n + T) - W(t_{n-1} + T), \ldots, W(t_1 + T) - W(t_0 + T)$$

of $W(t)$ *are independent. For any* $0 \leq s < t$ *the increment* $V(t) - V(s)$ *has the normal distribution with mean zero and variance* $t - s$, *since* $W(t + T) - W(s + T)$ *does. Moreover, the paths* $t \mapsto V(t) = W(t + T) - W(T)$ *are continuous and*

$$V(0) = W(T) - W(T) = 0.$$

By definition 3.18 $V(t)$ *is a Wiener process.*

4.28. Solution to Exercise 3.28. *It is clear that* $V(0) = \frac{1}{c}W(0) = 0$ *a.s. and the paths* $t \mapsto V(t) = \frac{1}{c}W(c^2 t)$ *are a.s. continuous. We shall verify that* $V(t)$ *and* $|V(t)|^2 - t$ *are martingales with respect to the filtration*

$$\begin{aligned} \mathcal{G}_t &= \sigma\{V(s) : 0 \leq s \leq t\} \\ &= \sigma\{W(c^2 s) : 0 \leq s \leq t\} \\ &= \sigma\{W(s) : 0 \leq s \leq c^2 t\} \\ &= \mathcal{F}_{c^2 t}. \end{aligned} \tag{77}$$

Indeed, if $s < t$, then $c^2 s < c^2 t$, so

$$\begin{aligned} E(V(t) \mid \mathcal{G}_s) &= E\left(\frac{1}{c}W(c^2t) \mid \mathcal{F}_{c^2s}\right) \\ &= \frac{1}{c}E\left(W(c^2t) \mid \mathcal{F}_{c^2s}\right) \\ &= \frac{1}{c}W(c^2s) = V(s), \end{aligned} \tag{78}$$

and

$$\begin{aligned} E(|V(t)|^2 - t \mid \mathcal{G}_s) &= E\left(\frac{1}{c^2}|W(c^2t)|^2 - t \mid \mathcal{F}_{c^2s}\right) \\ &= \frac{1}{c^2}E(|W(c^2t)|^2 - c^2t \mid \mathcal{F}_{c^2s}) \\ &= \frac{1}{c^2}(|W(c^2s)|^2 - c^2s) \\ &= |V(s)|^2 - s, \end{aligned} \tag{79}$$

since $W(t)$ and $|W(t)|^2 - t$ are martingales with respect to the filtration $\mathcal{F}_t$. It follows by Lévy's martingale characterization that $V(t)$ is a Wiener process.

4.29. Solution to Exercise 3.29. *Since the increments $\Delta_i^n W$ are independent and*

$$E(\Delta_i^n W) = 0, \quad E\left((\Delta_i^n W)^2\right) = \frac{T}{n}, \quad E\left((\Delta_i^n W)^4\right) = \frac{3T^2}{n^2},$$

it follows that

$$\begin{aligned} E\left(\left[\sum_{i=0}^{n-1}(\Delta_i^n W)^2 - T\right]^2\right) &= E\left(\left[\sum_{i=0}^{n-1}\left((\Delta_i^n W)^2 - \frac{T}{n}\right)\right]^2\right) \\ &= \sum_{i=0}^{n-1} E\left[\left((\Delta_i^n W)^2 - \frac{T}{n}\right)^2\right] \\ &= \sum_{i=0}^{n-1}\left[E\left((\Delta_i^n W)^4\right) - \frac{2T}{n}E\left((\Delta_i^n W)^2\right) + \frac{T^2}{n^2}\right] \\ &= \sum_{i=0}^{n-1}\left[\frac{3T^2}{n^2} - \frac{2T^2}{n^2} + \frac{T^2}{n^2}\right] = \frac{2T^2}{n} \to 0, \end{aligned} \tag{80}$$

as $n \to \infty$.

4.30. Solution to Exercise 3.30. *We claim that, with probability 1, for any positive integer n there is a $t \in [0, \frac{1}{n^4}]$ such that $\frac{|W(t)|}{t} > n$. This condition implies that $W(t)$ is not differentiable at $t = 0$.*

Let us put

$$A_n = \left\{\frac{|W(t)|}{t} > n \text{ for some } t \in [0, \frac{1}{n^4}]\right\}.$$

By Exercise 3.28

$$V_n(t) = \frac{1}{n^2}W(n^4t)$$

is a Brownian motion for any n. Therefore

$$
\begin{aligned}
P(A_n) &\geq P\left\{\frac{|W(1/n^4)|}{1/n^4} > n\right\} \\
&= P\left\{\frac{|V_n(1/n^4)|}{1/n^4} > n\right\} \qquad (81)\\
&= P\{|W(1)| > \frac{1}{n}\} \to 1 \quad \text{as } n \to \infty.
\end{aligned}
$$

Since $A_1, A_2, \ldots$ is a contracting sequence of events,

$$
P\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} P(A_n) = 1,
$$

which proves the claim.

4.31. Solution to Exercise 3.31. *By Exercise 3.27 $V_t(s) = W(s+t) - W(t)$ is a Wiener process for any $t \geq 0$. Therefore, by Exercise 3.30 $V_t(s)$ is a.s. non-differentiable at $s = 0$. But this implies that $W(t)$ is a.s. non-differentiable at t.*

4.32. Solution to Exercise 3.32. *Differentiating*

$$
p(t,x,y) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}},
$$

we obtain

$$
\frac{\partial}{\partial t} p(t,x,y) = \frac{y^2 - 2yx + x^2 - t}{2t^2} p(t,x,y),
$$

$$
\frac{\partial}{\partial y} p(t,x,y) = \frac{x-y}{t} p(t,x,y),
$$

$$
\frac{\partial^2}{\partial y^2} p(t,x,y) = \frac{y^2 - 2yx + x^2 - t}{t^2} p(t,x,y),
$$

so

$$
\frac{\partial p}{\partial t} = \frac{1}{2}\frac{\partial^2 p}{\partial y^2},
$$

as required.

4.33. Solution to Exercise 3.33. *Clearly, $Z(t) = -W(t)$ has a.s. continuous trajectories and $Z(0) = -W(0) = 0$ a.s. If $W(t)$ has stationary independent increments, then so does $Z(t) = -W(t)$. Finally,*

$$
Z(t) - Z(s) = -(W(t) - W(s))
$$

has the same distribution as $W(t) - W(s)$, i.e. normal with mean 0 and variance $t - s$. By definition 3.18 $Z(t)$ is a Wiener process.

4.34. Solution to Exercise 3.34. *Let $0 \le s < t$. Then*

$$
\begin{aligned}
\int_{\{W(s)\in B\}} 1_A(W(t))\,dP &= P\{W(s)\in B, W(t)\in A\} \\
&= \int_B \int_A p(s,0,x)p(t-s,x,y)\,dx\,dy \\
&= \int_B \left(\int_A p(t-s,x,y)\,dy\right) p(s,0,x)\,dx \\
&= \int_{\{W(s)\in B\}} \left(\int_A p(t-s,W(s),y)\,dy\right) dP,
\end{aligned} \tag{82}
$$

for any Borel set $B \subset \mathbb{R}$. It follows that

$$
P\{W(t)\in A \mid W(s)\} = E(1_A(W(t)) \mid W(s)) = \int_A p(t-s,W(s),y)\,dy.
$$

4.35. Solution to Exercise 3.35. *We shall prove that $e^{W(t)}e^{-\frac{t}{2}}$ is a martingale with respect to the filtration $\mathcal{F}_t$. Clearly, it is adapted to the filtration $\mathcal{F}_t$, since $W(t)$ is. Let $0 \le s < t$. Because $W(t)-W(s)$ is independent of $\mathcal{F}_s$ and $W(s)$ is $\mathcal{F}_s$-measurable,*

$$
\begin{aligned}
E\left(e^{W(t)} \mid \mathcal{F}_s\right) &= E\left(e^{W(t)-W(s)}e^{W(s)} \mid \mathcal{F}_s\right) \\
&= e^{W(s)}E\left(e^{W(t)-W(s)} \mid \mathcal{F}_s\right) \\
&= e^{W(s)}E\left(e^{W(t)-W(s)}\right).
\end{aligned} \tag{83}
$$

The increment $W(t)-W(s)$ has the normal distribution with mean 0 and variance $t-s$, so the expectation of $e^{W(t)-W(s)}$ is equal to

$$
\begin{aligned}
E\left(e^{W(t)-W(s)}\right) &= \int_{-\infty}^{+\infty} e^x p(t-s,0,x)\,dx \\
&= e^{\frac{t-s}{2}} \int_{-\infty}^{+\infty} p(t-s,0,x-t)\,dx \\
&= e^{\frac{t-s}{2}}.
\end{aligned} \tag{84}
$$

It follows that

$$
E\left(e^{W(t)}e^{-\frac{t}{2}} \mid \mathcal{F}_s\right) = e^{W(s)}e^{-\frac{s}{2}}.
$$

It also follows that $e^{W(t)}e^{-\frac{t}{2}}$ is integrable. Therefore $e^{W(t)}e^{-\frac{t}{2}}$ is a martingale.

4.36. Solution to Exercise 3.36. *Let $0 \le s < t$. We are looking for a Borel function F such that $E(W(s) \mid W(t)) = F(W(t))$, i.e.*

$$
\int_{\{W(t)\in A\}} W(s)\,dP = \int_{\{W(t)\in A\}} F(W(t))\,dP
$$

for any Borel set A in $\mathbb{R}$. The integral on the right-hand side can be written as

$$
\int_{\{W(t)\in A\}} F(W(t))\,dP = \int_A F(y)p(t,0,y)\,dy,
$$

and the integral on the left-hand side as

$$\int_{\{W(t)\in A\}} W(s)\,dP = \int_A \left(\int_{-\infty}^{+\infty} xp(s,0,x)p(t-s,x,y)\,dx \right) dy,$$

using the expression for the joint density of $W(s)$ and $W(t)$ in Solution 3.20. Let us compute the inner integral:

$$\begin{aligned} \int_{-\infty}^{+\infty} xp(s,0,x)p(t-s,x,y)\,dx &= p(t,0,y) \int_{-\infty}^{+\infty} xp\left(\frac{s(t-s)}{t}, \frac{s}{t}y, x \right) dx \\ &= \frac{s}{t} yp(t,0,y). \end{aligned} \tag{85}$$

(To see that the first equality holds, just use formula **??** *for $p(t,x,y)$.) Therefore*

$$\int_{\{W(t)\in A\}} W(s)\,dP = \int_A \frac{s}{t} yp(t,0,y)\,dy.$$

It follows that $F(y) = \frac{s}{t}y$, i.e.

$$E(W(s) \mid W(t)) = \frac{s}{t} W(t).$$

CHAPTER 4

Itô Stochastic Calculus

1. Itô Stochastic Integral: Definition

We shall follow a construction resembling that of the Riemann integral. First, the integral will be defined for a class of piecewise constant processes called random step processes. Then it will be extended to a larger class by approximation.

There are, however, at least two major differences between the Riemann and Itô integrals. One is the type of convergence. The approximations of the Riemann integral converge in $\mathbb{R}$, while the Itô integral will be approximated by sequences of random variables converging in L^2. The other difference is this. The Riemann sums approximating the integral of a function $f : [0,T] \to \mathbb{R}$ are of the form

$$\sum_{j=0}^{n-1} f(s_j)(t_{j+1} - t_j), \tag{86}$$

where $0 = t_0 < t_1 < \cdots < t_n = T$ and s_j is an arbitrary point in $[t_j, t_{j+1}]$ for each j. The value of the Riemann integral does not depend on the choice of the points $s_j \in [t_j, t_{j+1})$. In the stochastic case the approximating sums will have the form

$$\sum_{j=0}^{n-1} f(s_j)(W(t_{j+1}) - W(t_j)). \tag{87}$$

It turns out that the limit of such approximations does depend on the choice of the intermediate points s_j in $[t_j, t_{j+1}]$. In the next exercise we take $f(t) = W(t)$ and consider two different choices of intermediate points.

Let $0 = t_0^n < t_1^n < \cdots < t_n^n = T$, where $t_j^n = \frac{jT}{n}$, be a partition of the interval $[0,T]$ into n equal parts. Find the following limits in L^2:

$$\lim_{n\to\infty} \sum_{j=0}^{n-1} W(t_j^n)\left(W(t_{j+1}^n) - W(t_j^n)\right) \tag{88}$$

and

$$\lim_{n\to\infty} \sum_{j=0}^{n-1} W\left(t_{j+1}^n\right)\left(W\left(t_{j+1}^n\right) - W\left(t_j^n\right)\right). \tag{89}$$

Remark 4.1. ***Hint:*** *Apply Exercise 6.29. You will need to transform the sums to make this possible. The identities*

$$a(b-a)=\frac{1}{2}(b^2-a^2)-\frac{1}{2}(a-b)^2, \tag{90}$$

$$b(b-a)=\frac{1}{2}(b^2-a^2)+\frac{1}{2}(a-b)^2 \tag{91}$$

may be of help.

The ambiguity resulting from different choices of the intermediate points s_j in each subinterval $[t_j,t_{j+1}]$ can be removed by insisting that the approximations of the integrand should consist only of processes adapted to the underlying filtration $\mathcal{F}_t$. This amounts to taking $s_j=t_j$ for each j. The choice is motivated by the interpretation of $\mathcal{F}_t$: the value of the approximation at t may depend only on what has happened up to time t, but not on any future events.

Definition 4.1. *We shall call $f(t), t\geq 0$ a* ***random step process*** *if there is a finite sequence of numbers $0=t_0<t_1<\cdots<t_n$ and square integrable random variables $\eta_0,\eta_1,\ldots,\eta_{n-1}$ such that*

$$f(t)=\sum_{j=0}^{n-1}\eta_j\mathbf{1}_{[t_j,t_{j+1})}(t), \tag{92}$$

where η_j is $\mathcal{F}_{t_j}$-measurable for $j=0,1,\ldots,n-1$. The set of random step processes will be denoted by M^2_{step}.

Observe that the assumption that the η_j are to be $\mathcal{F}_{t_j}$-measurable ensures that $f(t)$ is adapted to the filtration $\mathcal{F}_t$. The assumption that the η_j are square integrable ensures that $f(t)$ is square integrable for each t. Also, M^2_{step} is a vector space, that is, $af+bg\in M^2_{\text{step}}$ for any $f,g\in M^2_{\text{step}}$ and $a,b\in\mathbb{R}$.

Definition 4.2. *The stochastic integral of a random step process $f\in M^2_{\text{step}}$ of the form (92) is defined by*

$$I(f)=\sum_{j=0}^{n-1}\eta_j\left(W(t_{j+1})-W(t_j)\right). \tag{93}$$

Proposition 4.3. *For any random step process $f\in M^2_{\text{step}}$ the stochastic integral $I(f)$ is a square integrable random variable, i.e. $I(f)\in L^2$, such that*

$$\mathbb{E}\left(|I(f)|^2\right)=\mathbb{E}\left(\int_0^\infty |f(t)|^2\,dt\right). \tag{94}$$

Proof. *Let us denote the increment $W(t_{j+1})-W(t_j)$ by Δ_jW and $t_{j+1}-t_j$ by Δ_jt for brevity. Then*

$$\mathbb{E}(\Delta_jW)=0 \quad \textit{and} \quad \mathbb{E}\left((\Delta_jW)^2\right)=\Delta_jt. \tag{95}$$

First, we shall compute the expectation of

$$|I(f)|^2 = \sum_{j=0}^{n-1}\sum_{k=0}^{n-1} \eta_j\eta_k \Delta_j W \Delta_k W = \sum_{j=0}^{n-1} \eta_j^2 (\Delta_j W)^2 + 2\sum_{k<j} \eta_j\eta_k \Delta_j W \Delta_k W. \tag{96}$$

Since η_j and $\Delta_j W$ are independent,

$$\mathbb{E}\left(\eta_j^2(\Delta_j W)^2\right) = \mathbb{E}\left(\eta_j^2\right)\mathbb{E}\left((\Delta_j W)^2\right) = \mathbb{E}\left(\eta_j^2\right)\Delta_j t. \tag{97}$$

If $k < j$, then $\eta_j\eta_k\Delta_k W$ and $\Delta_j W$ are independent, so

$$\mathbb{E}\left(\eta_j\eta_k\Delta_j W \Delta_k W\right) = \mathbb{E}\left(\eta_j\eta_k\Delta_k W\right)\mathbb{E}\left(\Delta_j W\right) = 0. \tag{98}$$

Therefore

$$\mathbb{E}\left(|I(f)|^2\right) = \sum_{j=0}^{n-1} \mathbb{E}\left(\eta_j^2\right)\Delta_j t. \tag{99}$$

It follows that $I(f) \in L^2$, since $\eta_0, \eta_1, \ldots, \eta_{n-1} \in L^2$.
On the other hand,

$$|f(t)|^2 = \sum_{j=0}^{n-1}\sum_{k=0}^{n-1} \eta_j\eta_k \mathbf{1}_{[t_j,t_{j+1})}(t)\mathbf{1}_{[t_k,t_{k+1})}(t) = \sum_{j=0}^{n-1} \eta_j^2 \mathbf{1}_{[t_j,t_{j+1})}(t), \tag{100}$$

implying that

$$\mathbb{E}\left(\int_0^\infty |f(t)|^2\,dt\right) = \sum_{j=0}^{n-1} \mathbb{E}\left(\eta_j^2\right)\Delta_j t. \tag{101}$$

This means that

$$\mathbb{E}\left(|I(f)|^2\right) = \mathbb{E}\left(\int_0^\infty |f(t)|^2\,dt\right), \tag{102}$$

as required. □

Verify that for any random step processes $f, g \in M^2_{\text{step}}$

$$\mathbb{E}(I(f)I(g)) = \mathbb{E}\left(\int_0^\infty f(t)g(t)\,dt\right). \tag{103}$$

REMARK 4.2. ***Hint:*** *Try to adapt the proof of Proposition 4.3. Use a common partition $0 = t_0 < t_1 < \cdots < t_n$ in which to represent both f and g in the form* (92).

Show that $I : M^2_{\text{step}} \to L^2$ is a linear map, i.e. for any $f, g \in M^2_{\text{step}}$ and any $\alpha, \beta \in \mathbb{R}$

$$I(\alpha f + \beta g) = \alpha I(f) + \beta I(g). \tag{104}$$

REMARK 4.3. ***Hint:*** *As in Exercise 4.2, use a common partition* $0 = t_0 < t_1 < \cdots < t_n$ *in which to represent both* f *and* g *in the form* (92).

The stochastic integral $I(f)$ *has been defined for any random step process* $f \in M^2_{\text{step}}$. *The next stage is to extend* I *to a larger class of processes by approximation. This larger class can be defined as follows.*

DEFINITION 4.4. *We denote by* M^2 *the class of stochastic processes* $f(t), t \geq 0$ *such that*

$$\mathbb{E}\left(\int_0^\infty |f(t)|^2\,dt\right) < \infty \tag{105}$$

and there is a sequence $f_1, f_2, \ldots \in M^2_{\text{step}}$ *of random step processes such that*

$$\lim_{n\to\infty} \mathbb{E}\left(\int_0^\infty |f(t) - f_n(t)|^2\,dt\right) = 0. \tag{106}$$

In this case we shall say that the sequence of random step processes $f_1, f_2, \ldots$ *approximates* f *in* M^2.

DEFINITION 4.5. *We call* $I(f) \in L^2$ *the* ***Itô stochastic integral*** *(from* 0 *to* ∞*) of* $f \in M^2$ *if*

$$\lim_{n\to\infty} \mathbb{E}\left(|I(f) - I(f_n)|^2\right) = 0 \tag{107}$$

for any sequence $f_1, f_2, \ldots \in M^2_{\text{step}}$ *of random step processes that approximates* f *in* M^2, *i.e. such that* (106) *is satisfied. We shall also write*

$$\int_0^\infty f(t)\,dW(t) \tag{108}$$

in place of $I(f)$.

Proposition 4.6. *For any* $f \in M^2$ *the stochastic integral* $I(f) \in L^2$ *exists, is unique (as an element of* L^2*, i.e. to within equality a.s.) and satisfies*

$$\mathbb{E}\left(|I(f)|^2\right) = \mathbb{E}\left(\int_0^\infty |f(t)|^2\,dt\right). \tag{109}$$

PROOF. *It will be convenient to write*

$$\|f\|_{M^2} = \sqrt{\mathbb{E}\left(\int_0^\infty |f(t)|^2\,dt\right)} \quad \text{and} \quad \|\eta\|_{L^2} = \sqrt{\mathbb{E}(\eta^2)} \tag{110}$$

for any $f \in M^2$ *and* $\eta \in L^2$. *These are norms in* M^2 *and* L^2, *respectively.*

Let $f_1, f_2, \ldots \in M^2_{\text{step}}$ *be a sequence of random step processes approximating* $f \in M^2$, *i.e. satisfying* (106), *which can be written as*

$$\lim_{n\to\infty} \|f - f_n\|_{M^2} = 0. \tag{111}$$

We claim that $I(f_1), I(f_2), \dots$ is a Cauchy sequence in L^2. Indeed, for any $\varepsilon > 0$ there is an N such that $\|f - f_n\|_{M^2} < \frac{\varepsilon}{2}$ for all $n > N$. By Proposition 4.3

$$\|I(f_m) - I(f_n)\|_{L^2} = \|I(f_m - f_n)\|_{L^2} \tag{112}$$
$$= \|f_m - f_n\|_{M^2} \tag{113}$$
$$\leq \|f - f_m\|_{M^2} + \|f - f_n\|_{M^2} \tag{114}$$
$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \tag{115}$$

for any $m, n > N$, which proves the claim.

Because L^2 with the norm $\|\cdot\|_{L^2}$ is a complete space (in fact a Hilbert space), every Cauchy sequence in L^2 has a limit. It follows that $I(f_1), I(f_2), \dots$ has a limit in L^2 for any sequence $f_1, f_2, \dots$ of random step processes approximating f. It remains to show that the limit is the same for all such sequences. Suppose that $f_1, f_2, \dots$ and $g_1, g_2, \dots$ are two sequences of random step processes approximating f. Then the interlaced sequence $f_1, g_1, f_2, g_2, \dots$ approximates f too, so the sequence $I(f_1), I(g_1), I(f_2), I(g_2), \dots$ has a limit in L^2. But then all subsequences of the latter sequence, in particular, $I(f_1), I(f_2), \dots$ and $I(g_1), I(g_2), \dots$ have the same limit, which we denote by $I(f)$. We have shown that

$$\lim_{n\to\infty} \|I(f) - I(f_n)\|_{L^2} = 0, \tag{116}$$

i.e. (107) *holds for any sequence $f_1, f_2, \dots$ of random step processes approximating f.*
Finally, by Proposition 4.3

$$\|I(f_n)\|_{L^2} = \|f_n\|_{M^2} \tag{117}$$

for each n, since the f_n are random step processes. By taking the limit as $n \to \infty$ we obtain

$$\|I(f)\|_{L^2} = \|f\|_{M^2}. \tag{118}$$

But this is equality (109). □

Show that for any $f, g \in M^2$

$$\mathbb{E}(I(f)I(g)) = \mathbb{E}\left(\int_0^\infty f(t)g(t)\,dt\right). \tag{119}$$

REMARK 4.4. ***Hint:*** *Write the left-hand side in terms of $\mathbb{E}\left(|I(f)+I(g)|^2\right)$ and $\mathbb{E}\left(|I(f)-I(g)|^2\right)$, the right-hand side in terms of $\mathbb{E}\left(\int_0^\infty |f(t)+g(t)|^2\,dt\right)$ and $\mathbb{E}\left(\int_0^\infty |f(t)-g(t)|^2\,dt\right)$ and then use* (109).

Having defined the Itô stochastic integral from 0 to ∞, we are now in a position to consider stochastic integrals over any finite time interval $[0, T]$.

DEFINITION 4.7. *For any $T > 0$ we shall denote by M_T^2 the space of all stochastic processes $f(t), t \geq 0$ such that*

$$\mathbf{1}_{[0,T)} f \in M^2. \tag{120}$$

The Itô stochastic integral (from 0 to T) of $f \in M_T^2$ is defined by

$$I_T(f) = I(\mathbf{1}_{[0,T)} f). \tag{121}$$

We shall also write

$$\int_0^T f(t)\, dW(t) \tag{122}$$

in place of $I_T(f)$.

Show that each random step process $f \in M^2_{\text{step}}$ belongs to M_t^2 for any $t > 0$ and

$$I_t(f) = \int_0^t f(s)\, dW(s) \tag{123}$$

is a martingale.

Remark 4.5. ***Hint:*** *The stochastic integral of a random step process f is given by the sum (93). What is the conditional expectation of the j-th term of this sum given $\mathcal{F}_s$ if $s < t_j$? What is it when $s \geq t_j$?*

The processes for which the stochastic integral exists have been defined as those that can be approximated by random step processes. However, it is not always easy to check whether or not such an approximation exists. For practical purposes it is important to have a straightforward sufficient condition for a process to have a stochastic integral. In calculus there is a well-known result of this kind: the Riemann integral exists for any continuous function. Here is a theorem of this kind for the Itô integral.

Theorem 4.8. *Let $f(t), t \geq 0$ be a stochastic process with a.s. continuous paths adapted to the filtration $\mathcal{F}_t$. Then:*

(1) *$f \in M^2$, i.e. the Itô integral $I(f)$ exists, whenever*

$$\mathbb{E}\left(\int_0^\infty |f(t)|^2\, dt\right) < \infty; \tag{124}$$

(2) *$f \in M_T^2$, i.e. the Itô integral $I_T(f)$ exists, whenever*

$$\mathbb{E}\left(\int_0^T |f(t)|^2\, dt\right) < \infty. \tag{125}$$

Proof. (1) *Suppose that $f(t), t \geq 0$ is an adapted process with a.s. continuous paths. If (124) holds, then*

$$f_n(t) = \begin{cases} n\int_{\frac{k-1}{n}}^{\frac{k}{n}} f(s)\, ds & \frac{k}{n} \leq t < \frac{k+1}{n} \textit{ for } k = 1, 2, \ldots, n^2 - 1, \\ 0 & \textit{otherwise}, \end{cases} \tag{126}$$

is a sequence of random step processes in M^2_{step}. *Observe that for any* $k = 1, 2, \dots$

$$\int_{\frac{k}{n}}^{\frac{k+1}{n}} |f_n(t)|^2\,dt = n\left|\int_{\frac{k-1}{n}}^{\frac{k}{n}} f(t)\,dt\right|^2 \le \int_{\frac{k-1}{n}}^{\frac{k}{n}} |f(t)|^2\,dt \quad a.s. \tag{127}$$

by Jensen's inequality. We claim that

$$\lim_{n\to\infty}\int_0^\infty |f(t)-f_n(t)|^2\,dt = 0 \quad a.s. \tag{128}$$

This will imply that

$$\lim_{n\to\infty}\mathbb{E}\left(\int_0^\infty |f(t)-f_n(t)|^2\,dt\right) = 0 \tag{129}$$

by the dominated convergence theorem and condition (124) *because*

$$\int_0^\infty |f(t)-f_n(t)|^2\,dt \le 2\int_0^\infty \left(|f(t)|^2+|f_n(t)|^2\right)\,dt \tag{130}$$

$$\le 4\int_0^\infty |f(t)|^2\,dt. \tag{131}$$

The last inequality follows, since

$$\int_0^\infty |f_n(t)|^2\,dt \le \int_0^\infty |f(t)|^2\,dt \quad a.s. \tag{132}$$

for any n, *by taking the sum from* $k = 0$ *to* ∞ *in* (127).
To verify the claim observe that

$$\int_0^\infty |f(t)-f_n(t)|^2\,dt = \int_0^N |f(t)-f_n(t)|^2\,dt + \int_N^\infty |f(t)-f_n(t)|^2\,dt \tag{133}$$

$$\le \int_0^N |f(t)-f_n(t)|^2\,dt + 2\int_N^\infty \left(|f(t)|^2+|f_n(t)|^2\right)\,dt \tag{134}$$

$$\le \int_0^N |f(t)-f_n(t)|^2\,dt + 4\int_{N-1}^\infty |f(t)|^2\,dt \quad a.s. \tag{135}$$

The last inequality holds because

$$\int_N^\infty |f_n(t)|^2\,dt \le \int_{N-\frac{1}{n}}^\infty |f(t)|^2\,dt \le \int_{N-1}^\infty |f(t)|^2\,dt \quad a.s. \tag{136}$$

for any n *and* N, *by taking the sum from* $k = nN$ *to* ∞ *in* (127). *The claim follows because*

$$\lim_{N\to\infty}\int_{N-1}^\infty |f(t)|^2\,dt = 0 \quad a.s. \tag{137}$$

by (124) *and*

$$\lim_{n\to\infty}\int_0^N |f(t)-f_n(t)|^2\,dt = 0 \quad a.s. \tag{138}$$

for any fixed N by the continuity of paths of f.

The above means that the sequence $f_1, f_2, \ldots \in M^2_{\text{step}}$ approximates f in the sense of Definition 4.4, so $f \in M^2$.

(2) *If f satisfies* (125) *for some $T > 0$, then $\mathbf{1}_{[0,T)}f$ satisfies* (124)*. Since f is adapted and has a.s. continuous paths, $\mathbf{1}_{[0,T)}f$ is also adapted and its paths are a.s. continuous, except perhaps at T. But the lack of continuity at the single point T does not affect the argument in (1), so $\mathbf{1}_{[0,T)}f \in M^2$. This in turn implies that $f \in M^2_T$, completing the proof.*

Show that the Wiener process $W(t)$ belongs to M^2_T for each $T > 0$.

Remark 4.6. ***Hint:*** *Apply part (2) of Theorem 4.8.*

Show that $W(t)^2$ belongs to M^2_T for each $T > 0$.

Remark 4.7. ***Hint:*** *Once again, apply part (2) of Theorem 4.8.*

The next theorem, which we shall state without proof, provides a characterization of M^2 and M^2_T, i.e. a necessary and sufficient condition for a stochastic process f to belong to M^2 or M^2_T. It involves the notion of a progressively measurable process.

Definition 4.9. *A stochastic process $f(t), t \geq 0$ is called **progressively measurable** if for any $t \geq 0$*

$$(s,\omega) \mapsto f(s,\omega) \tag{139}$$

is a measurable function from $[0,t]\times\Omega$ with the σ-field $\mathcal{B}[0,t]\bar{\times}\mathcal{F}$ to $\mathbb{R}$. Here $\mathcal{B}[0,t]\bar{\times}\mathcal{F}$ is the product σ-field on $[0,t]\times\Omega$, that is, the smallest σ-field containing all sets of the form $A\times B$, where $A\subset[0,t]$ is a Borel set and $B\in\mathcal{F}$.

Theorem 4.10. (1) *The space M^2 consists of all progressively measurable stochastic processes $f(t), t\geq 0$ such that*

$$\mathbb{E}\left(\int_0^\infty |f(t)|^2\,dt\right) < \infty. \tag{140}$$

(2) *The space M^2_T consists of all progressively measurable stochastic processes $f(t), t \geq 0$ such that*

$$\mathbb{E}\left(\int_0^T |f(t)|^2\,dt\right) < \infty. \tag{141}$$

2. Examples

According to Exercise 4.6, the Wiener process $W(t)$ belongs to M_T^2 for any $T > 0$. Therefore the stochastic integral in the next exercise exists.

Verify the equality

$$\int_0^T W(t)\,dW(t) = \frac{1}{2}W(T)^2 - \frac{1}{2}T \tag{142}$$

by computing the stochastic integral from the definition, that is, by approximating the integrand by random step functions.

Remark 4.8. ***Hint:*** *It is convenient to use a partition of the interval $[0,T]$ into n equal parts. The limit of the sums approximating the integral has been found in Exercise 4.1.*

Verify the equality

$$\int_0^T t\,dW(t) = TW(T) - \int_0^T W(t)\,dt, \tag{143}$$

by computing the stochastic integral from the definition. (The integral on the right-hand side is understood as a Riemann integral defined pathwise, i.e. separately for each $\omega \in \Omega$.)

Remark 4.9. ***Hint:*** *You may want to use the same partition of $[0,T]$ into n equal parts as in Solution 4.8. The sums approximating the stochastic integral can be transformed with the aid of the identity*

$$c(b-a) = (db - ca) - b(d-c). \tag{144}$$

Show that $W(t)^2$ belongs to M_T^2 for each $T > 0$ and verify the equality

$$\int_0^T W(t)^2\,dW(t) = \frac{1}{3}W(T)^3 - \int_0^T W(t)\,dt, \tag{145}$$

where the integral on the right-hand side is a Riemann integral.

Remark 4.10. ***Hint:*** *As in the exercises above, it is convenient to use the partition of $[0,T]$ into n equal parts. The identity*

$$a^2(b-a) = \frac{1}{3}(b^3 - a^3) - a(b-a)^2 - \frac{1}{3}(b-a)^3 \tag{146}$$

can be applied to transform the sums approximating the stochastic integral. You may also need the following identity:

$$\left(a^2 - b^2\right)^2 = (a-b)^4 + 4(a-b)^3 b + 4(a-b)^2 b^2. \tag{147}$$

3. Properties of the Stochastic Integral

The basic properties of the Itô integral are summarized in the theorem below.

Theorem 4.11. *The following properties hold for any $f, g \in M_t^2$, any $\alpha, \beta \in \mathbb{R}$, and any $0 \leq s < t$:*

(1) ***Linearity:***

$$\int_0^t (\alpha f(r) + \beta g(r))\, dW(r) = \alpha \int_0^t f(r)\, dW(r) + \beta \int_0^t g(r)\, dW(r); \tag{148}$$

(2) ***Isometry:***

$$\mathbb{E}\left(\left|\int_0^t f(r)\, dW(r)\right|^2\right) = \mathbb{E}\left(\int_0^t |f(r)|^2\, dr\right); \tag{149}$$

(3) ***Martingale property:***

$$\mathbb{E}\left(\int_0^t f(r)\, dW(r) \mid \mathcal{F}_s\right) = \int_0^s f(r)\, dW(r). \tag{150}$$

Proof. (1) *If f and g belong to M_t^2, then $\mathbf{1}_{[0,t)} f$ and $\mathbf{1}_{[0,t)} g$ belong to M^2, so there are sequences $f_1, f_2, \ldots$ and $g_1, g_2, \ldots$ in M_{step}^2 approximating $\mathbf{1}_{[0,t)} f$ and $\mathbf{1}_{[0,t)} g$. It follows that $\mathbf{1}_{[0,t)}(\alpha f + \beta g)$ can be approximated by $\alpha f_1 + \beta g_1, \alpha f_2 + \beta g_2, \ldots$. By Exercise 4.3*

$$I(\alpha f_n + \beta g_n) = \alpha I(f_n) + \beta I(g_n) \tag{151}$$

for each n. Taking the L^2 limit on both sides of this equality as $n \to \infty$, we obtain

$$I\left(\mathbf{1}_{[0,t)}(\alpha f + \beta g)\right) = \alpha I\left(\mathbf{1}_{[0,t)} f\right) + \beta I\left(\mathbf{1}_{[0,t)} g\right), \tag{152}$$

which proves (1).

(2) *This follows by approximating $\mathbf{1}_{[0,t)} f$ by random step processes in M_{step}^2 and using Proposition 4.3.*

(3) *If f belongs to M_t^2, then $\mathbf{1}_{[0,t)} f$ belongs to M^2. Let $f_1, f_2, \ldots$ be a sequence of processes in M_{step}^2 approximating $\mathbf{1}_{[0,t)} f$. By Exercise 4.5*

$$\mathbb{E}\left(I\left(\mathbf{1}_{[0,t)} f_n\right) \mid \mathcal{F}_s\right) = I\left(\mathbf{1}_{[0,s)} f_n\right) \tag{153}$$

for each n. By taking the L^2 limit of both sides of this equality as $n \to \infty$, we shall show that

$$\mathbb{E}\left(I\left(\mathbf{1}_{[0,t)} f\right) \mid \mathcal{F}_s\right) = I\left(\mathbf{1}_{[0,s)} f\right), \tag{154}$$

which is what needs to be proved. Indeed, observe that $\mathbf{1}_{[0,s)} f_1, \mathbf{1}_{[0,s)} f_2, \ldots$ is a sequence in M_{step}^2 approximating $\mathbf{1}_{[0,s)} f$, so

$$I\left(\mathbf{1}_{[0,s)} f_n\right) \to I\left(\mathbf{1}_{[0,s)} f\right) \quad \text{in } L^2 \text{ as } n \to \infty. \tag{155}$$

Similarly, $\mathbf{1}_{[0,t)}f_1, \mathbf{1}_{[0,t)}f_2, \ldots$ *is also a sequence in* M^2_{step} *approximating* $\mathbf{1}_{[0,t)}f$, *which implies that*

$$I\left(\mathbf{1}_{[0,t)}f_n\right) \to I\left(\mathbf{1}_{[0,t)}f\right) \quad \text{in } L^2 \text{ as } n \to \infty. \tag{156}$$

The lemma below implies that

$$\mathbb{E}\left(I\left(\mathbf{1}_{[0,t)}f_n\right) \mid \mathcal{F}_s\right) \to \mathbb{E}\left(I\left(\mathbf{1}_{[0,t)}f\right) \mid \mathcal{F}_s\right) \quad \text{in } L^2 \text{ as } n \to \infty, \tag{157}$$

completing the proof.

Lemma 4.12. *If* ξ *and* $\xi_1, \xi_2, \ldots$ *are square integrable random variables such that* $\xi_n \to \xi$ *in* L^2 *as* $n \to \infty$, *then*

$$\mathbb{E}\left(\xi_n \mid \mathcal{G}\right) \to \mathbb{E}\left(\xi \mid \mathcal{G}\right) \quad \text{in } L^2 \text{ as } n \to \infty \tag{158}$$

for any σ*-field* $\mathcal{G}$ *on* Ω *contained in* $\mathcal{F}$.

Proof. *By Jensen's inequality, see Theorem 2.2,*

$$\left|\mathbb{E}\left(\xi_n \mid \mathcal{G}\right) - \mathbb{E}\left(\xi \mid \mathcal{G}\right)\right|^2 = \left|\mathbb{E}\left(\xi_n - \xi \mid \mathcal{G}\right)\right|^2 \leq \mathbb{E}\left(\left|\xi_n - \xi\right|^2 \mid \mathcal{G}\right), \tag{159}$$

which implies that

$$\mathbb{E}\left(\left|\mathbb{E}\left(\xi_n \mid \mathcal{G}\right) - \mathbb{E}\left(\xi \mid \mathcal{G}\right)\right|^2\right) \leq \mathbb{E}\left(\mathbb{E}\left(\left|\xi_n - \xi\right|^2 \mid \mathcal{G}\right)\right) \tag{160}$$

$$= \mathbb{E}\left(\left|\xi_n - \xi\right|^2\right) \to 0 \tag{161}$$

as $n \to \infty$. □

In the next theorem we consider the stochastic integral $\int_0^t f(s)\,dW(s)$ *as a function of the upper integration limit* t. *Similarly as for the Riemann integral, it is natural to ask if this is a continuous function of* t. *The answer to this question involves the notion of a modification of a stochastic process.*

Definition 4.13. *Let* $\xi(t)$ *and* $\zeta(t)$ *be stochastic processes defined for* $t \in T$, *where* $T \subset \mathbb{R}$. *We say that the processes are* ***modifications*** *(or versions) of one another if*

$$P\{\xi(t) = \zeta(t)\} = 1 \quad \text{for all } t \in T. \tag{162}$$

Remark 4.11. *If* $T \subset \mathbb{R}$ *is a countable set, then* (162) *is equivalent to the condition*

$$P\{\xi(t) = \zeta(t) \text{ for all } t \in T\} = 1. \tag{163}$$

However, this is not necessarily so if T *is uncountable.*

The following result is stated without proof.

Theorem 4.14. *Let* $f(s)$ *be a process belonging to* M^2_t *and let*

$$\xi(t) = \int_0^t f(s)\,dW(s) \tag{164}$$

for every $t \geq 0$. Then there exists an adapted modification $\zeta(t)$ of $\xi(t)$ with a.s. continuous paths. This modification is unique up to equality a.s.

From now on we shall always identify $\int_0^t f(s)\,dW(s)$ with the adapted modification having a.s. continuous paths. This convention works beautifully together with Theorem 4.8 whenever there is a need to show that a stochastic integral can be used as the integrand of another stochastic integral, i.e. belongs to M_T^2 for $T \geq 0$. This is illustrated by the next exercise.

Show that

$$\xi(t) = \int_0^t W(s)\,dW(s) \tag{165}$$

belongs to M_T^2 for any $T \geq 0$.

REMARK 4.12. ***Hint:*** *By Theorem 4.14, $\xi(t)$ can be identified with an adapted modification having a.s. continuous trajectories. Because of this, it suffices to verify that $\xi(t)$ satisfies condition* (125) *of Theorem 4.8.*

4. Stochastic Differential and Itô Formula

Any continuously differentiable function $x(t)$ such that $x(0) = 0$ satisfies the formulae

$$x(T)^2 = 2\int_0^T x(t)\,dx(t), \tag{166}$$

$$x(T)^3 = 3\int_0^T x(t)^2\,dx(t), \tag{167}$$

where $dx(t)$ can simply be understood as a shorthand notation for $x'(t)\,dt$, the integrals on the right-hand side being Riemann integrals. Similar formulae have been obtained in Exercises 4.8 and 2 for the Wiener process:

$$W(T)^2 = \int_0^T dt + 2\int_0^T W(t)\,dW(t), \tag{168}$$

$$W(T)^3 = 3\int_0^T W(t)\,dt + 3\int_0^T W(t)^2\,dW(t). \tag{169}$$

*Here the stochastic integrals resemble the corresponding expressions for a smooth function $x(t)$, but there are also the intriguing terms $\int_0^T dt$ and $3\int_0^T W(t)\,dt$. The formulae for $W(T)^2$ and $W(T)^3$ are examples of the much more general Itô formula, a crucial tool for transforming and computing stochastic integrals. Terms such as $\int_0^T dt$ and $3\int_0^T W(t)\,dt$, which have no analogues in the classical calculus of smooth functions, are a feature inherent in the Itô formula and referred to as the **Itô correction**. The class of processes appearing in the Itô formula is defined as follows.*

DEFINITION 4.15. *A stochastic process $\xi(t), t \geq 0$ is called an **Itô process** if it has a.s. continuous paths and can be represented as*

$$\xi(T) = \xi(0) + \int_0^T a(t)\,dt + \int_0^T b(t)\,dW(t) \quad a.s., \tag{170}$$

where $b(t)$ is a process belonging to M_T^2 for all $T > 0$ and $a(t)$ is a process adapted to the filtration $\mathcal{F}_t$ such that

$$\int_0^T |a(t)|\,dt < \infty \quad a.s. \tag{171}$$

for all $T \geq 0$. The class of all adapted processes $a(t)$ satisfying (171) *for some $T > 0$ will be denoted by $\mathcal{L}_T^1$.*

For an Itô process ξ it is customary to write (170) *as*

$$d\xi(t) = a(t)\,dt + b(t)\,dW(t) \tag{172}$$

and to call $d\xi(t)$ the ***stochastic differential*** *of $\xi(t)$. This is known as the Itô differential notation. It should be emphasized that the stochastic differential has no well-defined mathematical meaning on its own and should always be understood in the context of the rigorous equation* (170)*. The Itô differential notation is an efficient way of writing this equation, rather than an attempt to give a precise mathematical meaning to the stochastic differential.*

EXAMPLE 4.1. *The Wiener process $W(t)$ satisfies*

$$W(T) = \int_0^T dW(t). \tag{173}$$

(The right-hand side is the stochastic integral $I(f)$ of the random step process $f = \mathbf{1}_{[0,T)}$.) This is an equation of the form (170) *with $a(t) = 0$ and $b(t) = 1$, which belong, respectively, to $\mathcal{L}_T^1$ and M_T^2 for any $T \geq 0$. It follows that the Wiener process is an Itô process.*

EXAMPLE 4.2. *Every process of the form*

$$\xi(T) = \xi(0) + \int_0^T a(t)\,dt, \tag{174}$$

where $a(t)$ is a process belonging to $\mathcal{L}_T^1$ for any $T \geq 0$, is an Itô process. In particular, every deterministic process of this form, where $a(t)$ is a deterministic integrable function, is an Itô process.

EXAMPLE 4.3. *Since $a(t) = 1$ and $b(t) = 2W(t)$ belong, respectively, to the classes $\mathcal{L}_T^1$ and M_T^2 for each $T \geq 0$,*

$$W(T)^2 = \int_0^T dt + 2\int_0^T W(t)\,dW(t) \tag{175}$$

is an Itô process; see Exercise 4.8. The last equation can also be written as

$$d\left(W(t)^2\right) = dt + 2W(t)\,dW(t), \tag{176}$$

providing a formula for the stochastic differential $d(W(t)^2)$ *of* $W(t)^2$.

Show that $W(t)^3$ *is an Itô process and find a formula for the stochastic differential* $d(W(t)^3)$.

REMARK 4.13. ***Hint:*** *Refer to Exercise 2.*

Show that $tW(t)$ *is an Itô process and find a formula for the stochastic differential* $d(tW(t))$.

REMARK 4.14. ***Hint:*** *Use Exercise 2.*

The above examples and exercises are particular cases of an extremely important general formula for transforming stochastic differentials established by Itô. To begin with, we shall state and prove a simplified version of the formula, followed by the general theorem. The proof of the simplified version captures the essential ingredients of the somewhat tedious general argument, which will be omitted. In fact, many of the essential ingredients of the proof are already present in the examples and exercises considered above.

Theorem 4.16 (Itô formula, simplified version)**.** *Suppose that* $F(t,x)$ *is a real-valued function with continuous partial derivatives* $F'_t(t,x)$, $F'_x(t,x)$ *and* $F''_{xx}(t,x)$ *for all* $t \geq 0$ *and* $x \in \mathbb{R}$. *We also assume that the process* $F'_x(t,W(t))$ *belongs to* M^2_T *for all* $T \geq 0$. *Then* $F(t,W(t))$ *is an Itô process such that*

$$F(T,W(T)) - F(0,W(0)) = \int_0^T \left(F'_t(t,W(t)) + \frac{1}{2} F''_{xx}(t,W(t)) \right) dt + \int_0^T F'_x(t,W(t))\, dW(t) \quad a.s. \tag{177}$$

In differential notation this formula can be written as

$$dF(t,W(t)) = \left(F'_t(t,W(t)) + \frac{1}{2} F''_{xx}(t,W(t)) \right) dt + F'_x(t,W(t))\, dW(t). \tag{178}$$

REMARK 4.15. *Compare the latter with the chain rule*

$$dF(t,x(t)) = F'_t(t,x(t))\, dt + F'_x(t,x(t))\, dx(t) \tag{179}$$

for a smooth function $x(t)$, *where* $dx(t)$ *is understood as a shorthand notation for* $x'(t)\,dt$. *The additional term* $\frac{1}{2}F''_{xx}(t,W(t))\,dt$ *in* (178) *is called the* ***Itô correction***.

PROOF. *First we shall prove the Itô formula under the assumption that* F *and the partial derivatives* F'_x *and* F''_{xx} *are bounded by some* $C > 0$.

Consider a partition $0 = t^n_0 < t^n_1 < \cdots < t^n_n = T$, *where* $t^n_i = \frac{iT}{n}$, *of* $[0,T]$ *into* n *equal parts. We shall denote the increments* $W(t^n_{i+1}) - W(t^n_i)$ *by* $\Delta^n_i W$ *and* $t^n_{i+1} - t^n_i$ *by* $\Delta^n_i t$. *We shall also write* W^n_i *instead of* $W(t^n_i)$ *for brevity. According to the Taylor formula, there is a point* $\tilde{W}^n_i$ *in each interval* $[W(t^n_i), W(t^n_{i+1})]$ *and a point* $\tilde{t}^n_i$ *in each interval* $[t^n_i, t^n_{i+1}]$ *such that*

$$
\begin{aligned}
&F(T,W(T)) - F(0,W(0)) \\
&= \sum_{i=0}^{n-1} \left(F\left(t_{i+1}^n, W_{i+1}^n\right) - F\left(t_i^n, W_i^n\right)\right) \\
&= \sum_{i=0}^{n-1} \left(F\left(t_{i+1}^n, W_{i+1}^n\right) - F\left(t_i^n, W_{i+1}^n\right)\right) + \sum_{i=0}^{n-1} \left(F\left(t_i^n, W_{i+1}^n\right) - F\left(t_i^n, W_i^n\right)\right) \qquad (180)\\
&= \sum_{i=0}^{n-1} F_t'\left(\tilde{t}_i^n, W_{i+1}^n\right)\Delta_i^n t + \sum_{i=0}^{n-1} F_x'\left(t_i^n, W_i^n\right)\Delta_i^n W + \frac{1}{2}\sum_{i=0}^{n-1} F_{xx}''\left(t_i^n, \tilde{W}_i^n\right)\left(\Delta_i^n W\right)^2 \qquad (181)\\
&= \sum_{i=0}^{n-1} F_t'\left(\tilde{t}_i^n, W_{i+1}^n\right)\Delta_i^n t + \frac{1}{2}\sum_{i=0}^{n-1} F_{xx}''\left(t_i^n, W_i^n\right)\Delta_i^n t + \sum_{i=0}^{n-1} F_x'\left(t_i^n, W_i^n\right)\Delta_i^n W \qquad (182)\\
&\quad + \frac{1}{2}\sum_{i=0}^{n-1} F_{xx}''\left(t_i^n, W_i^n\right)\left(\left(\Delta_i^n W\right)^2 - \Delta_i^n t\right) \qquad (183)\\
&\quad + \frac{1}{2}\sum_{i=0}^{n-1} \left[F_{xx}''(t_i^n, \tilde{W}_i^n) - F_{xx}''(t_i^n, W_i^n)\right]\left(\Delta_i^n W\right)^2. \qquad (184)
\end{aligned}
$$

We shall deal separately with each sum in the last expression, splitting the proof into several steps.

Step 1. *We claim that*

$$\lim_{n\to\infty}\sum_{i=0}^{n-1} F_t'\left(\tilde{t}_i^n, W_{i+1}^n\right)\Delta_i^n t = \int_0^T F_t'(t, W(t))\,dt \quad a.s. \tag{185}$$

This is because the paths of $W(t)$ are a.s. continuous, and $F_t'(t,x)$ is continuous as a function of two variables by assumption. Indeed, every continuous path of the Wiener process is bounded on $[0,T]$, i.e. there is an $M > 0$, which may depend on the path, such that

$$|W(t)| \le M \quad \text{for all } t \in [0,T]. \tag{186}$$

As a continuous function, $F_t'(t,x)$ is uniformly continuous on the compact set $[0,T] \times [-M,M]$ and W is uniformly continuous on $[0,T]$. It follows that

$$\lim_{n\to\infty}\sup_{i,t}\left|F_t'(\tilde{t}_i^n, W_{i+1}^n) - F_t'(t, W(t))\right| = 0 \quad a.s., \tag{187}$$

where the supremum is taken over all $i = 0, \ldots, n-1$ and $t \in [t_i^n, t_{i+1}^n]$. By the definition of the Riemann integral this proves the claim.

Step 2. *This is very similar to Step 1. By continuity*

$$\lim_{n\to\infty}\sup_{i,t}\left|F_{xx}''(t_i^n, W_i^n) - F_{xx}''(t, W(t))\right| = 0 \quad a.s., \tag{188}$$

where the supremum is taken over all $i = 0, \dots, n-1$ *and* $t \in [t_i^n, t_{i+1}^n]$. *By the definition of the Riemann integral*

$$\lim_{n\to\infty} \sum_{i=0}^{n-1} F''_{xx}(t_i^n, W_i^n)\Delta_i^n t = \int_0^T F''_{xx}(t, W(t))\, dt \quad a.s. \tag{189}$$

Step 3. *We shall verify that*

$$\lim_{n\to\infty} \sum_{i=0}^{n-1} F'_x(t_i^n, W_i^n)\Delta_i^n W = \int_0^T F'_x(t, W(t))\, dW(t) \quad in\ L^2. \tag{190}$$

If $F'_x(t,x)$ *is bounded by* $C > 0$, *then* $f(t) = F'_x(t, W(t))$ *belongs to* M_T^2 *by Theorem 4.8, and the sequence of random step processes*

$$f_n = \sum_{i=0}^{n-1} F'_x(t_i^n, W_i^n)\mathbf{1}_{[t_i^n, t_{i+1}^n)} \in M^2_{\text{step}} \tag{191}$$

approximates f. *Indeed, by continuity*

$$\lim_{n\to\infty} |f_n(t) - f(t)|^2 = 0 \quad for\ any\ t \in [0,T], \quad a.s. \tag{192}$$

Because $|f_n(t) - f(t)|^2 \le 4C^2$, *it follows that*

$$\lim_{n\to\infty} \int_0^T |f_n(t) - f(t)|^2\, dt = 0 \quad a.s. \tag{193}$$

by Lebesgue's dominated convergence theorem. But $\int_0^T |f_n(t) - f(t)|^2\, dt \le 4TC^2$, *so*

$$\lim_{n\to\infty} \mathbb{E}\left(\int_0^T |f_n(t) - f(t)|^2\, dt\right) = 0 \tag{194}$$

again by Lebesgue's dominated convergence theorem. This shows that f_n *approximates* f, *which in turn implies that* $I(f_n)$ *tends to* $I(f)$ *in* L^2, *concluding Step 3.*

Step 4. *If* F''_{xx} *is bounded by* $C > 0$, *then*

$$\lim_{n\to\infty} \sum_{i=0}^{n-1} F''_{xx}(t_i^n, W_i^n)\left((\Delta_i^n W)^2 - \Delta_i^n t\right) = 0 \quad in\ L^2, \tag{195}$$

since

$$\mathbb{E}\left|\sum_{i=0}^{n-1} F''_{xx}(t_i^n, W_i^n)\left((\Delta_i^n W)^2 - \Delta_i^n t\right)\right|^2$$

$$= \sum_{i=0}^{n-1} \mathbb{E}\left|F''_{xx}(t_i^n, W_i^n)\left((\Delta_i^n W)^2 - \Delta_i^n t\right)\right|^2 \tag{196}$$

$$= \sum_{i=0}^{n-1} \mathbb{E}\left|F''_{xx}(t_i^n, W_i^n)\right|^2 \mathbb{E}\left|(\Delta_i^n W)^2 - \Delta_i^n t\right|^2 \tag{197}$$

$$\leq C^2 \sum_{i=0}^{n-1} \mathbb{E}\left|(\Delta_i^n W)^2 - \Delta_i^n t\right|^2 = 2C^2 \sum_{i=0}^{n-1} (\Delta_i^n t)^2 \tag{198}$$

$$= 2C^2 \sum_{i=0}^{n-1} \frac{T^2}{n^2} = 2C^2 \frac{T^2}{n} \to 0 \quad \text{as } n \to \infty. \tag{199}$$

The first equality above holds because for any $i < j$

$$\mathbb{E}\left[F''_{xx}(t_i^n, W_i^n)\left((\Delta_i^n W)^2 - \Delta_i^n t\right) F''_{xx}(t_j^n, W_j^n)\left((\Delta_j^n W)^2 - \Delta_j^n t\right)\right] \tag{200}$$

$$= \mathbb{E}\left[F''_{xx}(t_i^n, W_i^n)\left((\Delta_i^n W)^2 - \Delta_i^n t\right) F''_{xx}(t_j^n, W_j^n)\right] \mathbb{E}\left[(\Delta_j^n W)^2 - \Delta_j^n t\right] \tag{201}$$

$$= 0. \tag{202}$$

This is because the expressions in the last two square brackets are independent and the last expectation is equal to zero.

Step 5. *By a similar continuity argument as in Steps 1 and 2*

$$\lim_{n\to\infty} \sup_i \left|F''_{xx}(t_i^n, \tilde{W}_i^n) - F''_{xx}(t_i^n, W_i^n)\right| = 0 \quad a.s., \tag{203}$$

where the supremum is taken over all $i = 0, 1, \ldots, n-1$. Since $\sum_{i=0}^{n-1} (\Delta_i^n W)^2 \to T$ in L^2 as $n \to \infty$, there is a subsequence $n_1 < n_2 < \ldots$ such that

$$\sum_{i=0}^{n_k-1} \left(\Delta_i^{n_k} W\right)^2 \to T \quad a.s. \tag{204}$$

as $k \to \infty$. It follows that

$$\left|\sum_{i=0}^{n_k-1} \left(F''_{xx}(t_i^{n_k}, \tilde{W}_i^{n_k}) - F''_{xx}(t_i^{n_k}, W_i^{n_k})\right) \left(\Delta_i^{n_k} W\right)^2\right| \tag{205}$$

$$\leq \sup_i \left|F''_{xx}(t_i^{n_k}, \tilde{W}_i^{n_k}) - F''_{xx}(t_i^{n_k}, W_i^{n_k})\right| \sum_{i=0}^{n_k-1} \left(\Delta_i^{n_k} W\right)^2 \to 0 \quad a.s. \tag{206}$$

as $k \to \infty$.

In those steps above where L^2 convergence was obtained, we also have convergence a.s. by taking a subsequence. This proves the Itô formula (177) *under the assumption that the partial* *derivatives $F'_x(t,x)$ and $F''_{xx}(t,x)$ are bounded. To complete the proof we need to remove this assumption.*

Let $F(t,x)$ be an arbitrary function satisfying the conditions of Theorem 4.16. For each positive integer n take a smooth function φ_n from $\mathbb{R}$ to $[0,1]$ such that $\varphi_n(x) = 1$ for any $x \in [-n,n]$ and $\varphi_n(x) = 0$ for any $x \notin [-n-1, n+1]$. Then

$$F_n(t,x) = \varphi_n(x)F(t,x) \tag{207}$$

also satisfies the conditions of Theorem 4.16 and has bounded partial derivatives $(F_n)'_x(t,x)$ and $(F_n)''_{xx}(t,x)$ for each n. Therefore, by the first part of the proof

$$\begin{aligned} &F_n(T, W(T)) - F_n(0, W(0)) \\ &= \int_0^T \left((F_n)'_t(t, W(t)) + \frac{1}{2}(F_n)''_{xx}(t, W(t)) \right) dt + \int_0^T (F_n)'_x(t, W(t))\, dW(t). \end{aligned} \tag{208}$$

Consider the expanding sequence of events

$$A_n = \left\{ \sup_{t\in[0,T]} |W(t)| < n \right\}. \tag{209}$$

Since $F(t,x) = F_n(t,x)$ for every $t \in [0,T]$ and $x \in [-n,n]$, it follows that (177) holds on A_n. It remains to show that

$$\lim_{n\to\infty} P(A_n) = 1 \tag{210}$$

to prove that (177) holds a.s. But the latter is true because of Doob's maximal L^2 inequality, Theorem 6.7, which implies that

$$n^2(1 - P(A_n)) = n^2 P\left\{ \sup_{t\in[0,T]} |W(t)| \geq n \right\} \tag{211}$$

$$\leq \mathbb{E}\left(\sup_{t\in[0,T]} |W(t)| \right)^2 \tag{212}$$

$$\leq 4\mathbb{E}|W(T)|^2 = 4T, \tag{213}$$

completing the proof. □

EXAMPLE 4.4. *For $F(t,x) = x^2$ we have $F'_t(t,x) = 0$, $F'_x(t,x) = 2x$ and $F''_{xx}(t,x) = 2$. The Itô formula gives*

$$d\left(W(t)^2\right) = dt + 2W(t)\, dW(t), \tag{214}$$

which is the same equality as in Exercise 4.8.

EXAMPLE 4.5. *For $F(t,x) = x^3$ we have $F'_t(t,x) = 0$, $F'_x(t,x) = 3x^2$ and $F''_{xx}(t,x) = 6x$. By the Itô formula we obtain the same equality*

$$d\left(W(t)^3\right) = 3W(t)\,dt + 3W(t)^2\,dW(t) \tag{215}$$

as in Exercise 2.

[Exponential martingale] Show that the exponential martingale $X(t) = e^{W(t)}e^{-t/2}$ is an Itô process and verify that it satisfies the equation

$$dX(t) = X(t)\,dW(t). \tag{216}$$

REMARK 4.16. ***Hint:*** *Use the Itô formula with $F(t,x) = e^x e^{-t/2}$.*

As compared with the simplified version just proved, in the general Itô formula below $W(t)$ will be replaced by an arbitrary Itô process $\xi(t)$ such that

$$d\xi(t) = a(t)\,dt + b(t)\,dW(t), \tag{217}$$

where a belongs to $\mathcal{L}_t^1$ and b to M_t^2 for all $t \geq 0$. In the general case the proof will be omitted.

Theorem 4.17 (Itô formula, general case)**.** *Let $\xi(t)$ be an Itô process as above. Suppose that $F(t,x)$ is a real-valued function with continuous partial derivatives $F_t'(t,x)$, $F_x'(t,x)$ and $F_{xx}''(t,x)$ for all $t \geq 0$ and $x \in \mathbb{R}$. We also assume that the process $b(t)F_x'(t,\xi(t))$ belongs to M_T^2 for all $T \geq 0$. Then $F(t,\xi(t))$ is an Itô process such that*

$$\begin{aligned} dF(t,\xi(t)) = {} & \left(F_t'(t,\xi(t)) + F_x'(t,\xi(t))a(t) + \frac{1}{2}F_{xx}''(t,\xi(t))b(t)^2\right)\,dt \\ & + F_x'(t,\xi(t))b(t)\,dW(t). \end{aligned} \tag{218}$$

A convenient way to remember the Itô formula is to write down the Taylor expansion for $F(t,x)$ up to the terms with partial derivatives of order two, substituting $\xi(t)$ for x and the expression on the right-hand side of (217) for $d\xi(t)$, and using the so-called Itô multiplication table

$$\begin{array}{ll} dt\,dt = 0, & dt\,dW(t) = 0, \\ dW(t)\,dt = 0, & dW(t)\,dW(t) = dt. \end{array} \tag{219}$$

This informal procedure gives

$$\begin{aligned}
dF &= F_t'\,dt + F_x'\,d\xi + \frac{1}{2}F_{tt}''\,dt\,dt + F_{tx}''\,dt\,d\xi + \frac{1}{2}F_{xx}''\,d\xi\,d\xi && (220)\\
&= F_t'\,dt + F_x'(a\,dt + b\,dW) && (221)\\
&\quad + \frac{1}{2}F_{tt}''\,dt\,dt + F_{tx}''\,dt(a\,dt + b\,dW) + \frac{1}{2}F_{xx}''(a\,dt + b\,dW)(a\,dt + b\,dW) && (222)\\
&= F_t'\,dt + F_x'(a\,dt + b\,dW) + \frac{1}{2}F_{xx}''b^2\,dt && (223)\\
&= \left(F_t' + F_x'a + \frac{1}{2}F_{xx}''b^2\right)\,dt + F_x'b\,dW, && (224)
\end{aligned}$$

which is the expression in (218). *Here we have omitted the arguments* $(t, \xi(t))$ *and, respectively,* (t) *in all functions for brevity.*

Applying the Itô formula to $F(t,x) = x^n$, *show that*

$$dW(t)^n = \frac{n(n-1)}{2} W(t)^{n-2}\, dt + nW(t)^{n-1}\, dW(t). \tag{225}$$

Remark 4.17. ***Hint:*** *This is a direct application of the Itô formula, but be careful with the assumptions, in particular make sure that* $nW(t)^{n-1}$ *belongs to* M_T^2 *for all* $T > 0$.

[Ornstein-Uhlenbeck process] Suppose that $\alpha > 0$ *and* $\sigma \in \mathbb{R}$ *are fixed. Define* $Y(t), t \geq 0$ *to be an adapted modification of the Itô integral*

$$Y(t) = \sigma e^{-\alpha t} \int_0^t e^{\alpha s}\, dW(s) \tag{226}$$

with a.s. continuous paths. Show that $Y(t)$ *satisfies*

$$dY(t) = -\alpha Y(t)\, dt + \sigma\, dW(t). \tag{227}$$

Remark 4.18. ***Hint:*** $Y(t) = F(t, \xi(t))$ *with* $\xi(t) = \sigma \int_0^t e^{\alpha s}\, dW(s)$ *and* $F(t,x) = e^{-\alpha t} x$.

5. Stochastic Differential Equations

This section will be devoted to stochastic differential equations of the form

$$d\xi(t) = f(\xi(t))\, dt + g(\xi(t))\, dW(t). \tag{228}$$

Solutions will be sought in the class of Itô processes $\xi(t)$ *with a.s. continuous paths. As in the theory of ordinary differential equations, we need to specify an initial condition*

$$\xi(0) = \xi_0. \tag{229}$$

Here ξ_0 *can be a fixed real number or, in general, a random variable. Being an Itô process,* $\xi(t)$ *must be adapted to the filtration* $\mathcal{F}_t$ *of* $W(t)$, *so* ξ_0 *must be* $\mathcal{F}_0$*-measurable.*

Example 4.6. *The stochastic differential equation*

$$dX(t) = X(t)\, dW(t) \tag{230}$$

was used as a motivation for developing Itô stochastic calculus at the beginning of the present chapter. In Exercise 4 it was verified that the exponential martingale

$$X(t) = e^{W(t)} e^{-t/2} \tag{231}$$

satisfies (230). *It also satisfies the initial condition* $X(0) = 1$. *This is an example of a linear stochastic differential equation. For the solution of a general equation of this type with an arbitrary initial condition, see Exercise 5.*

EXAMPLE 4.7. *In Exercise 4 it was shown that the Ornstein-Uhlenbeck process*

$$Y(t) = \sigma e^{-\alpha t} \int_0^t e^{\alpha s}\, dW(s) \tag{232}$$

satisfies the stochastic differential equation

$$dY(t) = -\alpha Y(t)\, dt + \sigma\, dW(t) \tag{233}$$

with initial condition $Y(0) = 0$. This is an example of an inhomogeneous linear stochastic differential equation. See Exercise 5 for a solution with an arbitrary initial condition.

DEFINITION 4.18. *An Itô process $\xi(t), t \geq 0$ is called a* ***solution*** *of the initial value problem*

$$d\xi(t) = f(\xi(t))\, dt + g(\xi(t))\, dW(t), \tag{234}$$

$$\xi(0) = \xi_0 \tag{235}$$

if ξ_0 is an $\mathcal{F}_0$-measurable random variable, the processes $f(\xi(t))$ and $g(\xi(t))$ belong, respectively, to $\mathcal{L}^1_T$ and M^2_T, and

$$\xi(T) = \xi_0 + \int_0^T f(\xi(t))\, dt + \int_0^T g(\xi(t))\, dW(t) \quad a.s. \tag{236}$$

for all $T \geq 0$.

REMARK 4.19. *In view of this definition, the notion of a stochastic differential equation is a fiction. In fact, only stochastic integral equations of the form (236) have a rigorous mathematical meaning. However, it proves convenient to use stochastic differentials informally and talk of stochastic differential equations to draw on the analogy with ordinary differential equations. This analogy will be employed to solve some stochastic differential equations later on in this section.*

The existence and uniqueness theorem below resembles that in the theory of ordinary differential equations, where it is also crucial for the right-hand side of the equation to be Lipschitz continuous as a function of the solution.

Theorem 4.19. *Suppose that f and g are Lipschitz continuous functions from $\mathbb{R}$ to $\mathbb{R}$, i.e. there is a constant $C > 0$ such that for any $x, y \in \mathbb{R}$*

$$|f(x) - f(y)| \leq C|x - y|, \tag{237}$$

$$|g(x) - g(y)| \leq C|x - y|. \tag{238}$$

Moreover, let ξ_0 be an $\mathcal{F}_0$-measurable square integrable random variable. Then the initial value problem

$$d\xi(t) = f(\xi(t))\, dt + g(\xi(t))\, dW(t), \tag{239}$$

$$\xi(0) = \xi_0 \tag{240}$$

has a solution $\xi(t), t \geq 0$ *in the class of Itô processes. The solution is unique in the sense that if* $\eta(t), t \geq 0$ *is another Itô process satisfying* (239) *and* (240)*, then the two processes are identical a.s., that is,*

$$P\{\xi(t) = \eta(t) \text{ for all } t \geq 0\} = 1. \tag{241}$$

Proof (Outline). *Let us fix* $T > 0$*. We are looking for a process* $\xi \in M_T^2$ *such that*

$$\xi(s) = \xi_0 + \int_0^s f(\xi(t))\,dt + \int_0^s g(\xi(t))\,dW(t) \quad a.s. \tag{242}$$

for all $s \in [0, T]$*. Once we have shown that such a* $\xi \in M_T^2$ *exists, to obtain a solution to the stochastic differential equation* (239) *with initial condition* (240) *it suffices to take a modification of* ξ *with a.s. continuous paths, which exists by Theorem 4.14.*

To show that a solution to the stochastic integral equation (242) *exists we shall employ the Banach fixed point theorem in* M_T^2 *with the norm*

$$\|\xi\|_\lambda^2 = \mathbb{E}\int_0^T e^{-\lambda t}|\xi(t)|^2\,dt, \tag{243}$$

which turns M_T^2 *into a complete normed vector space. The number* $\lambda > 0$ *should be chosen large enough, see below. To apply the fixed point theorem define* $\Phi : M_T^2 \to M_T^2$ *by*

$$\Phi(\xi)(s) = \xi_0 + \int_0^s f(\xi(t))\,dt + \int_0^s g(\xi(t))\,dW(t) \tag{244}$$

for any $\xi \in M_T^2$ *and* $s \in [0, T]$*. We claim that* Φ *is a strict contraction, i.e.*

$$\|\Phi(\xi) - \Phi(\zeta)\|_\lambda \leq \alpha\|\xi - \zeta\|_\lambda \tag{245}$$

for some $\alpha < 1$ *and all* $\xi, \zeta \in M_T^2$*. Then, by the Banach theorem,* Φ *has a unique fixed point* $\xi = \Phi(\xi)$*. This is the desired solution to* (242)*.*

It remains to verify that Φ *is indeed a strict contraction. It suffices to show that the two maps* Φ_1 *and* Φ_2*, where*

$$\Phi_1(\xi)(s) = \int_0^s f(\xi(t))\,dt, \quad \Phi_2(\xi)(s) = \int_0^s g(\xi(t))\,dW(t), \tag{246}$$

are strict contractions with contracting constants α_1 *and* α_2 *such that* $\alpha_1 + \alpha_2 < 1$*. For* Φ_1 *this follows from the Lipschitz continuity of* f*. For* Φ_2 *we need to use the Lipschitz continuity of* g *and the isometry property of the Itô integral. Let us mention just one essential step in the latter case. For any* $\xi, \zeta \in M_T^2$

$$\|\Phi_2(\xi) - \Phi_2(\zeta)\|_\lambda^2 = \mathbb{E}\int_0^T e^{-\lambda s} \left| \int_0^s [g(\xi(t)) - g(\zeta(t))]\, dW(t) \right|^2 ds \tag{247}$$

$$= \mathbb{E}\int_0^T e^{-\lambda s} \int_0^s |g(\xi(t)) - g(\zeta(t))|^2 \, dt\, ds \tag{248}$$

$$\leq C^2 \mathbb{E}\int_0^T e^{-\lambda s} \int_0^s |\xi(t) - \zeta(t)|^2 \, dt\, ds \tag{249}$$

$$= C^2 \mathbb{E}\int_0^T \left(\int_t^T e^{-\lambda s} e^{\lambda t} \, ds \right) e^{-\lambda t} |\xi(t) - \zeta(t)|^2 \, dt \tag{250}$$

$$\leq \frac{C^2}{\lambda} \mathbb{E}\int_0^T e^{-\lambda t} |\xi(t) - \zeta(t)|^2 \, dt = \frac{C^2}{\lambda} \|\xi - \zeta\|_\lambda^2, \tag{251}$$

since $\int_t^T e^{-\lambda s} e^{\lambda t}\, ds = \frac{1}{\lambda}(1 - e^{-\lambda(T-t)}) \leq \frac{1}{\lambda}$. *Here* C *is the Lipschitz constant of* g. *If* $\lambda > C^2/\varepsilon$, *then* Φ_2 *is a strict contraction with contracting constant* $\leq \varepsilon$.

There remain some technical points to be settled, but the main idea of the proof is shown above. □

Find a solution of the stochastic differential equation

$$dX(t) = -\alpha X(t)\, dt + \sigma\, dW(t) \tag{252}$$

with initial condition $X(0) = x_0$, *where* x_0 *is an arbitrary real number. Show that the solution is unique.*

Remark 4.20. ***Hint:*** *Use the substitution* $Y(t) = e^{at} X(t)$.

A linear stochastic differential equation has the general form

$$dX(t) = aX(t)\, dt + bX(t)\, dW(t), \tag{253}$$

where a *and* b *are real numbers. In particular, for* $a = 0$ *and* $b = 1$ *we obtain the stochastic differential equation* $dX(t) = X(t)\, dW(t)$ *in Example 4.6. The solution to the initial value problem for any linear stochastic differential equation can be found by exploiting the analogy with ordinary differential equations, as presented in the exercises below.*

Suppose that $w(t), t \geq 0$ *is a deterministic real-valued function of class* C^1 *such that* $w(0) = 0$. *Solve the ordinary differential equation*

$$dx(t) = ax(t)\, dt + bx(t)\, dw(t), \tag{254}$$

with initial condition $x(0) = x_0$ *to find that*

$$x(t) = x_0 e^{at + bw(t)}. \tag{255}$$

(We write $dw(t)$ *in place of* $w'(t)\, dt$ *to emphasize the analogy with stochastic differential equations.)*

Remark 4.21. ***Hint:*** *The variables can be separated:*

$$\frac{dx(t)}{x(t)} = (a + bw'(t))\, dt. \tag{256}$$

By analogy with the deterministic solution (255)*, let us consider a process defined by*

$$X(t) = X_0 e^{at+bW(t)} \tag{257}$$

for any $t \geq 0$*, where* $W(t)$ *is a Wiener process.*

Show that $X(t)$ *defined by* (257) *is a solution of the linear stochastic differential equation*

$$dX(t) = \left(a + \frac{b^2}{2}\right) X(t)\, dt + bX(t)\, dW(t), \tag{258}$$

with initial condition $X(0) = X_0$*.*

Remark 4.22. ***Hint:*** *Use the Itô formula with* $F(t,x) = e^{at+bx}$*.*

Show that the linear stochastic differential equation

$$dX(t) = aX(t)\, dt + bX(t)\, dW(t) \tag{259}$$

with initial condition $X(0) = X_0$ *has a unique solution given by*

$$X(t) = X_0 e^{\left(a - \frac{b^2}{2}\right)t + bW(t)}. \tag{260}$$

Remark 4.23. ***Hint:*** *Apply the result of Exercise 5 with suitably redefined constants.*

Having solved the general linear stochastic differential equation (253)*, let us consider an example of a non-linear stochastic differential equation. Once again, we begin with a deterministic problem.*

Suppose that $w(t), t \geq 0$ *is a deterministic real-valued function of class* C^1 *such that* $w(0) = 0$*. Solve the ordinary differential equation*

$$dx(t) = \sqrt{1 + x(t)^2}\, dt + \sqrt{1 + x(t)^2}\, dw(t) \tag{261}$$

with initial condition $x(0) = x_0$*.*

Remark 4.24. ***Hint:*** *The variables in this differential equation can be separated.*

Show that the process defined by

$$X(t) = \sinh(C + t + W(t)), \tag{262}$$

where $W(t)$ *is a Wiener process and* $C = \sinh^{-1} X_0$*, is a solution of the stochastic differential equation*

$$dX(t) = \left(\sqrt{1 + X(t)^2} + \frac{1}{2} X(t)\right) dt + \left(\sqrt{1 + X(t)^2}\right) dW(t) \tag{263}$$

with initial condition $X(0) = X_0$*.*

REMARK 4.25. ***Hint:*** *Use the Itô formula with* $F(t,x) = \sinh(t+x)$.

We shall conclude this chapter with an example of a stochastic differential equation which does not satisfy the assumptions of Theorem 4.19. It turns out that the solution may fail to exist for all times $t \geq 0$. *This is a familiar phenomenon in ordinary differential equations. However, stochastic differential equations add a new effect, which does not even make sense in the deterministic case: the maximum time of existence of the solution, called the* ***explosion time*** *may be a (non-constant) random variable, in fact a stopping time.*

EXAMPLE 4.8. *Consider the stochastic differential equation*

$$dX(t) = X(t)^3\,dt + X(t)^2\,dW(t). \tag{264}$$

Then

$$X(t) = \frac{1}{1-W(t)} \tag{265}$$

is a solution, which can be verified, at least formally, by using the Itô formula with $F(t,x) = \frac{1}{1-x}$. *The solution* $X(t)$ *exists only up to the first hitting time*

$$\tau = \inf\{t \geq 0 : W(t) = 1\}. \tag{266}$$

This is the explosion time of $X(t)$. *Observe that*

$$\lim_{t \nearrow \tau} X(t) = \infty. \tag{267}$$

Strictly speaking, the Itô formula stated in Theorem 4.17 does not cover this case, since $F(t,x) = \frac{1}{1-x}$ *has a singularity at* $x = 1$. *Definition 4.18 does not apply either, as it requires the solution* $X(t)$ *to be defined for all* $t \geq 0$. *Suitable extensions of the Itô formula and the definition of a solution are required to study stochastic differential equations involving explosions. However, to prevent an explosion of this book, we have to refer the interested reader to a further course in stochastic analysis.*

6. Solutions